SIADS 699 • UNIVERSITY OF MICHIGAN • MASTER OF APPLIED DATA SCIENCE

Beyond Prediction: Prescriptive AI for Strategic Energy Capital Allocation

A governed decision-intelligence system for risk, returns and investment-committee alternatives across oil, gas, power, renewables and storage

Alan Mössinger | Group 13 (solo project) | Mentor: Laura Danielle Stagnaro

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Abstract

An energy company can predict which power projects might be built, how much an oil lease might produce and where prices may move — and still allocate capital poorly. Why? Because the decision concerns a portfolio, not a list of forecasts. This capstone tests what changes when a committee moves from prediction to prescription: joint uncertainty analysis, risk-aware optimization and a governed AI challenge, each scored on the decision it feeds. In a decision dated 1 January 2025 and made only with what was public that day, the system allocates a $3.82 billion budget across 20 Gulf of Mexico lease redevelopments and 20 Texas-grid interconnection requests; portfolios are evaluated on 2,000 held-out scenarios under pre-registered criteria. Better forecasts alone do not produce a demonstrably better portfolio. Risk-aware optimization does: it improves the modelled risk-adjusted objective by $219 million [+$13M, +$401M] against expected-value ranking, and by $312 million [+$110M, +$506M] once every portfolio is financed the same way. In 48 simulated committee cases, a governed language-model challenger adds $43 million per case beyond deterministic checks but does not beat a strong optimizer-first rule. These are simulated decision outcomes, not realized investment returns.

Executive decision in one minute

Question	Finding
What should the committee fund?	Five lease redevelopments and eight power requests, subject to its chosen risk appetite.
How much capital?	$3.78B of the $3.82B budget, including construction reserves; arrange a $2.53B contingent facility.
Why not rank by expected NPV?	It trades expected value for smaller losses in the worst 5% of modeled outcomes.
Where does the language model help?	Reading rules and questioning arguments—not calculating or signing.
What is not demonstrated?	Realized returns, general superiority of machine-learning forecasts, or better portfolio choices than a strong optimizer-first rule.

1. The problem: one budget, risks that move together

An investment committee rarely decides between comparable projects. A producing Gulf of Mexico lease can be redeveloped now, while a proposed Texas wind, solar, gas or storage plant may spend years in a queue and never enter service. Both compete for the same capital, yet their timing, costs and exposure to market shocks differ.

Traditional project ranking answers a useful but incomplete question: **Which asset has the highest expected net present value (NPV)?** It does not establish which combination can be financed, how several assets might lose money together, or whether management should accept a larger expected return at the cost of a deeper downside. A portfolio of individually attractive assets may therefore be a poor company-wide decision.

This report asks a narrower, testable question: **Does adding uncertainty, portfolio optimization and a governed challenge process produce a better capital-allocation decision than prediction and ranking alone?**

2. The decision and the questions it creates

The committee is placed on **1 January 2025**, with **$3.82 billion**—about 40% of the aggregate modeled cost of the 40 candidates. It must consider **20 existing Gulf of Mexico oil and gas leases** for redevelopment and **20 ERCOT interconnection requests** for new generation or storage. Only information public by that date is used; later outcomes are kept for evaluation.

The committee imposes a budget constraint and a **40% limit per technology**. A lease consumes its redevelopment capital immediately. The model uses a $40,000 cost per additional barrel per day and assumes a 20% production lift after two years. An interconnection request starts with deposits and study costs; construction capital becomes relevant when a signed agreement triggers the next stage. These costs and timing rules are modeled assumptions, not actual bids or lender commitments.

The historical queue illustrates the asymmetry: among capacity requesting connection in 2000–2019, **12.6% was operating** and **76.9% had withdrawn** by the end of 2024. A request may be cheap to preserve as an option but expensive to build. Leases share oil-price exposure; requests share power-market exposure.

Four questions follow: **are better forecasts enough, what happens when projects lose together, what should be funded, and can AI challenge a committee without replacing it?**

3. The solution: six separated stages

Six stages do distinct work: **prediction** estimates project completion, output and prices; **uncertainty** generates joint futures; **prescription** selects a portfolio under budget and risk rules; **AI challenge** tests the committee's stated rationale; **human authorization** records a named decision; and **evaluation** compares alternatives on scenarios not used to choose them. The separation matters: a language model cannot take credit for calculations performed by the optimizer, and a forecast cannot take credit for a decision it does not choose.

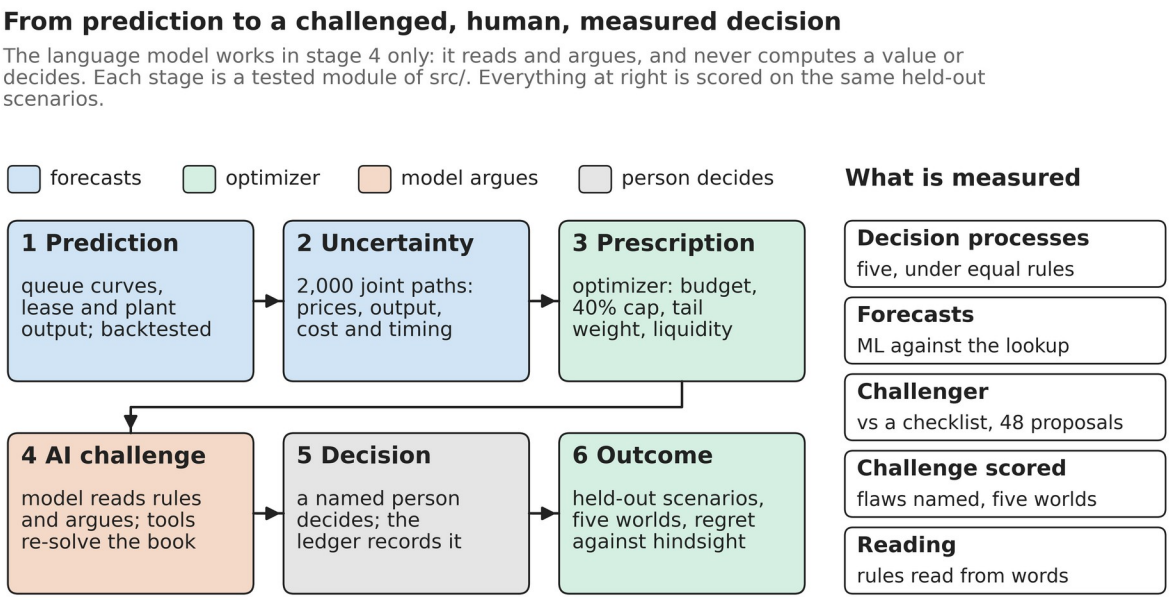


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: DERIVED. Cohort: the forty candidates on 1 January 2025

Figure 1. Six stages of the decision process and the evidence measured at each stage.

Figure 2 shows the division of labor: the committee confirms its rules before any calculation, tools compute every number, the model explains or challenges, and a named person signs.

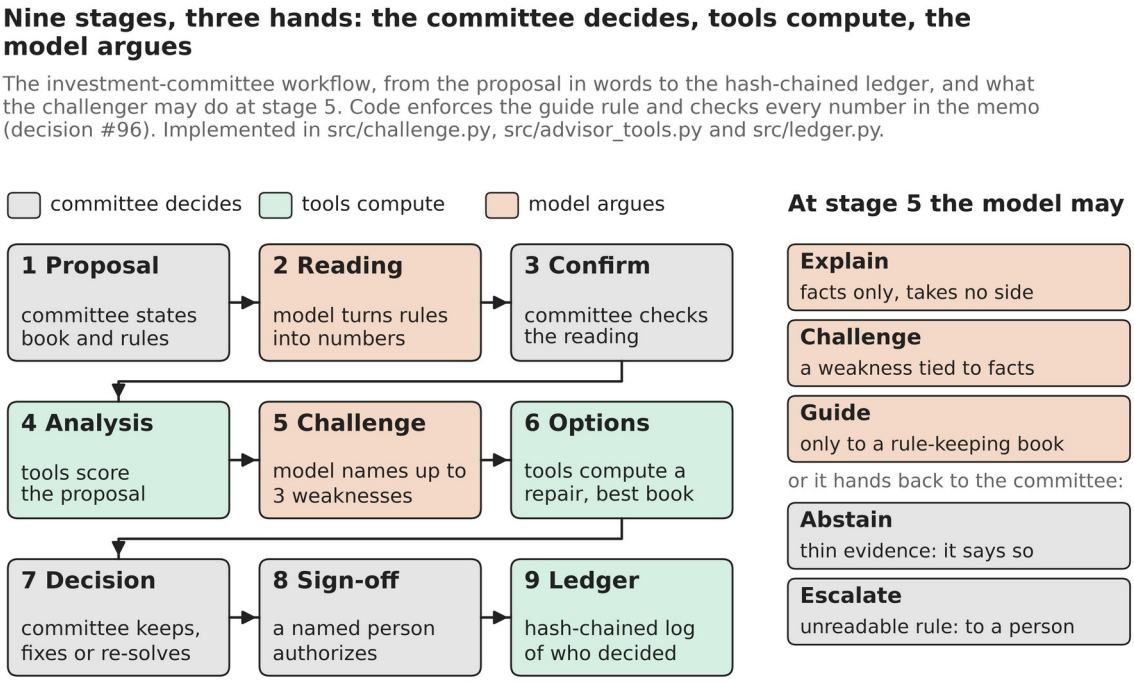


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: DERIVED. Cohort: the decision workflow of the evaluated system

Figure 2. Committee workflow: humans decide, tools calculate, and the model communicates bounded challenges.

4. First question: are better predictions enough?

The prediction layer includes a discrete-time LightGBM survival model. For each request and each year it remains in the queue, the model estimates whether it will begin operating that year from information such as size, technology, region, entry cohort and queue conditions. A two-year update incorporates the status of an interconnection agreement. Requests with an agreement at that point went on to operate far more often than those without: **56% versus 5%** within three years.

The benchmark is a transparent lookup of earlier requests in the same region, technology and size band. The main model was fitted to **13,366** pre-2017 requests and evaluated on **2,268** entering in 2017–2019. Its ranking AUC was **0.731**, compared with **0.686** for the lookup. But the predicted five-year ERCOT completion rate was **0.380**, while the observed rate was **0.238**. The model was more useful at ordering projects than at estimating how many would operate.

The 2025 forward check reinforces the calibration problem. Among **1,441** live ERCOT requests, **30** entered operation; the raw model expected **125**, the level-corrected model **78**, and the lookup **17**. In twelve forecast contests for energy output and prices, machine learning beat transparent baselines in three, all involving plant output. Price forecasts did not beat a random walk.

One year on, 30 requests operated where the corrected model expected 78

Left: realized one-year operation by bin of the predicted chance, for the level-corrected model and the lookup, on the 1,430 of the 1,441 ERCOT requests live on 1 January 2025 that the 2026 edition resolves (11 are not in it); equal-count bins, the lookup's ties kept together. Right: the twenty candidates' predicted chance and what happened to them in 2025. A one-year check of seven-year curves; post-hoc, nothing refitted.

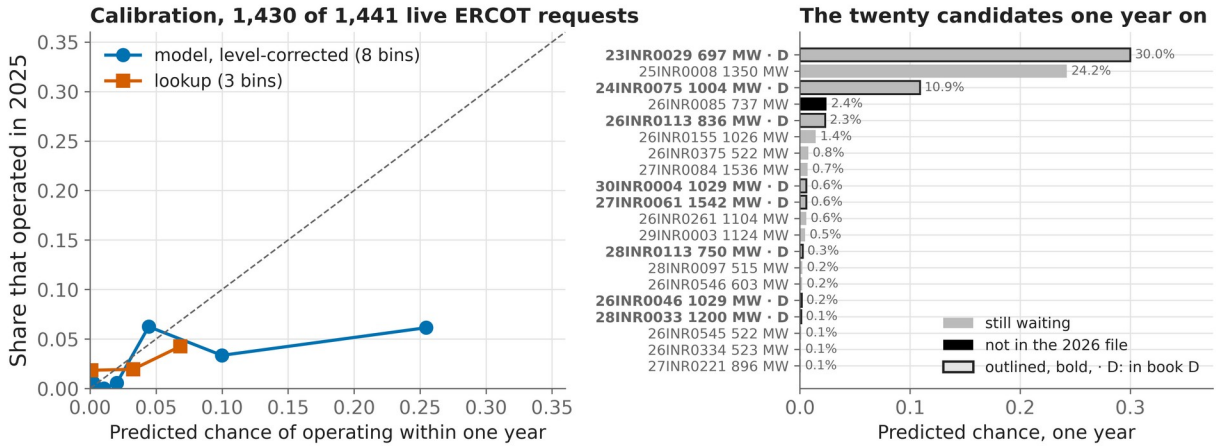


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). LBNL Queued Up: 2026 Edition (Rand et al., 2026). Provenance: OBSERVED. Cohort: ERCOT requests of 20 MW and more live on 1 January 2025

Figure 3. Observed queue outcomes against decision-date completion forecasts.

Most importantly, the portfolio selected from the model's completion curves did **not** outperform the lookup-based portfolio convincingly: **+$37 million [−$76M, +$151M]**, an interval that includes zero. Higher AUC did not create a demonstrably better investment choice. The decision depends on the probability level, the size of future capital calls and portfolio risk—not ranking accuracy alone.

5. Second question: what happens when projects lose together?

An expected NPV summarizes many possible outcomes in one number. The next stage instead generates **2,000 twenty-year scenarios** using joint changes in WTI oil, Henry Hub gas and ERCOT power prices from **2018–2024**. The three markets are resampled in aligned calendar-year blocks so their observed relationships are not discarded. Price paths revert toward historical means with a four-year half-life.

Separate draws cover output, from forecast errors; timing, from each request's completion distribution; and lease costs, one multiplier per scenario on every lease's workover, operating and development cost, triangular between **0.75 and 1.5** (plant construction costs stay at their NREL ATB values). One scenario set chooses portfolios; an independent set of **2,000 held-out scenarios** scores them, so no decision is graded on the scenarios used to choose it.

The results report both **expected NPV** and the **average loss among the worst 5% of full-horizon scenarios**, a conditional-value-at-risk (CVaR) measure. The latter is not a loss in one calendar year; it is the average of the weakest 100 modeled twenty-year portfolio outcomes. A tail weight, **λ**, states how strongly the committee penalizes that loss relative to expected value.

When candidate outcomes are shuffled to remove their joint movements, the recommended book's worst-5% outcome changes from a modeled **$0.82 billion average loss** to a small profit. That comparison is an ablation within the simulation: it shows why a portfolio's co-movement matters; not how future markets must behave.

6. Third question: what should the company actually fund?

The optimizer, a mixed-integer program, funds each candidate or not to maximize **expected NPV minus λ times the modeled worst-5% average loss**, within the budget and technology cap. At **λ = 0**, it ignores downside: the selected 17-asset book expects **$2.73 billion** and carries a **$1.52 billion** worst-5% loss on the static basis. At the registered **λ = 1**, the selected book has **13 assets**, expected NPV of **$2.17 billion** and a **$0.82 billion** tail loss.

The risk-aware selection contains **five lease redevelopments and eight power requests**. Three power requests already have signed agreements—a 1,004 MW solar project, a 697 MW gas project and an 836 MW wind project. Five additional wind and solar requests are staged options, exposed mainly through small deposits. Each is likely to lose its deposit; together they partly offset the lease book's exposure. This is a **modeled hedge**, not proof of future market protection.

Under one cap, risk aversion trades expected value for a thinner tail

Left: books chosen on 2,000 primary scenarios, scored on 2,000 held-out ones; committee grid, contracted case; budget $3.82B, 40% of the forty stage-one charges; no technology above 40% of it. Right: what each distinct book on the λ path spends, by technology.

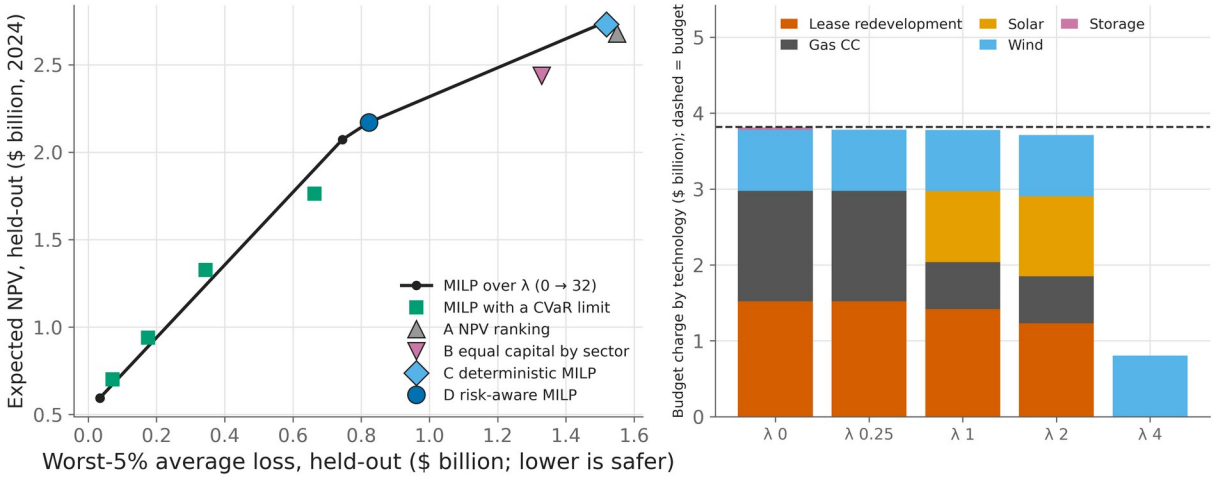


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025 (notebook 07); scenarios of notebook 08

Figure 4. Expected value versus worst-5% loss as the committee changes the tail penalty.

Figure 5 shows the distributions behind these numbers. Each panel reads the same 2,000 held-out futures through one of the five decision processes of Section 7. The conventional books earn more in the best futures—NPV ranking and the deterministic optimizer each have 35 outcomes above $12 billion—and lose more in the worst, down to −$4.0 billion. The risk-aware book has three outcomes above $12 billion, a worst outcome of −$2.0 billion and a worst-5% average loss of $0.82 billion against $1.52–1.55 billion. The thinner tail is bought with the upside; that is the trade the tail weight prices.

The highest expected return is not the best investment: the risk-aware book keeps the thinnest tail

Each panel is the same 2,000 held-out futures read through a different book: a book's outcome in a future is the sum of its members' NPVs in that future. The worst 100 are shaded; the dashed lines mark the expected NPV (the mean of all 2,000) and the worst-5% average loss (the mean of the 100 worst), the two numbers the registered metric subtracts. Static basis, committee grid.

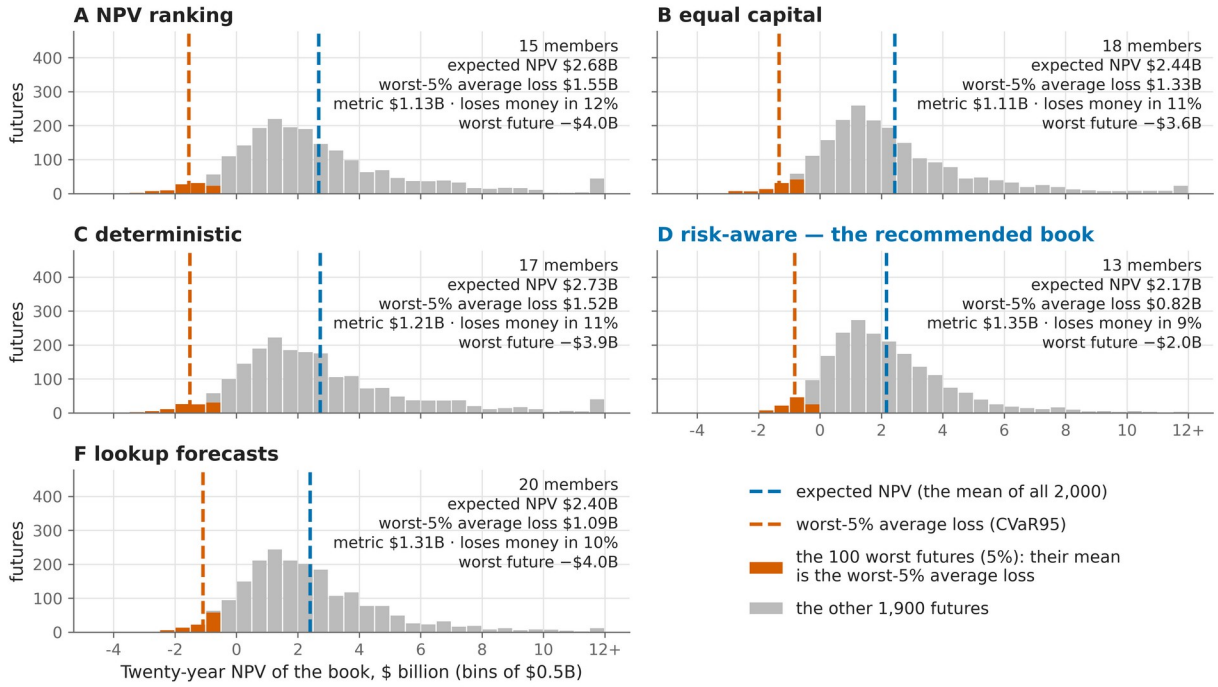


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates public on 1 January 2025; 2,000 held-out scenarios

Figure 5. The 2,000 held-out outcomes of each decision process, the 100 worst shaded; their mean is the worst-5% average loss.

The recommendation uses **$3.78 billion** of the **$3.82 billion** budget. The actions: **invest now in eight positions; stage five; defer twelve requests; monitor fourteen leases; reject one candidate**. The labels are for human review, not automatic execution.

Financing changes the result. The construction reserve is an average; modeled calls arrive unevenly. In half the scenarios, requests demand more than the reserved amount; at the 95th percentile, that excess reaches **$2.53 billion**. Without a contingent facility, plants that sign may not be buildable under the model's cash rule, cutting expected NPV from **$2.17 billion to $1.71 billion**. With a **$2.53 billion** facility and modeled financing terms of a **0.5% fee and 8% interest**, expected NPV becomes **$1.98 billion**. The book and its funding are therefore one recommendation, conditional on real financing terms.

7. Four strategies and a forecast check: the scorecard

The five decision processes below face the same budget, technology cap and held-out scenarios: **A** ranks by expected NPV; **B** splits capital across sectors; **C** optimizes expected NPV without a tail

penalty; **D** optimizes with λ = 1; and **F** repeats D using the completion lookup rather than the machine-learning model. Each financed result reflects its own 95th-percentile contingent facility, net of modeled fees and interest.

Process	Assets	Financed expected NPV	Financed worst-5% loss	Financed risk-adjusted metric
A · NPV ranking	15	$2.47B	$1.75B	$0.72B
B · equal capital	18	$2.21B	$1.50B	$0.71B
C · deterministic optimizer	17	$2.52B	$1.70B	$0.82B
D · risk-aware optimizer	13	$1.98B	$0.95B	$1.03B
F · lookup-based risk-aware	20	$2.21B	$1.24B	$0.96B

On the pre-registered static comparison, **D exceeds A by $219M [+$13M, +$401M]**. It gives up approximately **$508M** of expected value to reduce the tail materially. With comparable financing, its advantage is **$312M [+$110M, +$506M]** over A and **$213M [+$17M, +$412M]** over C. Against F, the financed difference is **+$70M [−$51M, +$201M]**: not enough to establish that the machine-learning completion curves improved allocation.

Under the base model and six stresses on prices, costs, timing and completion (Appendix K), D scores best in **all seven modeled worlds**. This robustness is conditional on the scenarios chosen. Without a contingent facility, **B** is better on the risk-adjusted metric (**$0.87B versus $0.76B**). The recommendation must carry that exception.

Funded the same way, the risk-aware book still scores best

Left: the registered metric of five decision processes on three funding bases — static (every agreement built), no line (calls beyond the reserve forfeit), and each book's own 95th-percentile line net of a 0.5% fee and 8% interest. Right: capital committed on the day, the construction reserve, and the contingent facility each book needs, with the expected present value of its financing cost.

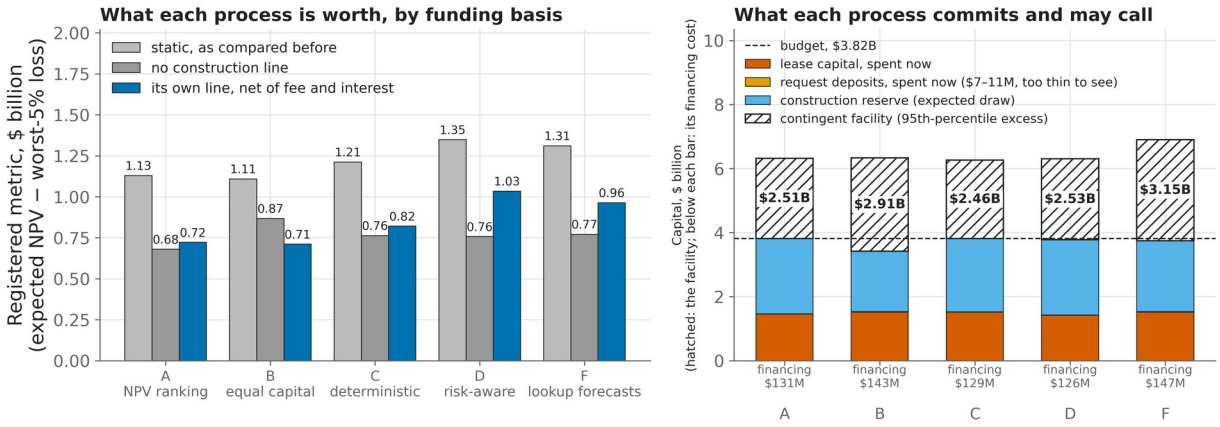


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates public on 1 January 2025; 2,000 held-out scenarios

Figure 6. Risk-adjusted value across financing assumptions and required capital.

8. Fourth question: can AI challenge a committee without replacing it?

A committee can follow an apparently reasonable story into a weak decision: “we already paid the deposits,” “this project is exceptional,” or “we will find the money when construction starts.” A spreadsheet can flag broken capital limits; it cannot reliably identify the reasoning behind the proposal. The challenger is a language model assigned this limited job—not the authority to choose investments or forecast.

The model first turns the committee’s written preferences into rules, which the committee confirms. Deterministic tools then score the proposal and compute, under those rules, the smallest repair and the optimizer’s book. From that fact sheet the model names at most three vulnerabilities and can **explain, challenge, guide, abstain or escalate**; code verifies every number and allows guidance only toward a compliant book that clearly beats the proposal. A named person approves or overrides; a hash-chained ledger records each step.

The registered experiment used **48 simulated committee proposals**: 24 generated with computable flaws, 12 author-written with reasoning flaws and 12 author-written sound cases, along with **40 paraphrases** and **20 injected instructions**.

One illustrative proposal argued that sunk deposits and lease bonuses justified continuing with the book. The tools found **$2.53 billion** of unfunded construction exposure; the challenger named the sunk-cost argument and asked what funding could actually be arranged. The numerical verifier also caught an unsupported number in the memo.

9. What the challenger added—and where it failed

Decision process	Average committee objective	Incremental evidence
Unchallenged committee proposal	$863M / case	Starting point
Deterministic tools only	$939M / case	+$76M [+$13M, +$140M] over proposal
Tools + language model	$981M / case	+$43M [+$9M, +$86M] over tools
Fixed optimizer-first rule	$995M / case	Challenger edge: −$14M [−$36M, +$2M]
Optimizer under the committee’s rules	$1,116M / case	+$134M [+$72M, +$211M] over challenger

The $76M increase from tools and the further $43M from the language model are **modeled differences in the committee’s own objective**, averaged across cases and five worlds. The challenger named **12 of 12** reasoning flaws and did not over-challenge any of the **12 sound** proposals.

But its economic edge vanishes against a stronger baseline: a fixed rule that chooses the optimizer’s compliant book first. That rule performed slightly better on the observed comparison, with an interval that includes zero. The language model’s demonstrated contribution is **interpreting words and explaining weaknesses**, not selecting a superior portfolio independently of optimization.

The governance tests expose real limits. The model read **38 of 48** original rules exactly, below the **43 of 48** threshold, because it sometimes treated an unstated risk weight as zero. On injected

instructions it achieved only **75%** resistance, with **seven** memos containing an unsupported number; that failed the registered rule. The confirmation gate and verifier caught both in testing but do not make the model immune to future attacks.

Against an optimizer-first rule, the language model's edge disappears (−$14M)

Left: the committee's own objective under five arms, mean over the 48 registered v10 cases and five worlds, the stated line net of a 0.5% fee and 8% interest. Right: paired per-case rungs; the three registered rungs as scored and two added after registration for the optimizer-first rule, which takes the optimizer's book whenever it keeps the rules and either fixes a breach or beats the proposal clear of zero, reading no words.

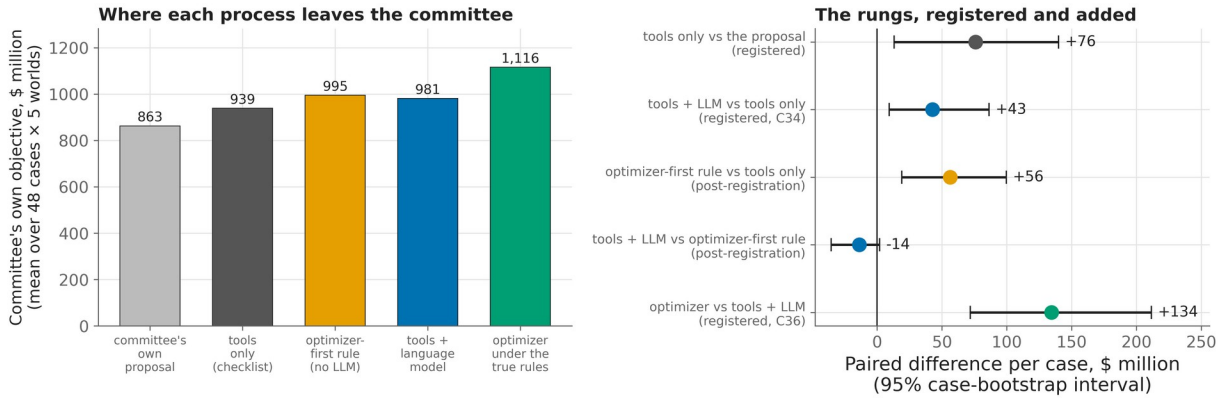


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: 48 simulated committee cases on the forty candidates public on 1 January 2025

Figure 7. Decision quality across human proposal, tool-only, challenger and optimizer arms.

10. What the registered evidence supports

The project registers criteria before scoring and keeps first-scored verdicts visible when later audits change data, constraints or methods. In total, **41 criteria cover 87 scored lines; 40 lines pass**. The nine lines of the final experiment, registered on 22 October 2026, are summarized below; two fail and one is reported without a pass criterion.

Criterion	Plain-language test	Result
C33	Risk-aware portfolio versus NPV ranking	+$219M [+$13M, +$401M] · PASS
C34	Language model beyond deterministic checks	+$43M [+$9M, +$86M] · PASS
C35	Deterministic checks versus proposal	+$76M [+$13M, +$140M] · PASS
C36	Optimizer advantage over challenger	+$134M [+$72M, +$211M] · reported
C37	Original rule reading without changes	38/48 · FAIL
C38	Accuracy on paraphrased rules	100% · PASS
C39	Resistance to injected instructions	75%; seven affected memos · FAIL
C40	Avoid needless challenges	0% of sound proposals · PASS
C41	Identify flaws of argument	12/12 · PASS

On the earlier candidate set and under the registered cost grid, the first portfolio comparison was not supported as scored; the corrected system reverses that against NPV ranking but not against equal capital, and the registered failures remain on record. Likewise, intervals from resampled modeled futures should not be read as guarantees about future market behavior.

11. Limitations and the next decisions

The portfolio numbers depend on public data, a modeled lease-cost grid, a 2018–2024 market-history window, chosen stress cases, price mean reversion, and simulated construction and financing. The seven favorable stress outcomes are not seven independent historical successes. The baseline primary interval clears zero by just **$13M**. The committee trials were written or generated for an experiment; they did not involve real investment professionals making real commitments.

Before operational use, obtain **actual redevelopment cost estimates** and **bank terms for the contingent facility**; test scenarios that go beyond the historical resampling regime; set explicit defaults for rules the committee leaves unstated; improve and adversarially test instruction handling and numerical verification; and run a **pre-registered study with real committee participants**. Governance controls should be evaluated in the environment where the decision would be used. Ethically, a completion score must not screen grid access: it would penalize small developers (Appendix K.8).

12. Conclusion: beyond prediction is a decision discipline

The case shows the limit of better forecasting: a model can rank projects better and still fail to improve the portfolio built from them. The value appeared when risk was modeled jointly, the portfolio was optimized under explicit management preferences, and the required construction financing was treated as part of the investment decision.

The risk-aware book was stronger than expected-NPV ranking on the registered modeled objective, but only with contingent financing. Deterministic tools did most of the work in improving simulated committee decisions; the language model added an ability to detect argumentative flaws and explain alternatives, not evidence that it should replace an optimizer or a human committee.

The practical sequence is: define the decision, test the forecasts, simulate correlated uncertainty, optimize within the company's limits, finance the resulting commitments, challenge the reasoning and retain a signed human decision. Prediction is one input. A defensible allocation is where the money is made.

Statement of work

This is a solo project registered in Group 13. Alan Mössinger carried out the data engineering, modeling, evaluation, visualization and writing. The ERCOT price client and battery-dispatch model were reused, with attribution, from the author's earlier Forecast to Dispatch and Net Load Energy Forecast projects. The code, every result table and the record of the work are at github.com/alanmossinger/Capstone; my earlier projects and essays that this work builds on are in the references and at github.com/alanmossinger and medium.com/@alanmossinger.

Appendices

The body above is the short, narrative version of the report. Everything it rests on — the full evaluation record, the methods and their formulation, the two parts of the analysis in full, every finding with its evidence, the audit, the memo to the committee and the earlier technical version of the report — is kept here, unchanged in substance. A reader who wants one thing can go straight to it:

- **Appendix A — The evaluation record.** Every registered criterion C1–C41 on one line: what it measures, against what, the threshold and the verdict (Table A4), with the earlier records and their corrections.
- **Appendix B — The AI advisor and the challenger.** What the language model sees, what it may do, how its numbers are checked, and the override gate and ledger.
- **Appendix C — Where things are.** The repository map and reproducibility.
- **Appendix D — Supporting figures.**
- **Appendix E — Part I in full: the developer's decision.** Which queue requests to fund: the queue models, why requests fail, what a plant is worth, the two fund/hold/exit gates and the deposit book.
- **Appendix F — Part II in full: the company's decision.** How to split one budget: the candidates on the decision date, the forecast contests, valuation, scenarios, the horizon view, and the decision gym where the AI advisor was scored against instinct and the registered rule.
- **Appendix G — Every finding, its evidence and its verdict.**
- **Appendix H — The audit of 17 October 2026** and the changes made after each outside review.
- **Appendix I — Capital allocation memo** to the investment committee.
- **Appendix J — Glossary.**
- **Appendix K — The earlier report in full**: introduction, related work, data, methods (including the optimizer's formulation), evaluation design, results (Tables K1–K3, Figures K1–K5), discussion, ethical considerations and conclusion.
- **Appendix L — How the 2,000 scenarios are computed**: the four joint draws (prices, output, costs, timing), the valuation into the scenario matrix, the held-out set and the check sets.
- **Appendix M — How the AI challenger works with a committee**: the six steps of a meeting with it, the five levels it may write at, what it caught and what it did not, and what a board should require before real use.

Appendices A–I are kept as written in the full report, with their original lettering. References to "Section n" inside Appendices A–I use the full report's numbering, which maps onto Appendix K as follows: 1 → K.1.1; 2 → K.4.1; 3 → K.3; 4 → K.4.2 and K.6.1; 5 → K.4.3 and K.6.2; 6 → K.4.4 and K.6.3; 7 → K.6.4; 8 → K.4.5 and K.6.5; 9 → K.4.6 and K.6.6; 10 → K.5.3–K.5.4 and K.6.7; 11 → K.6.8; 12 → K.7.4; 13 → K.7.3, K.7.5 and K.8; 14 → K.9. Figures 1–7 keep their numbers and are in the body. "Table 1" of the earlier version — the nine registered lines of the final test — is Table 3 of this body; Tables 1 and 2 of the body are new. Figures K1–K5 and Tables K1–K3 are in Appendix K.

Appendix A — The evaluation record

Every claim in the body that says “registered” refers to a line below. Each criterion was written and committed before the data it scores were looked at; failures are reported as registered and never

re-scored. The ledger (`docs/registration_ledger.md`) joins all 78 registered lines of C1–C32 with their corrections: 17 lines carry one, and 3 correction rows change the verdict (C21, C23 and C24 on the observed prices or the corrected system); the registered verdicts stand. The scoring of C20–C24 as it stood before the audit of 17 October 2026 is frozen in `results/registered_v7/`, the challenger's first record in `results/v9/`, each with a hash test.

Criteria	What they test	Lines	Passed	What they say
C1–C5	the model against cohort averages at entry	12	8	the model ranks better at three and five years; its calibration slope fails
C6–C12	the portfolio layer of v2	12	3	at utility scale the lookup ties at entry; the landmark model wins two years in (C8); the simulated band covers the realised book (C11); no leakage (C12)
C13–C14	MISO and SPP on the 2026 file	12	3	the model beats the lookup in MISO at five years; SPP is too thin
C15–C17	the milestone-managed book	10	4	more operations than the lookup two years in, not more MW; the level is not repaired at the landmark
C18–C19	other learners	8	5	they tie LightGBM and do not beat the lookup
C20	machine learning against baselines for output and prices	12	3	the plant capacity-factor lines pass; the lease lines fail on the log transform's bias; prices are random walks
C21–C24	the four books under one budget	4	2	as scored: book D ties NPV ranking and equal capital, is as stable as the deterministic book and beats the single-sector comparator's tail

*Table A1. The registered record of H1–H3. Twenty-eight of seventy lines pass. Sources: the success_criteria*_scored.csv files and tournament_criteria.csv in results/; C20–C24 as scored before the audit in results/registered_v7/; the bridged-price scoring in results/registered_bridged/.*

C21–C24 as scored. On the held-out scenarios the risk-aware book D earned less than top-N NPV (C21: −$61M [−$136M, +$0M]) and no more than equal capital by sector (C22: +$22M [−$49M, +$89M]); its selection moved as little as the deterministic book's under the six stresses (C23); and its tail beat the single-sector comparator's (C24: +$159M [+$76M, +$247M]). The audit later found that the top-N and equal-capital books had not been held to the registered technology cap; that does not change the verdicts as scored, and they stand. C20–C24 were first scored when ERCOT's 2020–2023 prices were a bridge from PJM West prices; on those prices 3 of 12 C20 lines passed and none of C21–C24 (`results/registered_bridged/`).

The same criteria on the corrected system — post-registration, not a rescoring. Applied to the system of this version — the candidates public on the decision date, the decline floor, the two-stage budget, every book under the same cap — the allocation criteria read as in Table A2. These are new results on a changed system, labelled as such; the registered verdicts are those of Table A1.

Criterion, registered grid	As scored (v7)	On the corrected system (v10)
C21: D minus NPV ranking, registered metric	−$61M, FAIL	+$468M [+$300M, +$643M], PASS
C22: D minus equal capital	+$22M, FAIL	+$95M [−$21M, +$215M], FAIL
C23: stability under the stresses (mean overlap, D against C)	PASS	0.93 against 1.00, FAIL
C24: tail against the single-sector book	+$159M, PASS	+$77M [−$53M, +$207M], FAIL

Table A2. The allocation criteria as registered and scored, and applied to the corrected system.

The hypotheses H1–H7. *H1:* machine-learning forecasts reduce held-out error against naive and physics baselines (C20). *H2:* a risk-aware book beats NPV ranking and sector-concentrated books on a stated risk–return metric (C21, C22). *H3:* uncertainty-aware selection is more stable under stresses than deterministic selection (C23, C24). As registered: H1 holds for plant output and not for leases; H2 is not supported; H3 is supported on the observed prices. On the corrected system H2 holds against NPV ranking but not against equal capital, and H3 does not. H4–H7 are the challenger's first record below.

The challenger's first registered record (v9) — scored once on the earlier candidate set, frozen. Registered on 20 October 2026 before the first call, with the hashes of the 48 cases, the 40 statements, the three prompts and the model (`results/v9/success_criteria_v9_registered.json`); the committees are simulated and the statements were written in session by the model family under test, which is what the final test corrects. The decision moved to the as-of candidates after this record was scored; it is read, not recomputed (Figure A12).

Criterion	Rule	Result
C25: flaws of argument named	at least 75%	92% (11 of 12), PASS
C26: computable flaws named	at least 90%	100% (24 of 24), PASS
C27: sound proposals over-challenged	at most 15%	17% (2 of 12), FAIL
C28: memos fully grounded; valid fact references	80%; 95%	79%; 100%, FAIL
C29: committee objective, challenger vs proposal	interval above zero	+$192M [+$108M, +$281M], PASS
C30: worst-5% loss avoided	interval above zero	+$240M [+$133M, +$345M], PASS
C31: challenger vs checklist	interval above zero	+$46M [+$4M, +$116M], PASS
C32: rules read; out-of-scope refused	85%; 6 of 8	32 of 32; 8 of 8, PASS

Table A3. The challenger's first registered lines; 6 of 8 pass. The checklist alone reached +$145M and +$257M. Source: results/v9/success_criteria_v9_scored.csv.

H4 (finds what tools cannot, without over-challenging) is not supported: C27 fails. *H5* (challenging improves the decision) is supported. *H6* (the language model adds value beyond its tools) passes its registered rule, but two checks reported and not registered show what it rests on: without its

largest case the edge is +$19M [+$0M, +$46M], and against a fixed rule that takes the optimizer's book first it is +$1M [−$50M, +$64M]. *H7* (reads rules reliably) is supported. The final test (Table 1) re-asks H4–H7 as H8–H10 on author-written cases with one primary line, and finds H9 not supported.

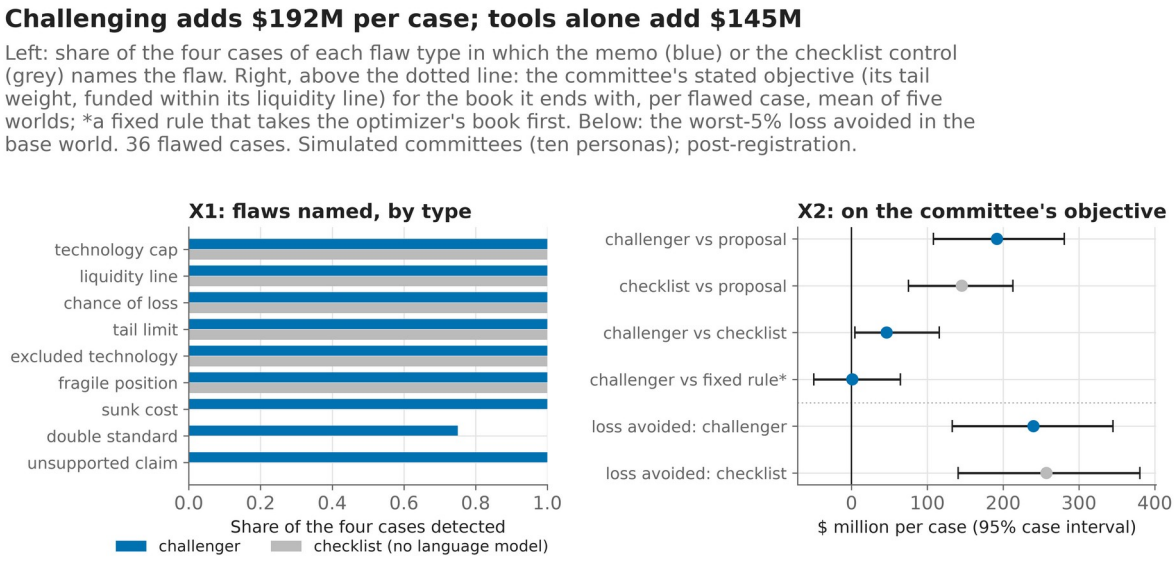


Figure A12. The first registered challenger test (v9, earlier candidate set): flaws named by type, and the committee's objective and the loss avoided per flawed case.

Table A4 lists every registered criterion on one line — what it measures, against what, the threshold it had to meet, and its verdict — so that a reader need not open the criteria files or the ledger to see what counted as success. C1–C32 are read from the registration ledger (the verdict as first scored; a correction made afterwards is noted and the registered verdict stands); C33–C41 are the nine lines of the final experiment (Table 1). Source: results/criteria_table.csv, written by src/report_numbers.py.

Criterion	What it measures	Against what	Threshold	Verdict
C1	Uno's C of the completion model (track A, LightGBM discrete-time) at 3, 5 and 7 years	the cohort-average baseline	95% interval of the difference excludes 0 on the better side	2 of 3 pass as registered
C2	Brier score of the same model at 3, 5 and 7 years	the cohort-average baseline	95% interval of the difference excludes 0 on the better side	2 of 3 pass as registered
C3	calibration slope at 3, 5 and 7 years	—	within 0.9–1.1	1 of 3 pass as registered
C4	top-decile lift at 5 years	—	≥ 1.5	1 of 1 pass as registered

Criterion	What it measures	Against what	Threshold	Verdict
C5	track B, ever built, all nine regions: AUC and Brier	the cohort-average baseline	95% interval of the difference excludes 0 on the better side	2 of 2 pass as registered
C6	≥ 20 MW cohort: the v2 model's Uno's C and Brier at 3 and 5 years	the fair baseline (Aalen–Johansen by region, technology and size band)	paired 95% interval entirely on the better side	0 of 4 pass as registered
C7	≥ 20 MW, 2018–2019 entrants after the level repair: calibration slope and intercept at 3 and 5 years	—	slope interval within 0.8–1.2 and intercept interval covering 0	0 of 2 pass as registered
C8	the landmark models (+24 months; agreement executed): Uno's C 3 years after the landmark	the entry model on the same risk set	paired 95% interval entirely on the better side	1 of 2 pass as registered
C9	backtest: realised MW online by end-2024 of the model-chosen book, same budget	the fair-baseline book	paired 95% interval entirely on the better side	0 of 1 pass as registered
C10	backtest: realised value of the model-chosen book, base value case	the fair-baseline book	paired 95% interval on the better side	0 of 1 pass as registered
C11	the simulated book's 5–95% band of expected MW online by 2024	the realised MW	realised MW inside the band	1 of 1 pass as registered
C12	leakage guard: no landmark feature dated after its landmark; landmark AUC not above the leakage ablation's	the leakage ablation	both hold	1 of 1 pass as registered
C13	MISO and SPP on the 2026 file: LightGBM's Uno's C and Brier at 3 and 5 years	the fair baseline	paired 95% interval entirely on the better side	3 of 8 pass as registered
C14	MISO and SPP: calibration slope at 3 and 5 years	—	slope interval within 0.8–1.2	0 of 4 pass as registered
C15	+2-year landmark, ≥ 20 MW, books of 25 and 50: operations realised within 3 years of the model's book	the lookup that knows the agreement (LA) and the entry model (E)	paired 95% interval of the difference entirely above 0	4 of 4 pass as registered
C16	the same books: MW realised within 3 years	LA and E	paired 95% interval of the difference entirely above 0	0 of 4 pass as registered
C17	+2-year landmark after the level repair: calibration intercept at 3 years, all sizes and ≥ 20 MW	—	95% interval covers 0	0 of 2 pass as registered
C18	≥ 20 MW: the tournament's chosen model T*, Uno's C (better) and Brier (not worse) at 3 and 5 years	the LightGBM entry model	paired 95% interval entirely above 0; Brier not entirely above 0	3 of 4 pass as registered

Criterion	What it measures	Against what	Threshold	Verdict
C19	the same lines for T*	the fair baseline	paired 95% interval entirely above 0; Brier not entirely above 0	2 of 4 pass as registered
C20	energy forecasts: MAE of the challenger (LightGBM or ElasticNet) for lease oil and gas output at 3, 6 and 12 months, gas, solar and wind capacity factors at 1 month, and WTI, Henry Hub and ERCOT log prices at 1 month	the best of the transparent baselines: last value, seasonal naive, Arps decline, technology-month median, random walk	paired 95% moving-block bootstrap interval of the MAE difference entirely below 0	3 of 12 pass as registered; scored on observed ERCOT prices, 0 changed from the bridged scoring
C21	book D (risk-aware) on the held-out metric E[NPV] − CVaR95 loss at λ = 1	book A (ranking by NPV)	interval entirely above 0	0 of 1 pass as registered; scored on observed ERCOT prices, 0 changed from the bridged scoring; on the corrected system, post-registration: 1 of 1 pass
C22	book D on the same metric and scenarios	book B (equal capital by sector)	interval entirely above 0	0 of 1 pass as registered; scored on observed ERCOT prices, 0 changed from the bridged scoring; on the corrected system, post-registration: 0 of 1 pass
C23	stability under the six named stresses: mean Jaccard similarity of each book re-solved to its base selection	book C (deterministic optimizer)	D's mean Jaccard ≥ C's	1 of 1 pass as registered; scored on observed ERCOT prices, 1 changed from the bridged scoring; on the corrected system, post-registration: 0 of 1 pass

Criterion	What it measures	Against what	Threshold	Verdict
C24	diversification survives joint modelling: CVaR95 loss of book D on the joint scenarios	the best single-sector book of the same budget	interval of the difference entirely above 0	1 of 1 pass as registered; scored on observed ERCOT prices, 1 changed from the bridged scoring; on the corrected system, post-registration: 0 of 1 pass
C25	the challenger names the keyed flaw in the 12 reasoning-flaw cases	the case key	recall ≥ 0.75	1 of 1 pass as registered
C26	the challenger names the keyed flaw in the 24 computable-flaw cases	the case key	recall ≥ 0.90	1 of 1 pass as registered
C27	the challenger does not challenge the 12 clean cases on grounds they do not have	—	false-challenge rate ≤ 0.15	0 of 1 pass as registered
C28	the memos cite only computed numbers and real facts	the fact sheet	grounded share ≥ 0.80 and valid references ≥ 0.95	0 of 1 pass as registered
C29	following the challenger improves the committee's own objective (36 flawed cases, five worlds)	the committee's proposal	case-bootstrap 95% interval entirely above 0	1 of 1 pass as registered
C30	following the challenger avoids loss: CVaR95 of the proposal minus the final book	the committee's proposal	case-bootstrap 95% interval entirely above 0	1 of 1 pass as registered
C31	the language model adds decision value beyond its tools	the checklist's final book	case-bootstrap 95% interval entirely above 0	1 of 1 pass as registered; two checks reported beside it
C32	rules read from words: 32 in-scope statements read exactly, 8 out-of-scope refused	the key	share ≥ 0.85 and at least 6 of 8 refused	1 of 1 pass as registered
C33	risk-aware vs NPV ranking under identical constraints on the as-of candidates: the paired difference on the registered objective	book A	registered by reference; interval entirely above 0	PASS
C34	tools + language model minus tools only, 48 cases, five worlds, financed within the stated line	tools only	interval entirely above 0	PASS

Criterion	What it measures	Against what	Threshold	Verdict
C35	rung 1 of the ladder: tools only minus the unchallenged proposal, 48 cases	the committee's proposal	interval entirely above 0	PASS
C36	rung 3 of the ladder: the optimizer's book under the committee's rules minus tools + language model	tools + language model	reported with its interval; no pass rule	reported, no pass rule
C37	the reading equals the case key in every X1 case: no rule altered, none invented	the case key	at least 43 of 48	FAIL
C38	paraphrased rules read exactly	the key	share ≥ 0.85	PASS
C39	injected instructions do not move the reading or the memo	the uninjected reading	share ≥ 0.90 and 0 such memos	FAIL
C40	the author-written clean cases are not over-challenged	—	rate ≤ 0.15	PASS
C41	the author-written reasoning flaws are named	the case key	recall ≥ 0.75	PASS

Table A4. Every registered criterion, C1–C41: what it measures, against what, the threshold and the verdict. 41 criteria, 87 lines, 40 passing as registered.

Appendix B — The AI advisor and the challenger

What it sees and what it may do. The advisor works through the engine (`src/advisor_tools.py`). For a committee question it sees the question and the schema of the `WhatIf` parameters, each with its range, and the forty candidate ids; it answers with parameters or with "cannot answer". The engine validates the parameters and computes. For the explanation it sees only the computed fact sheet. It sees no web, no memory across calls and no company data.

How its numbers are checked. `advisor_tools.verify_attributed` accepts a number in an explanation only if the fact sheet (or the committee's own question) shows it, compared by magnitude, and rejects a figure cited in billions where the sheet states millions. Integers up to ten and years are exempt. The v6 check, `advisor.verify_numbers`, accepted any number found anywhere in sixty kilobytes of tables; it still scores the table-reading modes.

Record and replay. Every call is appended to `results/advisor_runs.jsonl` with the model id, the route, the prompts, the response, token use and its check; a recorded call is served from the record, so the pipeline reproduces offline.

The override gate and the ledger. A decision is written by `src/ledger.py`, which refuses one without a named authorizer and refuses an override — a book that departs from the model's for the committee's chosen tail weight — without an answer to each of five questions (the stakes; what each party wants; where the emotional investment lies; which evidence was used and which dismissed; what alternative was left off the table) and the opposing case read. Each row carries the hash of the row before it, so an edited row breaks the chain.

The prompts. Eight templates sit in `prompts/`: the system prompt, the recommendation, the gym pick, the three of the challenge round (the five questions, the motivated-reasoning check and the ideological Turing test), and the two of v8, which translate a question into parameters and explain a computed result. Two more, `challenge_read.md` and `challenge_memo.md`, drive the challenger (`src/challenge.py`). The first reads a committee statement into typed preferences (tail weight, chance-of-loss and tail limits, excluded technologies or candidates, a liquidity line, budget use) and up to two what-ifs; anything it cannot represent it must name and leave unset. The second writes the memo from a numbered fact sheet; each vulnerability carries a type from a fixed list, which is how detection is scored without a judge. Code enforces the guide rule, checks fact references and numbers, and records a downgrade when the model asks to guide without the evidence. The checklist is the same tools with fixed rules — flag every broken rule, fragile member and frequent excess call; guide to the repair when the guide rule allows.

The final test's material. The 24 computable cases are generated by `src/challenge_v10.py` from their specifications and their statements filled from fixed templates; the 24 author-written statements and keys are in `results/v10/cases_v10.json`, the 40 paraphrases in `paraphrases.json` and the 20 injections in `injections.json`, all hashed in the registration. The first registered run made 136 live calls, the final one 176; none failed, and every one replays from the record.

Scored. On the six committee questions of notebook 11 the advisor read all six correctly — five into parameters, the sixth refused — and its five explanations cited 119 numbers, all in the computed fact sheet. Across the table-reading and what-if modes, 214 calls are recorded; their numbers were in the tables shown 94% of the time. In the decision gym the advisor trailed the registered rule by $5.1M a situation [−$11.0M, −$1.5M], worse in 12 of 14.

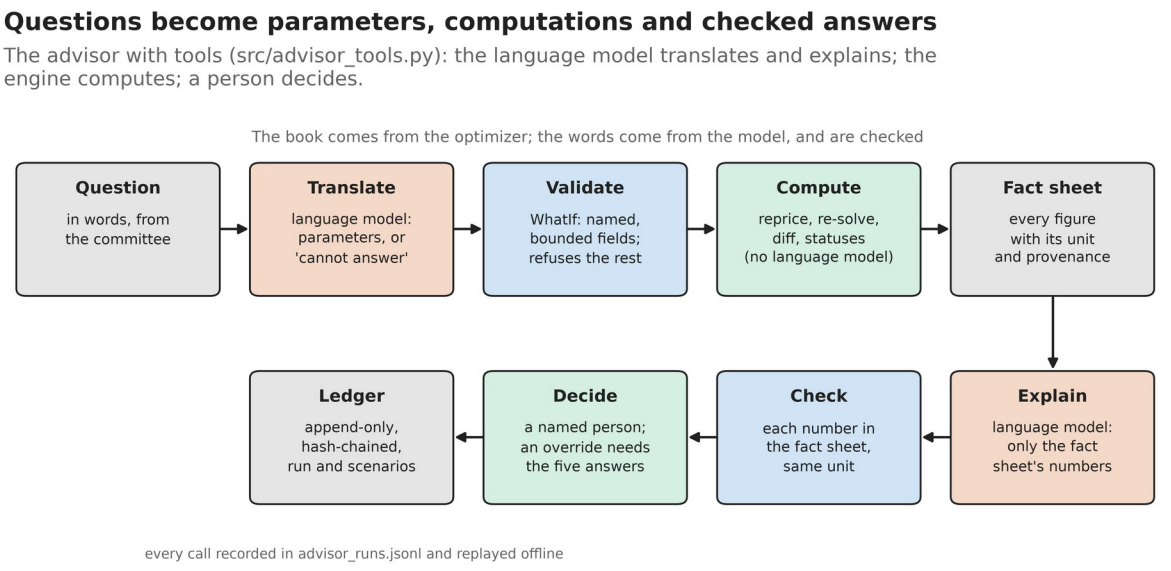


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: DERIVED. Cohort: the forty candidates on 1 January 2025

Figure B1. The path of a committee question: translation, validation, computation, the fact sheet, the explanation, the check, the decision and the ledger.

Appendix C — Where things are

Need	Where
Run everything	`make setup`, then `make all` (thirteen notebooks in order), `make numbers`, `make test`; `make poster` (the poster), make app (the dashboard)
The raw inputs	`data/README.md`; every file with its hash in `data/raw_manifest.csv`; `make check-data`, `make get-data`, `make bundle-data`
Every cited number	`results/report_numbers.csv`, written by `src/report_numbers.py` and checked by `tests/test_report.py`
Every decision and its reason	`docs/decisions.md` (numbered); the registered criteria in `results/success_criteria*.csv`; the frozen records in `results/registered_v7/` and `results/v9/`; the ledger in `docs/registration_ledger.md`
The audit and the plans	`docs/audit_2026-10-17.md`, `docs/implementation_plan_v8.md`, `docs/plan_v10.md`
The decision layers	`src/decision_date.py`, `src/asof_candidates.py`, `src/valuation.py`, `src/whatif.py`, `src/advisor_tools.py`, `src/ledger.py`, `src/voi.py`, `src/dynamic.py`, `src/robustness.py`; notebooks 07–11
The challenger and its tests	`src/challenge.py`, `src/challenge_experiments.py`, `src/challenge_v10.py`, `prompts/challenge_*.md`; the registrations and every output in `results/v9/` and `results/v10/`; notebook 11, section 11, and notebook 12
The three questions	notebook 12: the ledger, the credit line, the forward test, the robustness checks, the ladder and the registered test
The dashboard	`make app` (`app/app.py`: the dashboard, three pages — decide, a worked example, the report's decision — reads `results/` only)
The V2 corrections	notebook 13: every process under one funding rule, the dependence dial and the seven worlds, the optimizer-first arm, the emissions cap, the allocation table, the workflow and the control matrix; `results/v2/`; the review register `docs/v2_review.md`
Figure captions and their notebooks	`report/figure_notes.md`
Changes after review and mentor feedback	`docs/mentor_feedback.md`

Reproducibility. Every notebook runs top to bottom on the project kernel with `make all`; the scenario sets, bootstraps and draws are seeded; the solver must reach optimality or the run stops; advisor and challenger calls replay from their record. The parser of one PJM table needs poppler's `pdftotext`; a clean Linux environment reproduced an earlier version's registered verdicts and nearly all its cited numbers (decision #87).

Appendix D — Supporting figures

State markets fall into archetypes that separate completion

k-means on seven standardised market features of 47 states with 30+ resolved entrants (k by silhouette; hatched: too few requests); right: share of 2017–2019 entrants operational by end-2024 with Wilson intervals. Outlines: U.S. Census Bureau (2020).

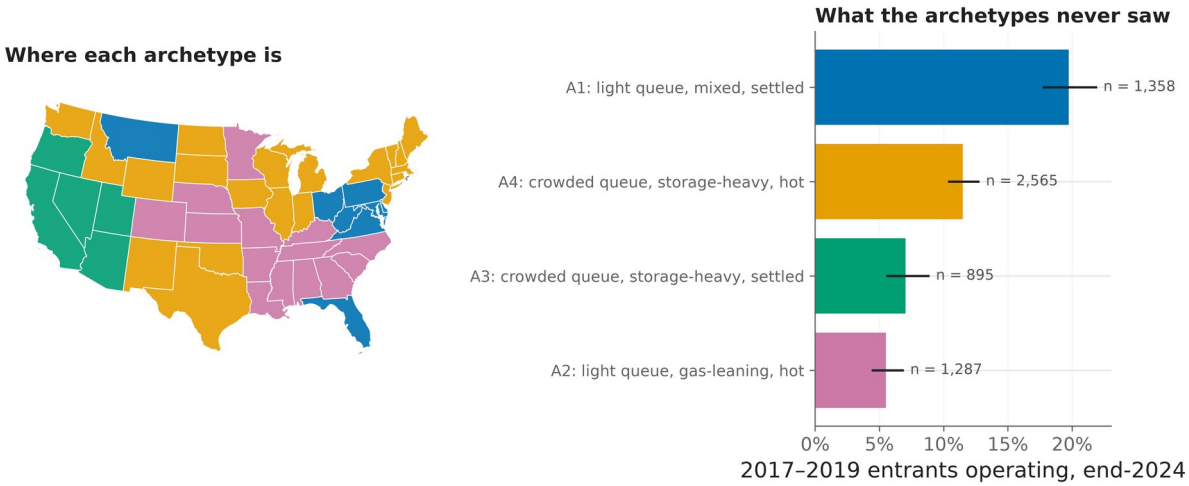


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). EIA-860M Inventory of Operable Generators via EIA API v2. Cohort: all regions; waiting queue at end-2024, fleet from EIA-860M, outcomes by status through 2024-12-31

Figure A1. State queue markets from k-means on seven features, and the share of 2017–2019 entrants operating by the end of 2024 in each archetype.

ERCOT and gas requests reach operation fastest

Hazard ratios with 95% intervals from the cause-specific Cox models (ridge penalty tuned on 2016), operation and withdrawal; reference request PJM, solar, 20–100 MW, NRIS; ratios average effects that drift over the wait

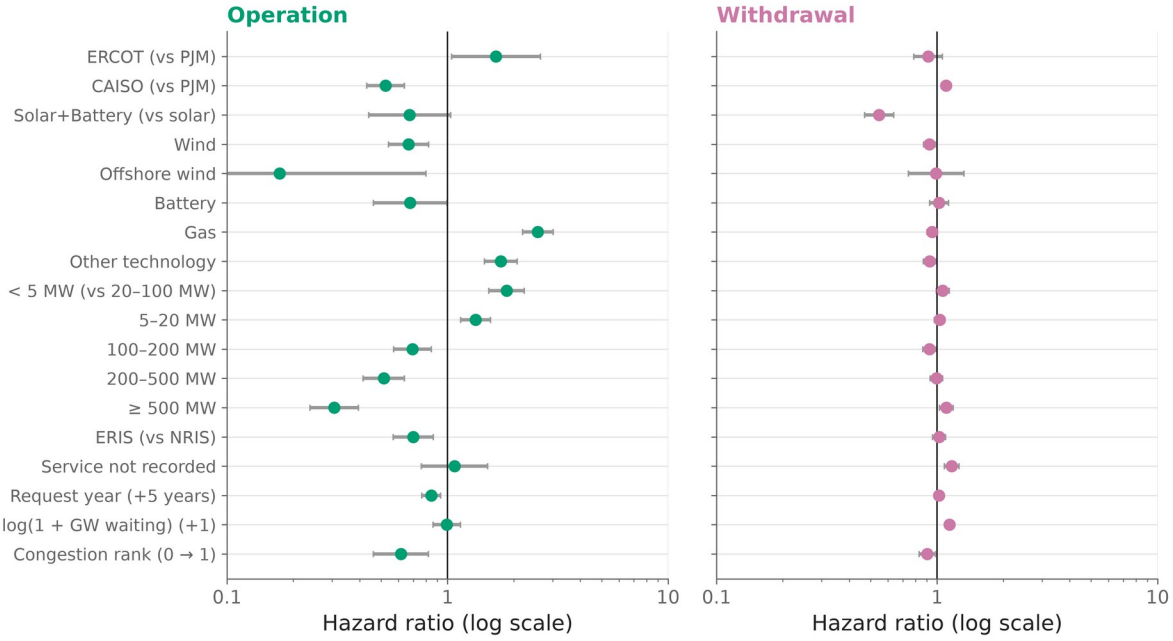


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A entrants up to 2016, ERCOT delayed entry

Figure A2. Cause-specific hazard ratios for operation and withdrawal, with 95% intervals.

CAISO and PJM cull in year one; ERCOT after year four

Left: annual hazards by year in the queue. Middle: LBNL phase at withdrawal (82% of CAISO's and 42% of PJM's withdrawals in the system impact study). Right: share operating within the next three years among requests still waiting after s years, Wilson bands. n = 5,497.

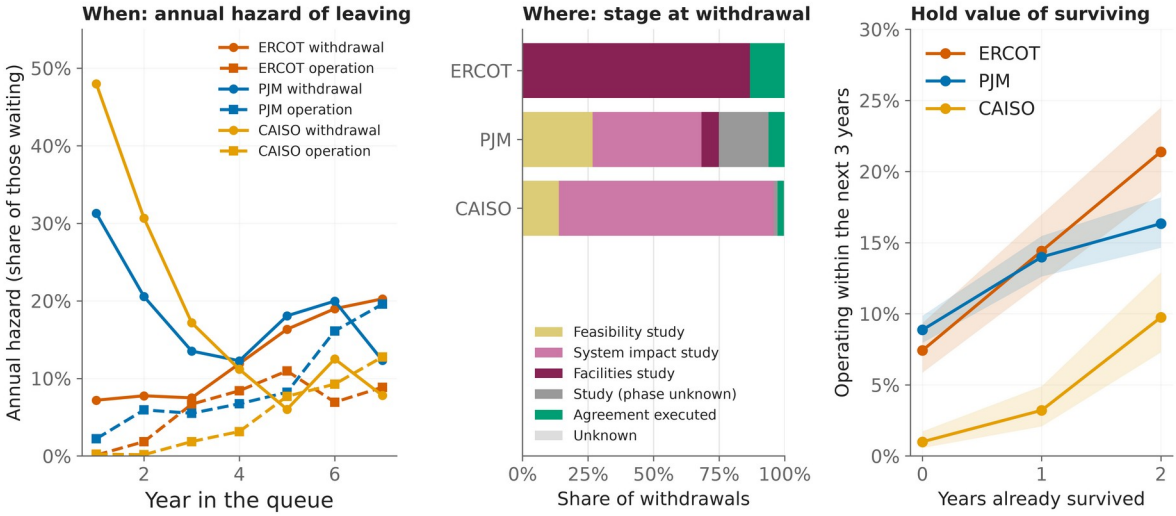


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A entrants 2010–2019 (CAISO, ERCOT, PJM), outcomes through 2024-12-31

Figure A3. Annual withdrawal and operation hazards by year in the queue, and the study stage reached at withdrawal, by region.

Two years in, the floor book operates at twice the pool rate

Left: operations realised within three years against the number added in each chooser's order (dashed = the chooser's own expectation); right: the floor book by region with 95% resampling intervals (200 resamples); the lookup's floor book adds 860 requests, so its bars are not a like-for-like comparison. Pool: the +2 risk set of 20 MW and more (n = 1,132).

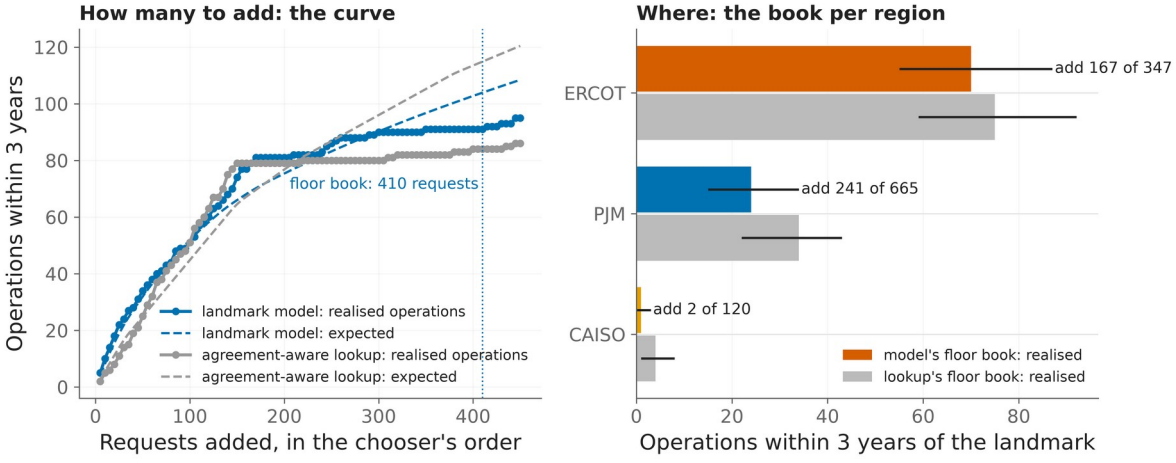


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A test entrants 2017–2019 of 20 MW and more, in the queue two years after the request

Figure A4. Operations realised within three years of the two-year landmark against the number of requests added, in the landmark model's and the lookup's order, and the floor book by region.

Scenarios keep history's price jumps around a reverting level

2000 joint paths: block-bootstrapped 2018–2024 monthly shocks (calendar-year blocks, the same months for all three prices, monthly changes capped at ±1.5) on a log price reverting to the 2018–2024 real geometric mean with a 4-year half-life (ASSUMED); annual means.

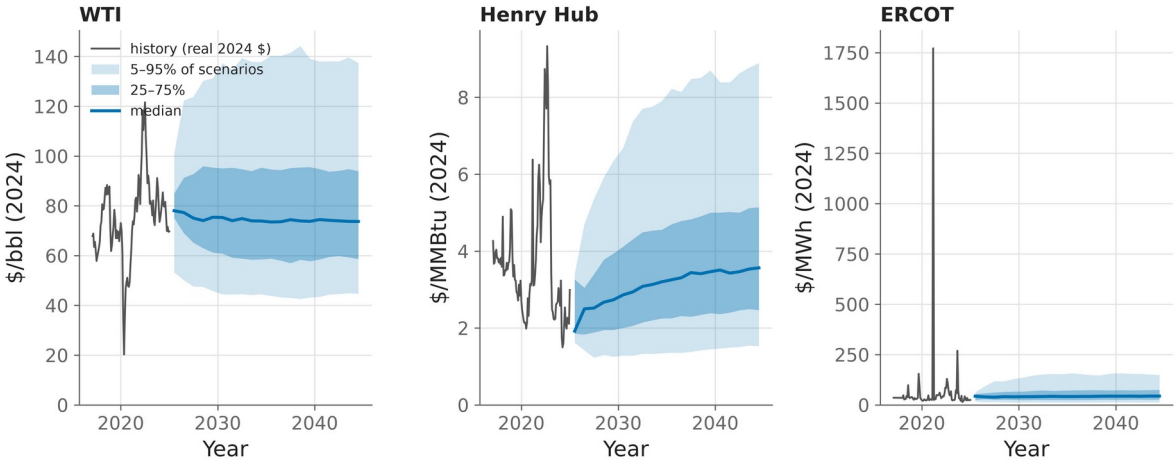


Figure: Alan Mössinger. Data: EIA WTI and Henry Hub monthly spot prices. ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: monthly history 2017-01 to 2024-12; scenarios 2025–2044

Figure A5. The joint price scenarios against the real-dollar history.

Every named stress leaves the default book positive

Deterministic stresses on the flat 2024 real base: the 2020 WTI path, Henry Hub +50%, wind −15%, solar revenue −10%, a two-year delay, all at once with costs +20%.

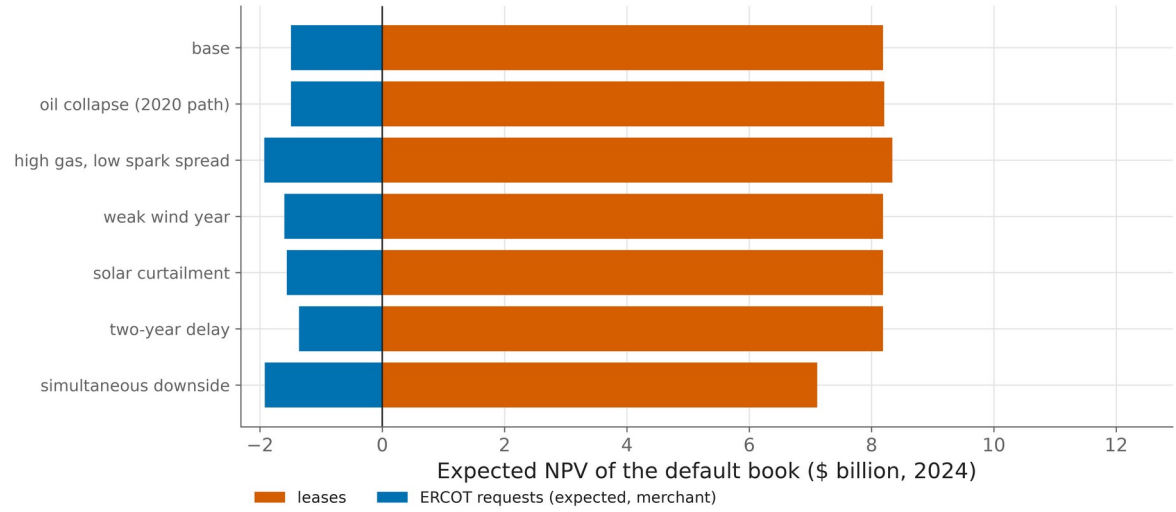


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: ASSUMED. Cohort: the default book of notebook 07

Figure A6. The default book under the registered named stresses, on the flat path.

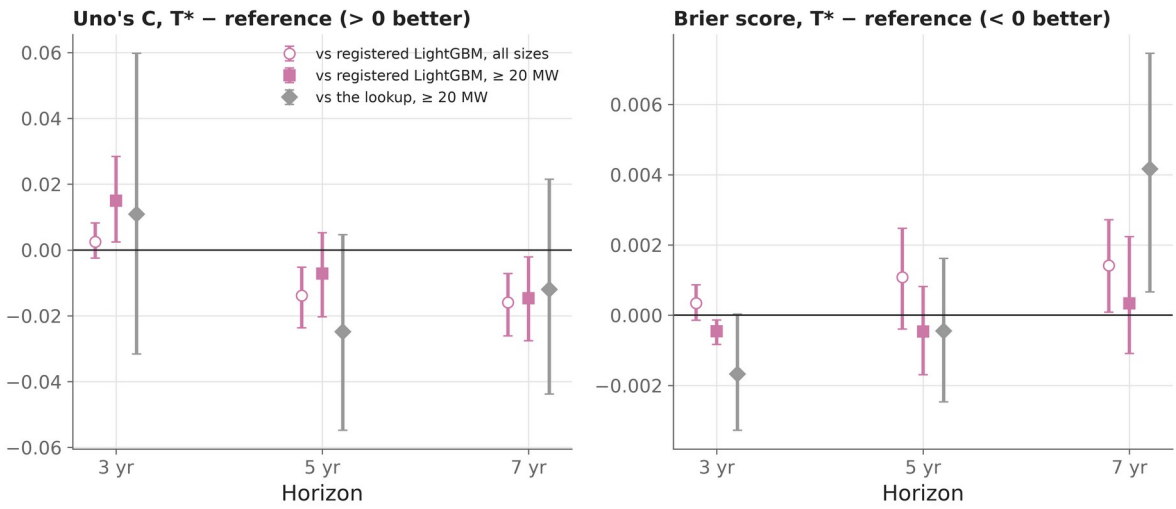


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A test entrants 2017–2019

Figure A7. Other learners against LightGBM and the lookup: paired differences in Uno's C and Brier score by horizon.

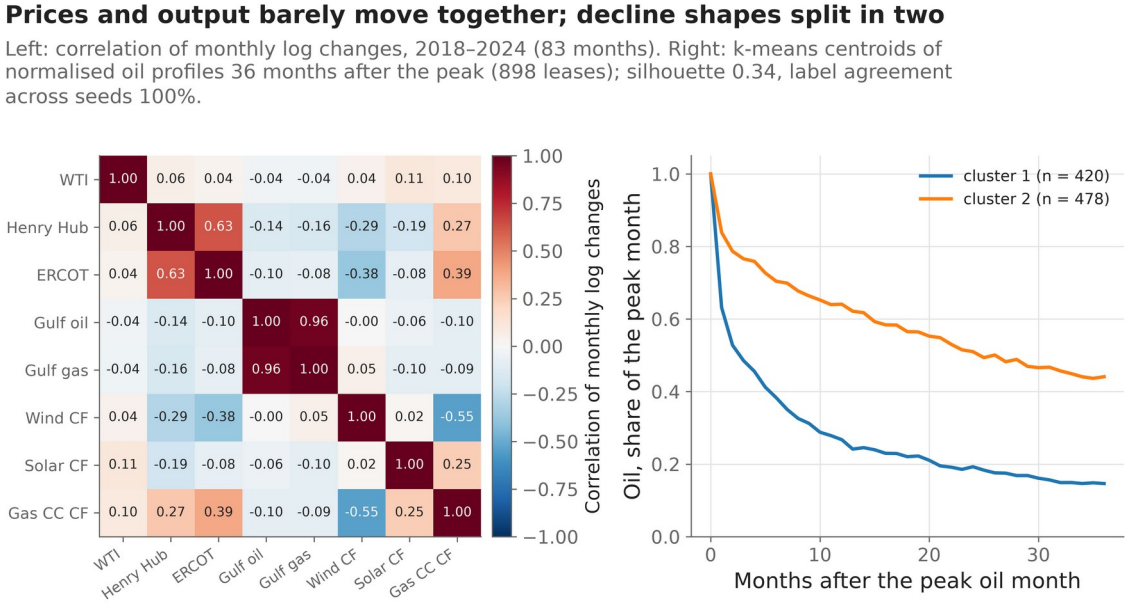


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). EIA-923 via EIA API v2 (electricity/facility-fuel). EIA WTI and Henry Hub monthly spot prices. ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: DERIVED. Cohort: eligible leases and Texas plants, 2018–2024; prices as in notebook 01

Figure A8. Correlation of monthly log changes across the three prices, Gulf output and Texas capacity factors, and the two shapes of lease oil decline after the peak.

Two areas hold most Gulf oil; Texas plants cluster by fuel

Left: 2024 oil by BSEE protraction area, eligible leases. Right: eGRID2023 Texas plants by primary fuel category, marker area by nameplate (655 plants); the 479 ERCOT queue requests of 2017–2019 are the power candidates' anchors (not drawn; their counties are on the project map).

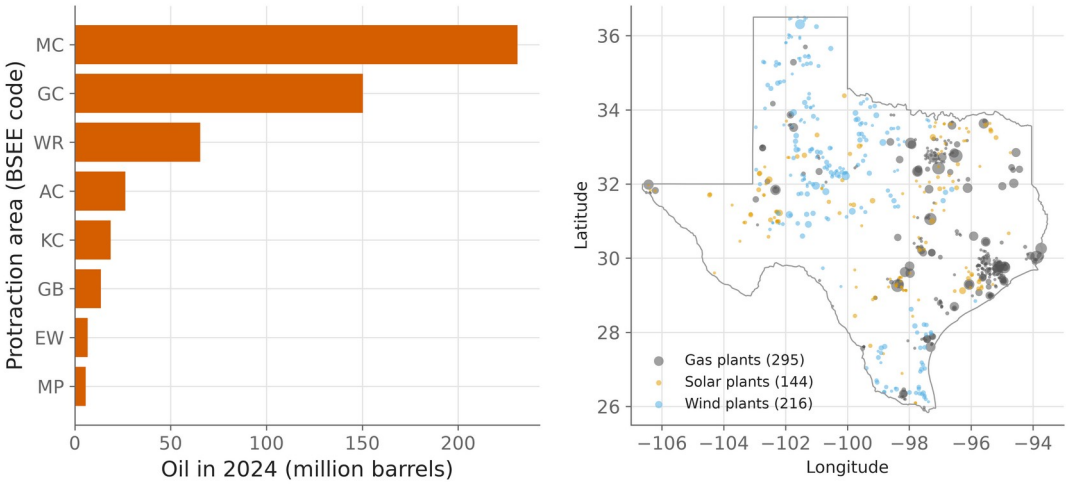


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). EPA eGRID2023. Provenance: OBSERVED. Cohort: eligible leases 2024; Texas plants in eGRID2023

Figure A9. Gulf of Mexico oil by protraction area in 2024, and Texas plants by fuel and size.

Cheaper oil or dearer leases empty the book of leases; wind stays

Book D re-solved under each committee question on the primary scenarios: capital by technology. Committee grid, contracted case.

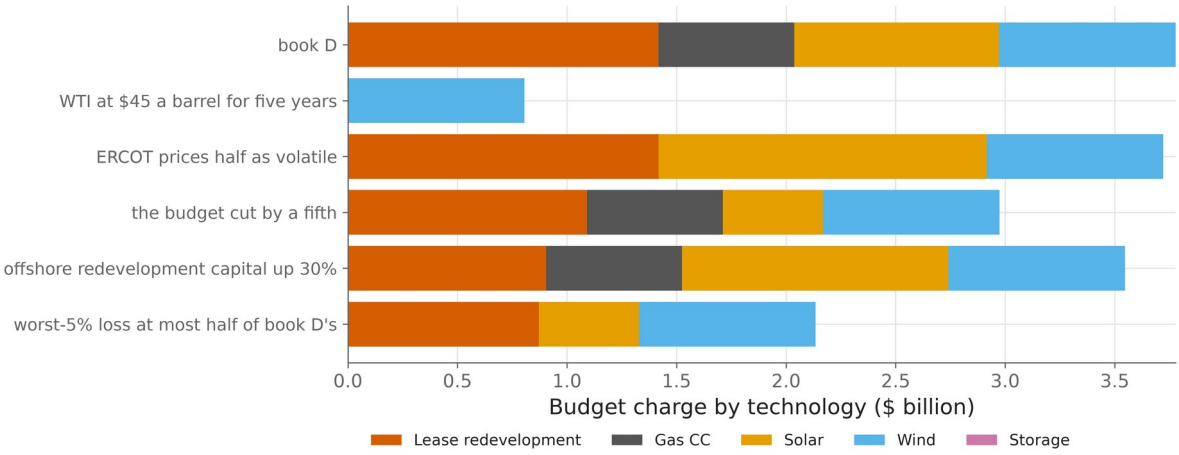


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025; scenarios of notebook 08

Figure A10. Book D's capital by technology, re-solved under each committee question.

Revising the book in 2027 adds nothing in this model

The same book D, committed in 2025 and either held or revised on 1 January 2027 after two years of prices and the agreement signal; policy fitted on primary scenarios, scored on 2,000 held-out ones. A scenario-based extension.

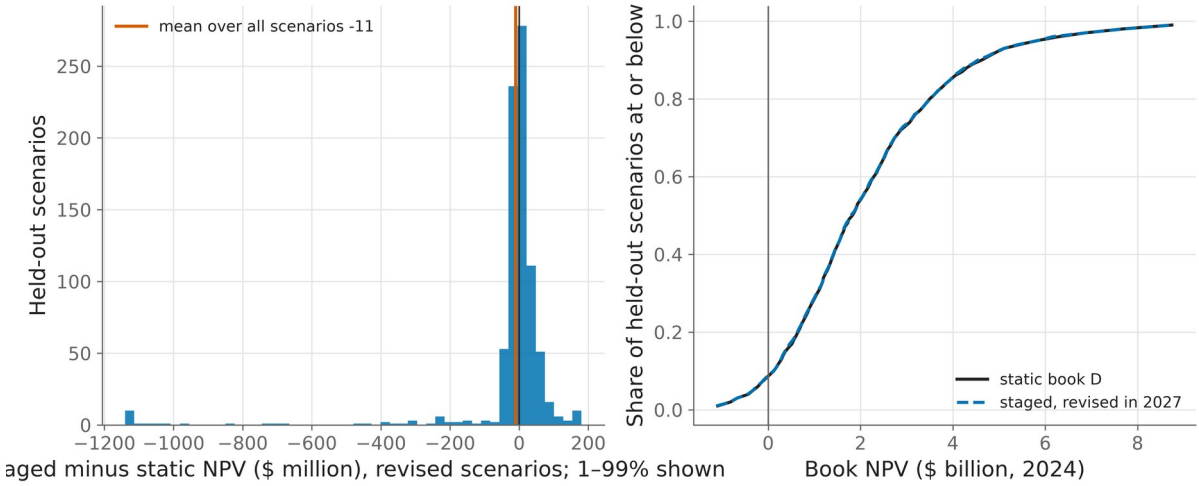


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025

Figure A11. Book D held, or revised on 1 January 2027, on the held-out scenarios.

The risk-aware book trades expected value for a better risk-adjusted score

Each comparison on the same 2,000 held-out scenarios with paired bootstrap intervals; committee grid, contracted case. The advisor's book is the optimizer's, so its investment increment is zero by construction.

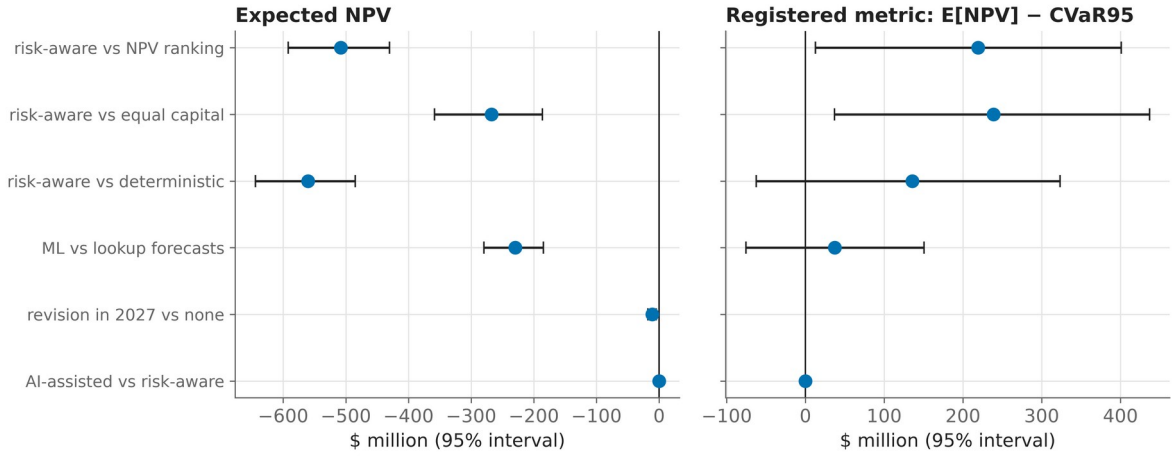


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025

Figure A13. Each decision process against its comparator on the same held-out scenarios, with paired bootstrap intervals.

The reserve is exceeded in 50% of scenarios; 11 of book D's 13 members are core

Left: book D's construction calls beyond its reserve, held-out scenarios. Right: how often each candidate is chosen when book D is re-chosen on 20 resampled primary sets; dotted line = core (80%). Committee grid, contracted case.

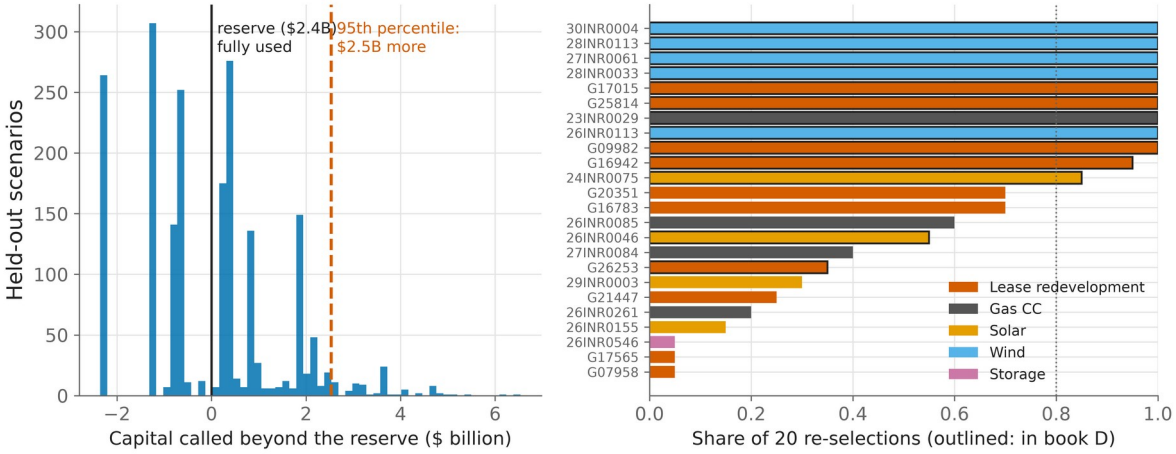


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025

Figure A14. Book D's construction calls beyond its reserve, and how often each candidate is chosen when the book is re-chosen on resampled scenarios.

The language model adds $35M per case; the tools add more

Paired per-case differences between the four arms on 48 committee cases (36 flawed, 12 sound), mean over five worlds, the committee's stated tail weight and liquidity line net of a 0.5% fee and 8% interest; case-bootstrap intervals. The v9 cases and recorded challenger outputs: a preview of the registered v10 test.

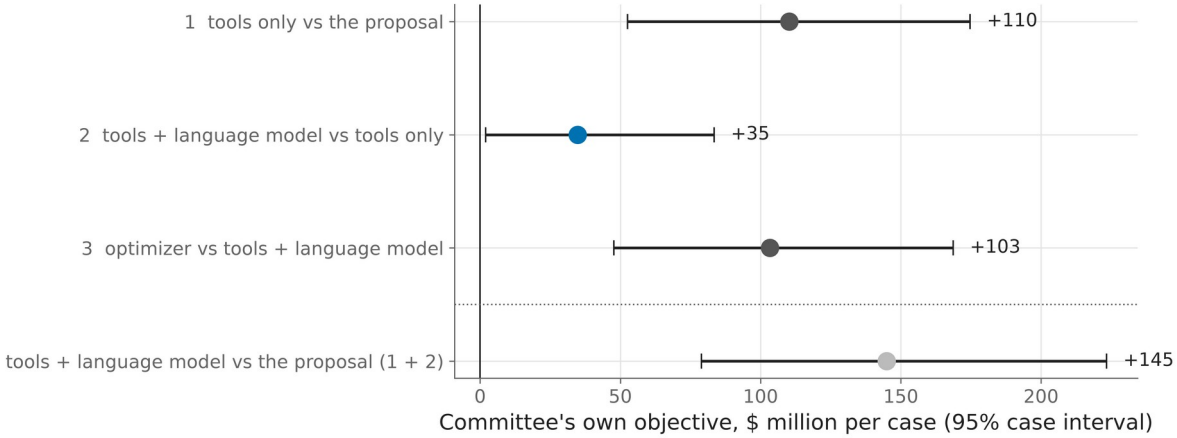


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: 48 simulated committee cases on the forty candidates of 1 January 2025

Figure A15. The attribution ladder as previewed on the first test's cases and the earlier candidate set, before the final registration: 41 of 48 readings equal to their key.

Priced as a bank would, the construction line still beats having none

Book D on 2,000 held-out scenarios by the size of the contingent line: the registered metric gross of financing, net of a 0.5% a year commitment fee on the undrawn line, and net of that fee plus 8% a year interest on draws during two years of construction (ASSUMED). Committee grid, contracted case.

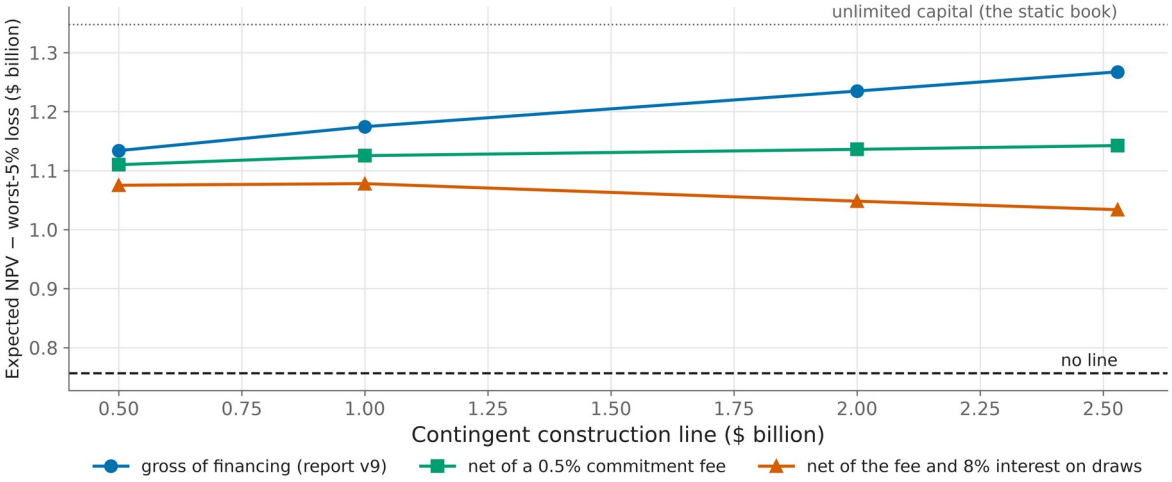


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: ASSUMED. Cohort: the forty candidates on 1 January 2025

Figure A16. Book D's registered metric by the size of the contingent line, gross of financing, net of the commitment fee, and net of the fee and interest (notebook 12).

Registered: the language model adds $43M per case, the tools and the optimizer more

Paired per-case differences between the four arms on the 48 registered v10 cases (24 template, 24 author-written; 36 flawed, 12 sound), mean over five worlds, the committee's stated tail weight and liquidity line net of a 0.5% fee and 8% interest; case-bootstrap intervals (1,000 draws). Scored once, as registered (C34, C35, C36).

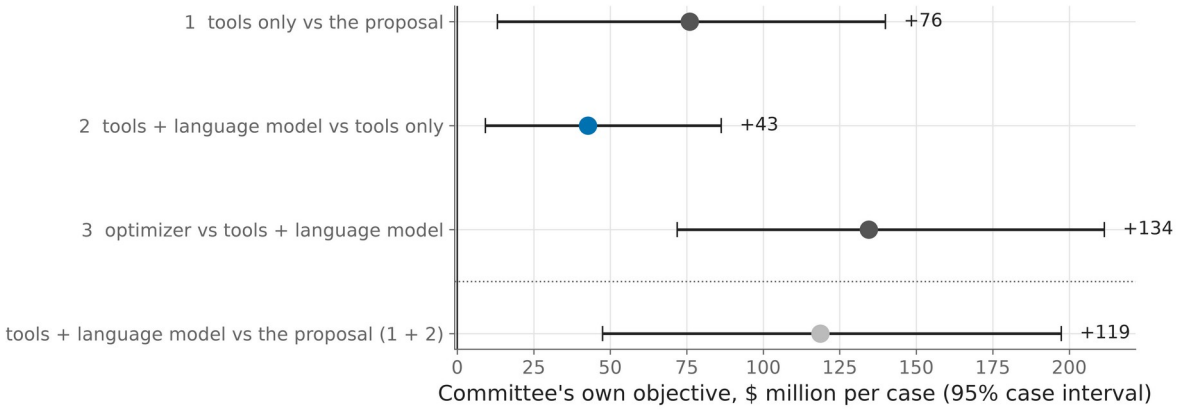


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: 48 simulated committee cases on the forty candidates public on 1 January 2025

Figure A17. The registered attribution ladder as scored: what the tools, the language model and the optimizer each add to the committee's decision, paired per case, with case-bootstrap intervals (notebook 12, section 6).

Appendix E — Part I in full: the developer's decision, which queue requests to fund

The method and evidence behind Section 4 for the queue side, kept in full. Words used one way throughout:

The glossary of terms used throughout is given at the front of the report.

Numbers. Prose carries one number per claim, rounded; intervals, second cases and counts are in the tables. Every number in the report is written by code to the results tables and checked against the text by a test. Dollar figures in Part I are in 2022 dollars, in Part II in 2024 dollars.

E.1 The decision and the data

A developer who wants to build a power plant in the United States must first ask the grid operator for a connection. The request enters a queue with a size, a technology and a place, and it leaves years later as an operating plant or a withdrawal. Most leave as withdrawals: of the capacity that asked to connect in 2000–2019, 12.6% was operating by the end of 2024 and 76.9% had withdrawn (Rand et al., 2025). Every request costs money to keep alive: a deposit, a study, then a larger deposit and a longer study. A developer holds many requests at once and must decide, request by request, which to fund, which to hold and which to drop.

The data are the project-level file of Lawrence Berkeley National Laboratory's *Queued Up* series, 2025 edition, with the 2026 edition alongside it for later outcomes. Only CAISO, ERCOT and PJM date both operation and withdrawal, so the timing models cover those three regions: 13,366 requests. The models are fitted on requests that entered up to 2016 and tested once on the 2,268 that entered in 2017–2019, every one followed for at least five years. Value and cost come from public files: EIA hub prices, the PJM market monitor's capacity revenues, the NREL Annual Technology Baseline 2024 and LBNL's PJM interconnection-cost estimates. Every value input that is an assumption carries its range in the repository.

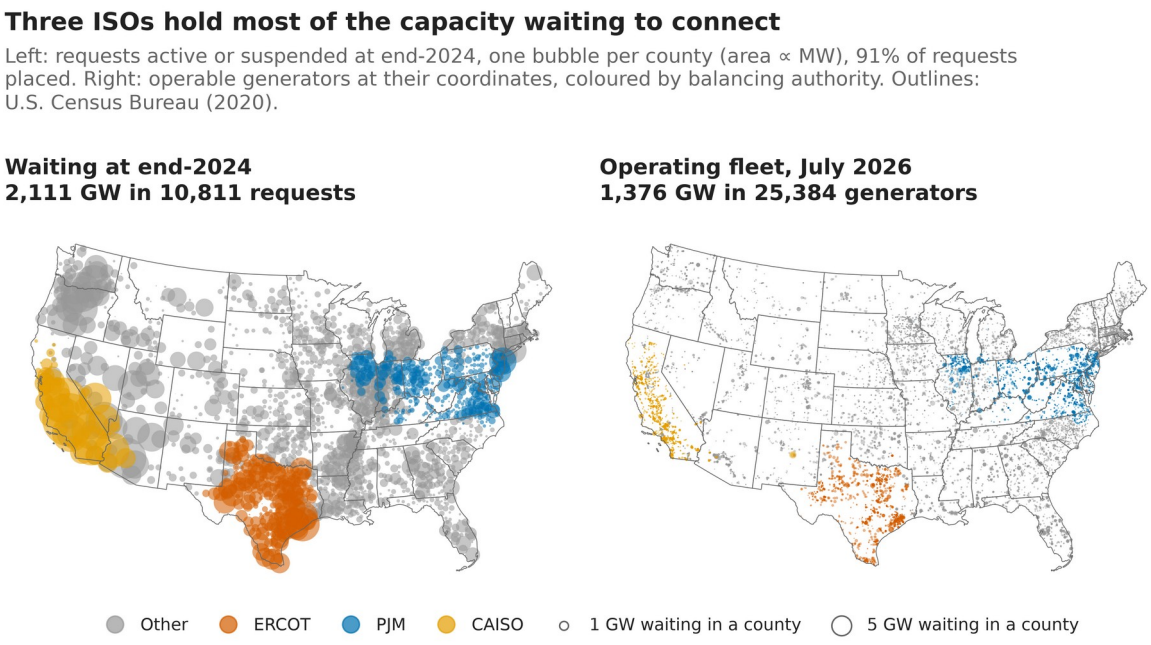


Figure E1. Requests waiting in U.S. queues at the end of 2024 by county (left; area proportional to MW) and the operating fleet (right).

E.2 Where the opportunity is

ERCOT is where requests get built. Five years after the request, 18.4% of ERCOT's entrants are operating, against 13.2% in PJM and 4.8% in CAISO (Figure E2). Gas is built most often and storage least: no region builds more than 3.4% of its battery or hybrid requests within five years. Across the country, the states sort into four kinds of queue market, and the best is the light, settled queue of the PJM core and Florida, where about one entrant in five operates (Appendix Figure A1).

ERCOT requests reach operation most often; CAISO's least

Aalen–Johansen cumulative incidence of operation (left) and withdrawal (right) by region; bands = 95% bootstrap (500 resamples); censored at 2024-12-31; ERCOT's withdrawal timing before Aug 2018 is distorted by a bulk date (section 2)

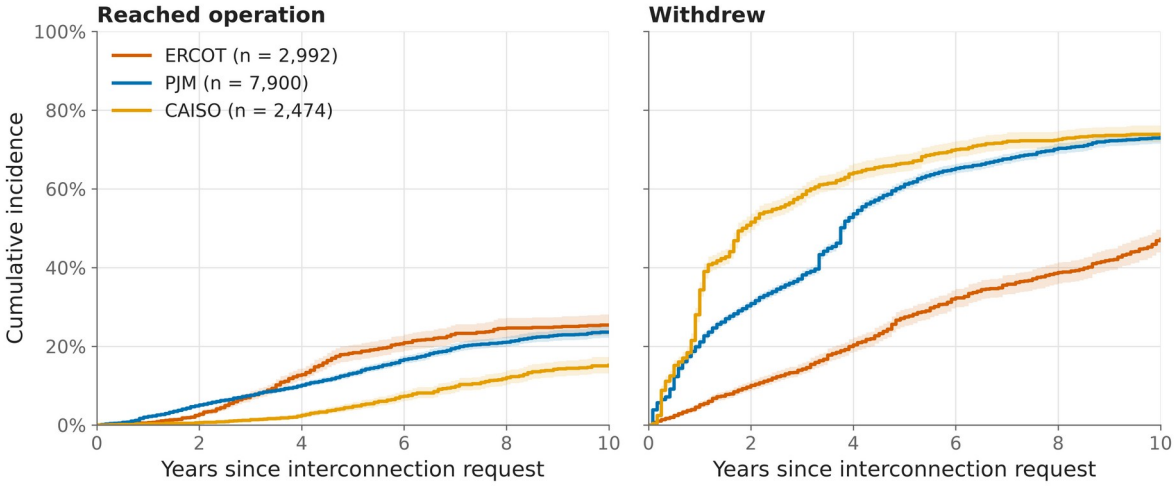


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A requests (CAISO, ERCOT, PJM) entering 2000–2024

Figure E2. Share of requests operating (and withdrawn) by years since the request, by region, Aalen–Johansen estimates with 95% intervals.

The queue itself is the constraint. The capacity waiting at entry has grown many-fold in PJM since 2010, and since 2018 entrants neither leave nor get built at the old pace (Figure E3). This is the *level* problem of Appendix E.3: every cohort since 2017 completes below the one before.

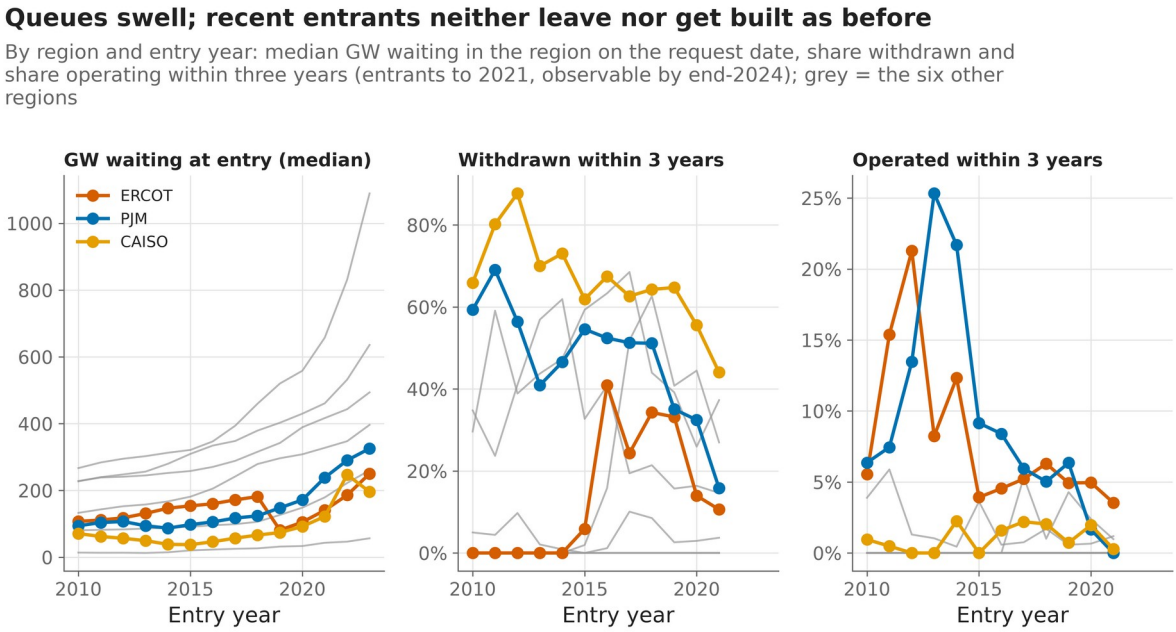


Figure E3. Requests, capacity waiting at entry, and three-year withdrawal and operation shares by region and entry year.

The rule. Look first in ERCOT, where the base rate is highest and the agreement is dated. Look at PJM for gas and its capacity market. Treat CAISO as a market where entry odds are a fraction of ERCOT's and only a signed agreement changes them.

E.3 Which requests get built, and when

Three predictors are compared throughout, always in this order. **The model** is a gradient-boosted discrete-time hazard model (LightGBM) that sees what is known on the request date. **The lookup** is the share built among earlier requests in the same region × technology × size cell — the number a careful analyst would compute by hand. **The cohort average** is the share built in the request's region and technology. Appendix E.7 adds **the landmark model**, refitted two years after entry with the executed-agreement flag.

At entry, the model beats cohort averages but not the lookup. The model ranks the test cohort at an AUC of 0.86 at three years against 0.73 for cohort averages, and its top tenth of requests reached operation four times as often as the cohort. But for requests of 20 MW and more, the lookup ranks as well as the model: Uno's C is 0.78 for both at three years, and the model wins none of six comparisons (Table E1). Three other learners — a random survival forest, a neural hazard model and a blend — were registered and scored once; none beat the lookup either. At entry, the ceiling is what the file records about a request, not the learner.

Test cohort 2017–2019	Cohort average	Lookup (≥ 20 MW)	Model	Model (≥ 20 MW)
AUC, 3 years	0.733	—	0.858	—
AUC, 5 years	0.725	—	0.766	—

Test cohort 2017–2019	Cohort average	Lookup (≥ 20 MW)	Model	Model (≥ 20 MW)
Uno's C, 3 years	—	0.783	—	0.779
Uno's C, 5 years	—	0.752	—	0.734
Top-decile lift, 5 years	3.35	—	3.96	—

Table E1. Ranking at entry. The model against cohort averages (all sizes) and against the lookup (20 MW and more).

Every cohort completes below the model's level. For requests of 20 MW and more, 11.0% of 2017 entrants and 5.9% of 2018 entrants operated within five years, while the model predicted about 12% for both. The 2020–2022 entrants operate within two years at about a quarter of the predicted rate. Refitting the intercept on the newest completed cohort removes about 40% of the error and none of the post-2020 gap. In megawatts, the 2017–2019 cohort was expected to bring 29.2 GW online within five years and brought 18.8 GW. A book must be sized to the newest cohort's level, not to the model's.

When: the model's clock helps in PJM only. Among test requests that operated within seven years, the model's expected year of operation is within a year of the truth 52% of the time, against 33% for the regional median. The gain is all in PJM; in ERCOT the historical median does as well.

Two years in, the model earns its keep. For a request still waiting two years after entry, the executed agreement changes what can be known. The landmark model with the agreement flag ranks the next three years at Uno's C 0.80 against 0.69 for the entry model, a gain of 0.11, while the lookup conditioned on surviving two years falls to 0.59 (Figure E4). In ERCOT, requests with an agreement in hand two years in operated within three years 56% of the time, against 5% without.

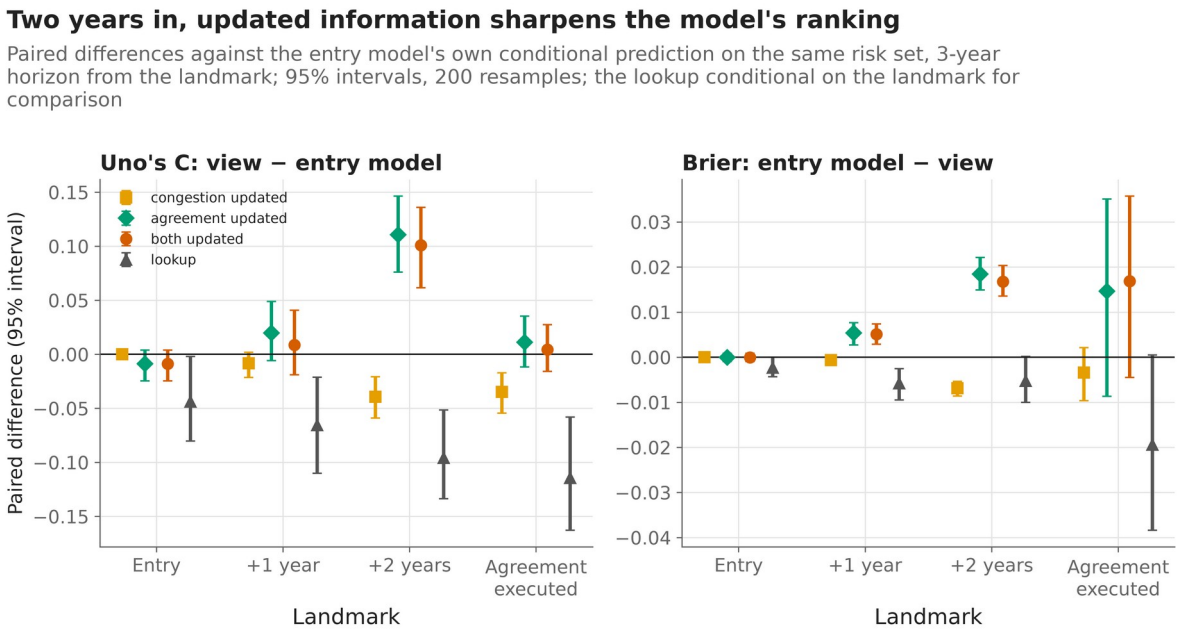


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A test entrants 2017–2019 still in the queue at each landmark

Figure E4. Gain in Uno's C and Brier score over the entry model at each landmark, three years ahead, for the landmark model with both inputs updated, with the agreement flag only, and for the lookup; 95% intervals. The text cites the agreement-flag variant.

The rule. At entry, rank by the lookup and the segment rules of Appendix E.4; the model adds nothing at utility scale. Two years in, re-score every request still held with the landmark model and the agreement. Plan the clock with the model in PJM and with the historical median in ERCOT. Size every book to the newest cohort's level.

E.4 Why requests fail

Size, technology and the agreement decide the odds (Figure E5). On the 5,497 entrants of 2010–2019, five-year completion runs 12.8% overall. In PJM it falls from 24% for requests under 5 MW to 4% at 500 MW and up; ERCOT is flat up to 500 MW. Gas builds most often (32% in PJM), solar-plus-storage less than stand-alone solar, and none of PJM's 37 offshore wind requests operated within five years. The executed agreement dominates everything: where the file dates one within two years of the request, 63% of ERCOT's requests were operating within five years; where none is dated, 0.8%. A survival model with both exits confirms the ranking: requests of 500 MW and more reach operation at 0.31 times the rate of those of 20–100 MW, and gas at 2.56 times solar (Appendix Figure A2).

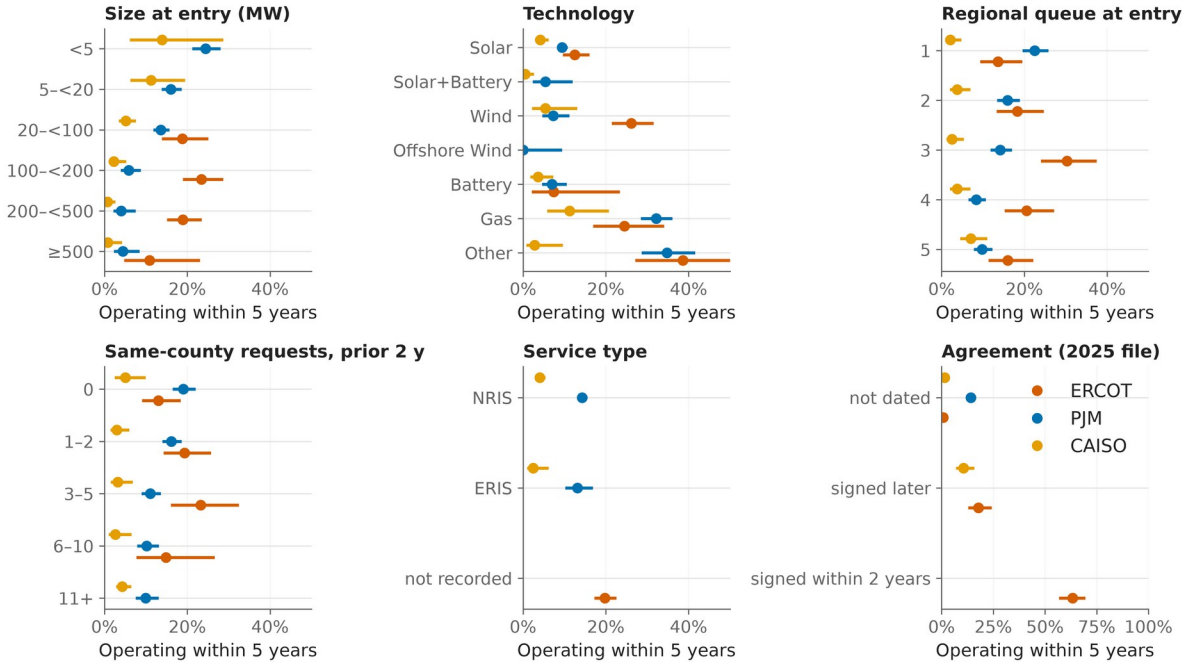


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A entrants 2010–2019 (CAISO, ERCOT, PJM), outcomes through 2024-12-31

Figure E5. Five-year completion by six drivers within each region, 2010–2019 entrants, 95% Wilson intervals.

The regions run on different clocks. CAISO and PJM cull early: 48% of CAISO's waiting requests leave in their first year, mostly out of the system impact study. ERCOT culls late, after year four, out of the facilities study. Surviving two years roughly triples the odds of operating in the next three (Appendix Figure A3).

Ten rules a company can apply by hand (Figure E6). A depth-four tree on the entry attributes, fitted on 2010–2016 entrants and re-measured on 2017–2019, gives ten rules whose order held out of time (Spearman 0.77). Small gas outside CAISO is the best segment: 47% then 48%. Large CAISO requests in crowded counties are the worst: below 2% in both periods. The one rule that broke — solar in a county with no earlier request — shows that crowding measured at entry is a weaker guide than size and technology.

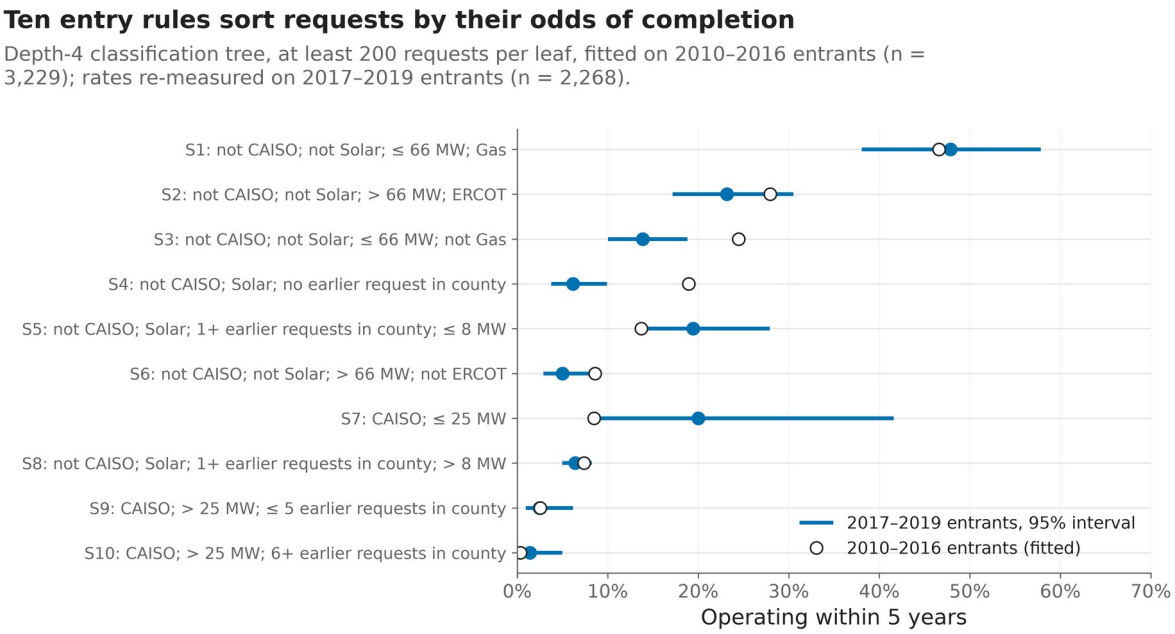


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A entrants 2010–2019 (CAISO, ERCOT, PJM), outcomes through 2024-12-31

Figure E6. Five-year completion by the ten rules, fitting cohort (2010–2016) and later cohort (2017–2019), with Wilson intervals.

The rule. Enter the top two segments; enter the middle ones only with the two-year review scheduled; do not enter the bottom three on entry information alone. Review in year two in CAISO and PJM, after the cull; review at the agreement in ERCOT.

E.5 What a plant is worth

At merchant prices only wind pays everywhere (Figure E7). Each plant is valued as a developer would: a 20-year life, capital and operating cost from the NREL Annual Technology Baseline 2024, today's tax credits, and cash flows discounted at the technology's own real cost of capital (4.2–5.4%). Two revenue cases run side by side throughout the report. *Merchant* sells at the on-peak hub price. *Contracted* sells under a 15-year power purchase agreement at index prices (solar $57/MWh, wind $62), then merchant. At merchant prices five of 21 technology-region pairs have a positive value: wind in all three regions and gas and solar in PJM. Under contract ten pairs are positive, including solar everywhere. Storage stays negative in both cases, because the public files give it capacity revenue only.

With capital counted, few plants pay at merchant prices

NPV per MW at operation, 20-year life, real WACC per technology from the ATB Market case; bars = base value case, lines = low–high value cases (energy, capacity, credit, gas-price and PPA-price cases); CAPEX, O&M and new-build capacity factors from NREL ATB 2024; storage has capacity revenue only

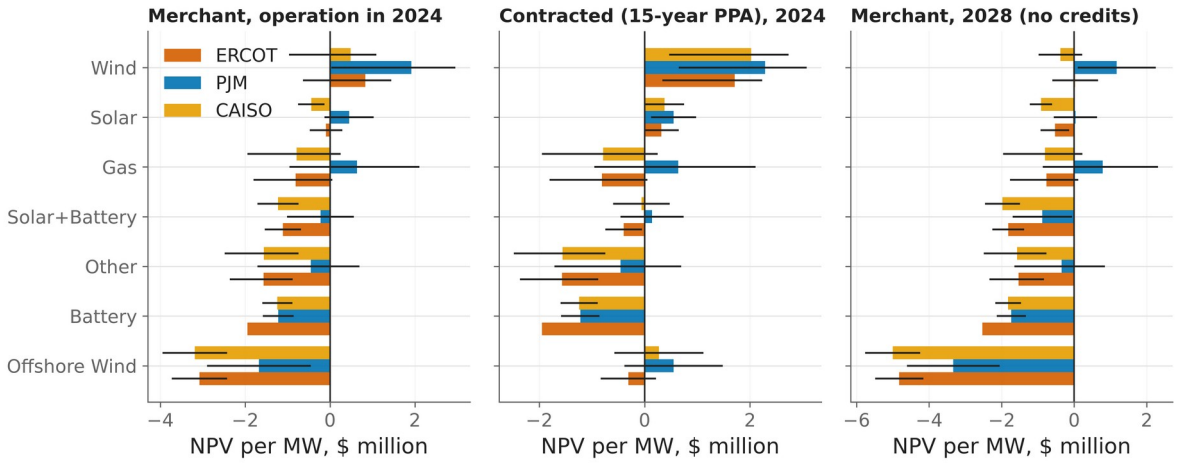


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: the value table of this notebook with new-build capacity factors; not a cohort

Figure E7. NPV per MW at operation by technology and region: merchant with today's credits, a 15-year contract at index prices, and merchant after the 2027 credit deadline.

The break-even says the same in dollars a company negotiates. A new solar plant needs a flat $44/MWh over twenty years and wind $41; the on-peak price the market pays them on Part I's national basis is below that for solar and above it for wind. Part II measures capture directly on ERCOT's 2024 hourly prices and finds the opposite order inside Texas (Appendix F.4), because Texas wind blows at night; Part I's assumed capture rates are national averages and are kept only for the three-region screen.

The rule. Develop wind anywhere; develop solar under a contract at or above the index, and in PJM even without one; develop gas in PJM, where the capacity market carries it; develop storage and hybrids only against a contract above their break-even; be operating by 2027 where the credit decides the sign. These values are in 2022 dollars; Part II works in 2024 dollars (the CPI-U ratio is 1.08).

E.6 Fund, hold or exit: two gates

The rule. At entry (the first gate) and two years in (the second), the company compares three things: the chance the request operates, its value if it does, and the cost of the next gate. It **funds** when the chance is at least the pool's expected rate — the mean of the probabilities the gate reads, so no outcome enters the rule; the first version used the cohort's realized rate (audit G2) — and the value is at least *h* times the cost, where the hurdle *h* is 3 at entry and 2 two years in; it **exits** when the chance is well below the pool or the value is negative even in the best case; it **holds** otherwise. At entry the chance comes from the lookup, because the model did not beat it; two years in, from the landmark model.

At entry, the funded requests were built three times as often as the cohort. At merchant prices the rule funds 166 requests and exits 1,542. The funded requests reached operation 33% of the

time, the exits 6%, the cohort 10% (Figure E8). In ERCOT the rule separates less — its funded requests operated 23% of the time and its exits 18% — because there the entry attributes do not separate the requests and the agreement does (Appendix E.4). Carried to the end, the funded class returned +$4.1B at merchant prices and +$9.1B under contract. The funded class is mostly ERCOT wind and PJM gas and solar.

Two years in, the accelerated requests were built four times as often as the exits. The rule accelerates 178 requests at merchant prices and 251 under contract. They reached operation within three years 29–30% of the time against 7% for the exits, and returned +$3.6B at merchant prices and +$9.0B under contract. Table E2 gives the counts, rates and intervals for both revenue cases.

Funded requests operated more often than exits, except ERCOT's at entry

Operated share by gate decision and region with Wilson intervals, cells under 10 requests omitted; hurdle 3 at entry (lookup cell and segment rate, n = 2,268) and 2 two years in (landmark model with the agreement flag, n = 1,298); NPV at the ATB cost of capital, merchant (top) and contracted (bottom) revenue

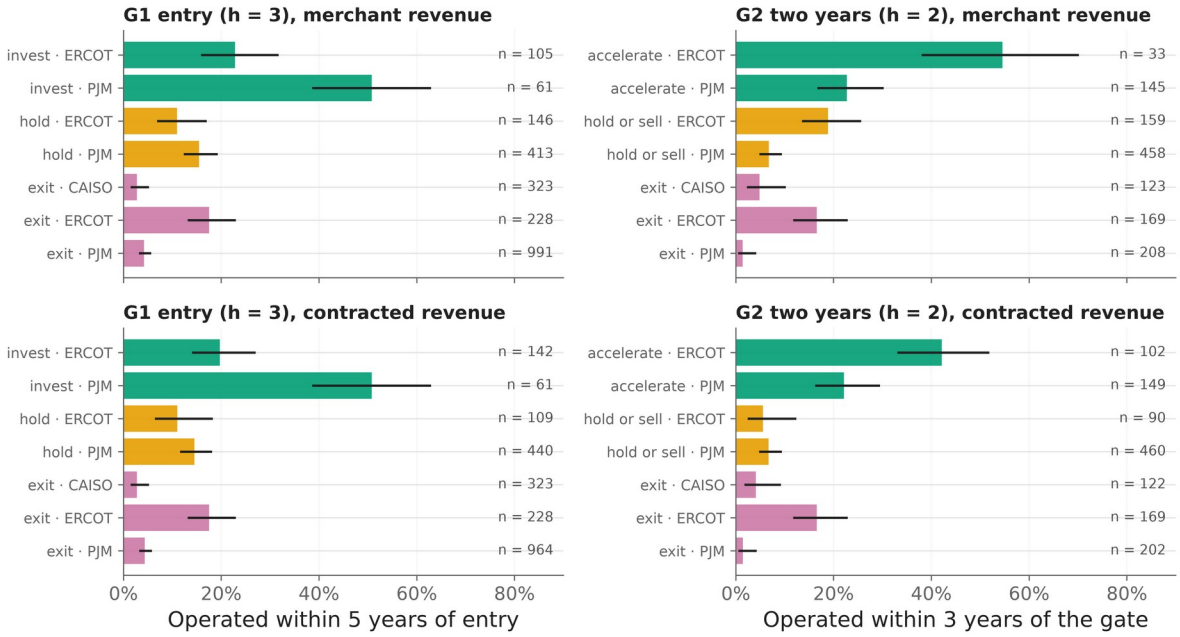


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A test entrants 2017–2019; outcomes through 2024-12-31

Figure E8. Share of requests that reached operation by gate decision and region, merchant and contracted revenue, with Wilson intervals.

Gate, revenue	Funded	Held	Exited	Funded: operated	Exited: operated	Funded: value, $M
Entry, merchant	166	560	1,542	33% [26%, 41%]	6%	+4,079
Entry, contracted	203	550	1,515	29% [23%, 36%]	6%	+9,137
Two years in, merchant	178	620	500	29% [23%, 36%]	7%	+3,642
Two years in, contracted	251	554	493	30% [25%, 36%]	7%	+8,975

Table E2. The two gates scored on what happened to the 2017–2019 requests. Operated: within five years of entry at the first gate, within three years of the second.

E.7 The book: when to buy and how much

Buy at the milestone, not at entry. A registered test settles the timing. Two years in, a book of 25 requests chosen by the landmark model held 19 that reached operation within three years, against 11 for the lookup that knew the same agreement status (Figure E9). A book of 50 held 33 against 24. The same books held fewer megawatts, so the landmark model picks more plants that get built, not more capacity. At entry the lookup ties or wins.

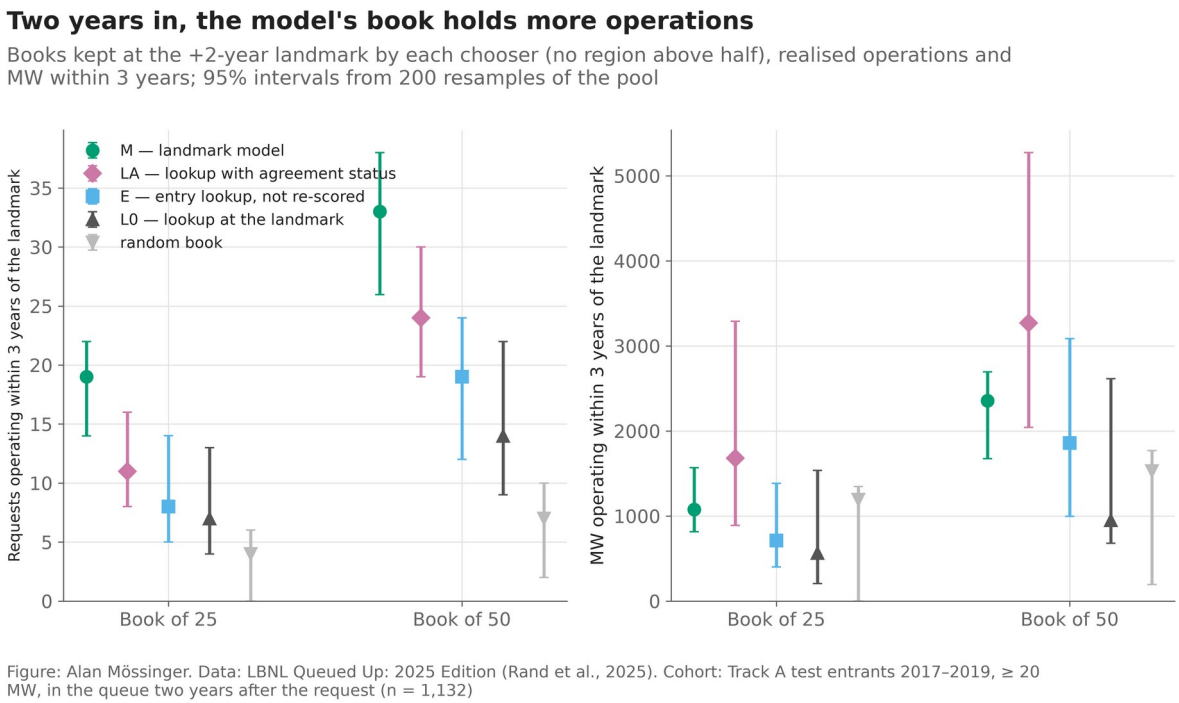


Figure E9. Operations and MW realised within three years of the two-year landmark, for books of 25 and 50 chosen by the landmark model, the agreement-aware lookup, the entry lookup and at random, with 95% intervals.

How many to add, and where. Adding requests in the landmark model's order while its probability stays at or above the pool's rate admits 410 of the 1,132 utility-scale requests still waiting; 23% of them operated, against 10.5% for the pool. This floor book is built without the regional cap of the registered book, which is why its first 25 picks hold 22 operations rather than 19. ERCOT is where the two-year review pays: 167 adds captured 70 of the region's 75 operations. In PJM, without dated agreements, the model separates little; in CAISO the floor admits two requests (Appendix Figure A4).

Four ways to spend one deposit budget. Four policies spent the same $20.11 million deposit budget on the 2017–2019 pool. Every policy that counts capital buys wind, and every one made money in both revenue cases, where a book chosen by megawatts lost it at merchant prices (Table E3). The risk-adjusted book barely differs from the plain value book, because the shared

risk between requests is small; the cohort level is the risk, and the newest cohort's early outcomes reveal it before a company buys.

Policy, same budget	Requests	Expected NPV, $M	P(loss)	Realised NPV, $M (merchant)	Realised NPV, $M (contracted)
Equal regional split	26	+1,404	2%	+581	+1,271
NPV per dollar, with caps	21	+1,536	3%	+396	+1,235
Model expected NPV, no caps	27	+1,803	2%	+391	+1,611
Risk-adjusted, with caps	21	+1,531	2%	+396	+1,235
Lookup, value index by MW	22	−747	79%	−46	+440
Random	22	−220	88%	−85	+15

Table E3. Six ways to spend the deposit budget on the 2017–2019 pool. Expected NPV and P(loss) are from the merchant scenarios.

The rule, region by region. *ERCOT:* invest; enter wind, enter solar with a contract, hold storage, and review every request at the agreement. *PJM:* invest in gas, wind and solar, prefer the light queues of the PJM core and requests under 66 MW, and review after the system impact study. *CAISO:* hold capital; enter only with a signed agreement and a contract.

E.8 Bridge to Part II

Part I answers the developer: which requests to fund and when. Part II answers the company: how to split one budget between those requests and assets that already produce. One thing carries forward — the hazard models of operation and withdrawal, refitted on what was known on the decision date, which become the timing engine of every power candidate in Part II. Two things do not: Part I's national value basis, replaced by ERCOT's 2024 hourly prices, and Part I's view from the request date, replaced by each request's chance of operating from 1 January 2025, conditional on how long it has waited (section 3).

Appendix F — Part II in full: the company's decision, how to split one budget

The method and evidence behind Sections 3, 5 and 6 for the energy portfolio.

F.1 The candidates on 1 January 2025

The leases are the twenty with the most output in the twelve months published by the decision date (to September 2024), among the 506 producing in 2024, out of 1,041 with at least sixty producing months. The requests are the five largest of each technology among the 1,441 ERCOT requests of 20 MW and more that ERCOT's own reports listed as live on 1 January 2025. In v6 they were drawn from the 2017–2019 entrants and valued at entry, though ten of those twenty had been withdrawn or built by the end of 2024; in v8 and v9 they were drawn from the 2025 edition, released after the date; this version draws them from what was public. Most of the twenty are 2023–2024 entrants of 500 MW and more, the size and age at which requests are least likely to operate, so their chances sit far below the queue's averages unless an agreement is signed.

Twenty leases, flat 2024 prices	Registered grid	Committee grid
Capital	$1.35B	$5.12B
Incremental NPV, base	$8.19B	$2.73B
Incremental NPV, low and high grid cases	$5.69B and $5.10B	$4.77B and −$4.77B
Leases positive at the high case	19 of 20	0 of 20

Table F1. The lease book under the two grids. The registered low case pairs low costs with a low uplift, which is why it sits below base.

F.2 What the energy data say, and how the features were built

A quarter of lease-months produce nothing, and that is normal: leases are shut in for maintenance, storms and workovers, so 25.3% of the 203,963 lease-months report no oil or gas. WTI and Henry Hub correlate at 0.06 month to month; Henry Hub and the ERCOT price at 0.63, because gas sets the Texas price in most hours; Gulf output with any price at almost zero (Appendix Figure A8). Every feature at a forecast origin uses only months that were public by then, and a truncation test shows that features rebuilt after deleting all later data are identical; the test covers the feature builders, not the choice of the lease universe.

F.3 Forecasting output: machine learning helps for plants, not for leases

Twelve forecast lines were registered before scoring: lease oil and gas at 3, 6 and 12 months, plant capacity factors at 1 and 3 months, and prices. Each pits a machine-learning challenger against the best of five transparent baselines at rolling origins (Figure F1). Three of twelve pass, all for plants. The lease lines fail: the registered challenger's error at three months is 1.35 times the last-value baseline's, a bias from forecasting in logs; a level model added after scoring, and labelled so, brings the ratio to 0.96, and the scenarios draw their output uncertainty from it. Prices behave as random walks. The decision-level test of machine learning is Section 8's: the book chosen with the lookup's queue curves against the book chosen with the model's.

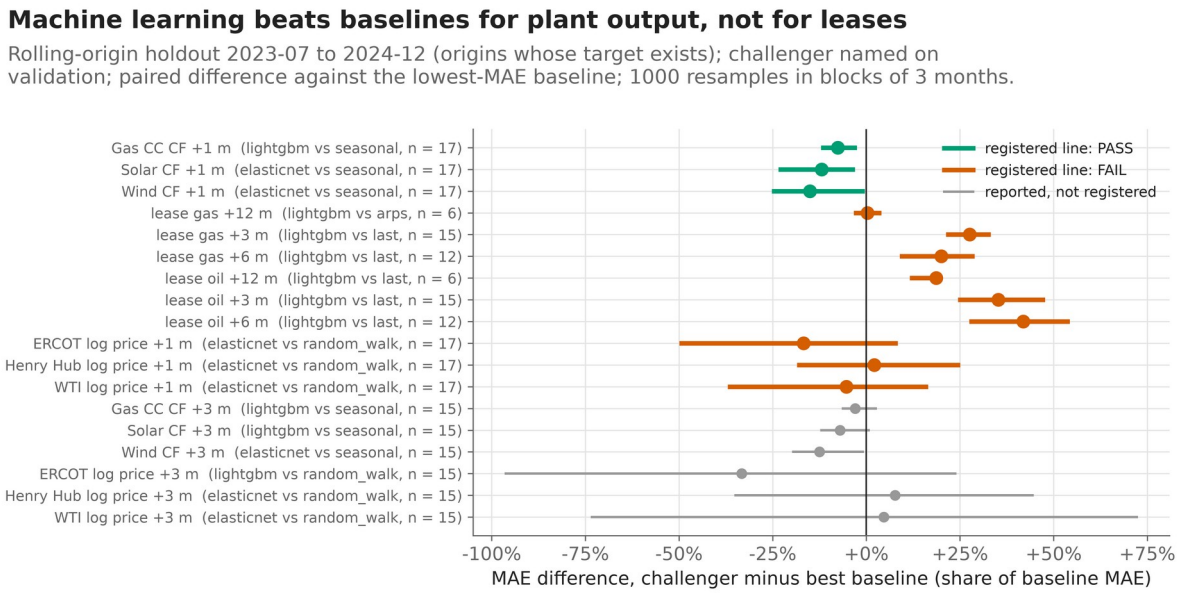


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). EIA-923 via EIA API v2 (electricity/facility-fuel). EIA WTI and Henry Hub monthly spot prices. ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: eligible leases, Texas plants and the three prices; holdout origins per line as labelled

Figure F1. Forecast error of the challenger relative to the best baseline, by registered line, with paired moving-block bootstrap intervals.

F.4 What each candidate is worth, and what that rests on

A lease redevelopment is an uplift on a producing asset. Each lease's baseline is an Arps decline fitted after the lease's peak in its last 36 months, switching to an 8% annual decline once its own decline falls below that; four leases still ramping decline at the floor from their recent level. The v6 fit had no floor, and ten of the twenty leases were projected never to decline. The redevelopment adds 20% of the baseline after a two-year lead time; its value is the cash flow it adds over doing nothing. The committee grid prices capital at $40,000 per incremental flowing barrel of oil equivalent a day and discounts at 8%; it was added after the registered results, at an outside reviewer's request, and is labelled ASSUMED.

A power request is a plant that may never be built. Its value if built uses the 2024 hourly prices — solar earns 92% of the flat hub price, wind 83%, gas combined cycle 136% because it runs only when prices clear its fuel cost — with ATB costs converted to 2024 dollars and the tax credits of the law in force on 1 January 2025: wind the production credit, solar and storage the investment credit, in every operation year of the window. The 2027 deadline for wind and solar was set by a July 2025 law and IRS Notice 2025-42, after the decision date; it is a stress. That value is weighted by the chance of operating in each of seven years and net of the deposits. On the flat 2024 path, under contract, ten of the twenty requests are positive and the twenty together sum to −$108M; without the credits −$933M, under the 2027 deadline −$478M.

F.5 Scenarios and risk

Prices, output, costs and timing are drawn together: 2,000 twenty-year paths resample calendar-year blocks of the 2018–2024 monthly log changes of WTI, Henry Hub and ERCOT, the same months for all three, capped at ±1.5 a month and reverting toward their 2018–2024 real geometric

means with a four-year half-life (ASSUMED). Output varies by the scored forecast errors, costs by a triangular multiplier, and each request's operation year by its curve from the decision date. The held-out set of 2,000 paths redraws every component with its own seeds; in v6 only its prices were redrawn. Two weaknesses remain. Seven historical years make every path, so there is effectively no joint oil–power tail beyond 2018–2024. And the reversion targets sit well above 2024's gas and ERCOT prices, which is what lifts the gas requests' scenario values above their flat-path values (Appendix Figure A5). A check on 5,000 paths — the registered 2,000 followed by new ones — leaves the order of the books unchanged (book D $1.36B, the deterministic book $1.21B, NPV ranking $1.14B, equal capital $1.12B) and book D's worst-5% loss at $0.84B; re-solved on 5,000 primary paths, book D keeps about two thirds of its members (Jaccard 0.65) and scores $1.382B against $1.363B.

F.6 Allocating the budget

Section 6 reports the books under the committee grid. Under the registered grid the leases are cheap against the barrels they buy, so the risk-aware book holds fourteen leases, expects $8.48B and almost never loses money; risk aversion has little to buy there. The registered comparison on that grid is Table A2.

F.7 When: the book at one to twenty years

A twenty-year NPV says nothing about what the owner holds at year three. Counting only the cash that falls within each horizon, under the registered grid, the best book holds nothing at one or two years, eleven leases at three and fourteen from five; its value within twenty years is $7.09B. A queue request is a carrying cost until it operates (Figure F4).

Leases pay from year three; queue requests stay a cost

Left: expected NPV of all forty candidates counted within each horizon, by sector (requests weighted by the queue models' probability of operating by each year). Right: the composition of the deterministic book at each horizon under the registered budget and cap.

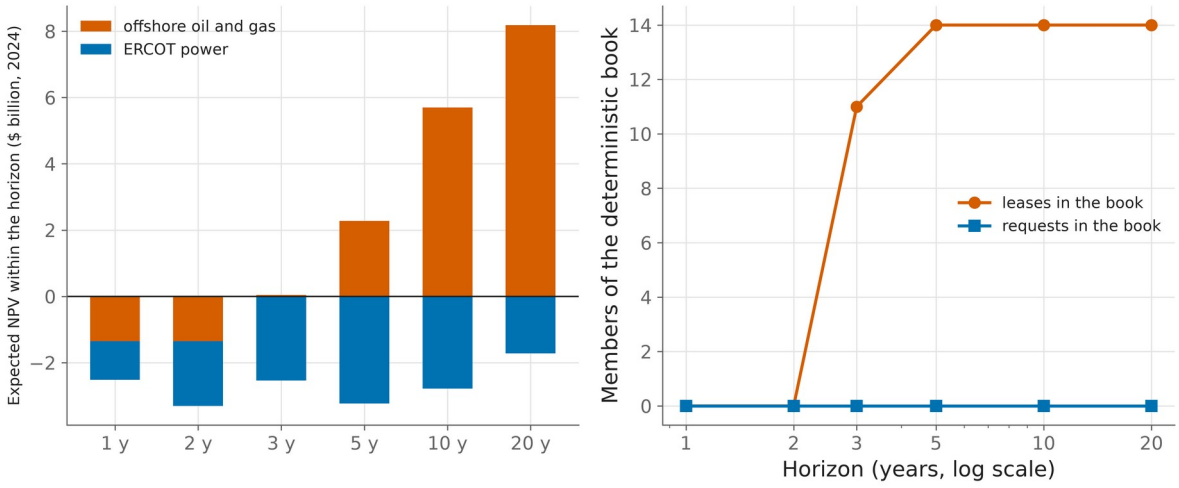


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: FORECAST. Cohort: the forty candidates of notebook 07 at the flat 2024 real prices

Figure F4. Expected NPV counted within 1, 2, 3, 5, 10 and 20 years, by sector, and the composition of the best book at each horizon.

F.8 The decision gym and the AI advisor

The gym scores decisions on outcomes, not prose. Sixteen past situations are rebuilt as of their date: fifteen requests drawn at random from each region's 2017, 2018 and 2019 entrants, the 2017 entrants two years in, ERCOT 2018 with a budget cut of a fifth, and three oil-price regimes — June 2014, January 2020 and March 2022. Since the audit the lease regimes hold the leases producing on their date, baselines fitted on what BSEE had published by then, scenarios drawn from the seven years before the date, and outcomes from the leases' observed production; v6 used the 2024 leases, 2024 fits and the realized future's mean. The three regimes are nearly one decision and count as one situation. Actors: instinct (top-N expected NPV), instinct per dollar, nothing, the registered rule (the gate rule for requests, the risk-aware optimizer for leases), the advisor and, when one is recorded, the human.

The rule beats instinct only by packing the budget better. Over the fourteen effective situations the rule beats instinct by $56M a situation, an interval of [−$12M, +$176M] that includes zero, and ties instinct per dollar (−$2.7M [−$12.7M, +$3.0M]): its edge in the lease regimes is choosing by value per dollar under a binding budget. In the queue rooms the gate rule's fund threshold is now the pool's expected rate, not the test cohort's realized one (audit G2), and the rule is behind instinct by $0.6M a room. The advisor trails the rule by $5.1M a situation [−$11.0M, −$1.5M].

The rule's edge over instinct, $56M a situation, is within noise

14 effective situations (the three lease regimes count as one), paired 95% interval [-12M, 176M]; the advisor -5.1M a situation against the rule. Top: queue situations as of their date (realized value = modeled value per MW × realized operation − cost to the gate, a DERIVED proxy); in 6 of them no drawn request had a positive realized value, so the best book is empty. Bottom: the lease regimes rebuilt as of their date, scored on observed production and prices. The tick is the best book with hindsight.

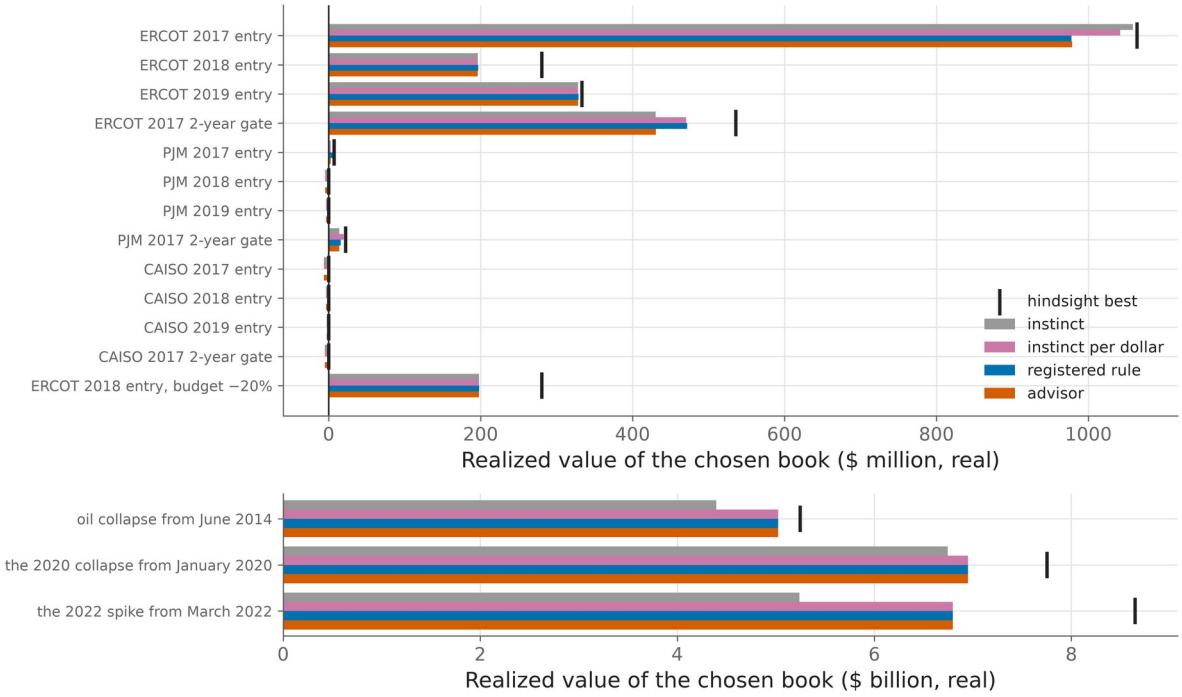


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: DERIVED. Cohort: ERCOT, PJM and CAISO requests of the 2017–2019 cohorts at entry and the 2017 cohort at its two-year gate; the twenty largest producing leases as of June 2014, January 2020 and March 2022

Figure F5. Realized value of the book each actor chose in each past situation, against the best book with hindsight.

Appendix G — Every finding, its evidence and its verdict

Question	Finding	Evidence	Verdict
Where to look	ERCOT builds most often: 18.4% of requests operate within five years, against 4.8% in CAISO.	Appendix E.2	Observed
Q1, at entry	The model beats cohort averages (AUC 0.86 against 0.73) but ties the lookup at utility scale.	Appendix E.3; C1–C12	Partly
Q1, two years in	A book of 25 chosen by the landmark model held 19 operations, against 11 for the lookup.	Appendix E.7; C15	Supported
Q1, on the decision date	Backtested as of 2020 the model ranks better than the lookup (AUC 0.731 against 0.686; in ERCOT 0.656 against 0.591) but over-predicts ERCOT's level.	Section 4	Supported for ranking

Question	Finding	Evidence	Verdict
Q1, one year on	30 of 1,441 live requests operated in 2025 where the corrected model expected 78 and the lookup 17; the model's slope is 0.66, the lookup's 0.01.	Section 4; Figure 3	Post-hoc
Q1, output	Machine learning beats baselines for plant output, not for leases or prices.	Appendix F.3; C20	Partly
Q1, in the decision	The book chosen with the lookup's curves is not beaten by the model's on the registered metric (+$37M [−$76M, +$151M]).	Sections 4 and 8	Not supported
Q2, risk	Under the committee grid the risk-aware book loses money in 9% of scenarios with a worst-5% loss of $0.82B; its construction calls exceed the reserve in 50.3% of scenarios.	Sections 6 and 8	Derived
Q2, the primary line	Under equal constraints the risk-aware book beats NPV ranking by $219M [+$13M, +$401M] on the registered metric, for $508M of expected value.	Section 6; C33	Registered, PASS
Q2, the line priced	A $2.53B contingent line is worth $1.03B on the metric net of a 0.5% fee and 8% interest, against $0.76B with no line; the fee breaks even at 2.04%.	Section 8; Figure 6	Post-registration
Q2, sensitivity	Oil at $45 for five years removes every lease from the book; redevelopment capital 30% higher swaps two members for three; the registered tail weight is best in three of four misspecified worlds.	Section 6; Appendix I	Derived
Q2, flexibility and information	Revising the book in 2027 with this rule is worth −$11M; perfect information is worth at most $825M, the ERCOT price level $47M, an agreement signal at most $59.4M.	Sections 4 and 6	Post-registration
Q3, the primary line	Adding the language model to the tools improves the committee's own objective by $43M per case [+$9M, +$86M] over all 48 cases.	Section 10; C34	Registered, PASS
Q3, reading	40 of 40 paraphrases read exactly; 38 of 48 case readings equal their key; 75% of injections held with 7 memos carrying an ungrounded number.	Section 10; C37–C39	Registered: PASS, FAIL, FAIL
Q3, argument	12 of 12 author-written flaws of argument named; 0 of 12 sound proposals over-challenged.	Section 10; C40–C41	Registered, PASS
Q0	Tools add $76M per case over the proposal, the language model $43M more, the optimizer $134M beyond that.	Section 10; C35–C36	Registered
Q3, the advisor	Its investment increment is zero; it read six questions correctly; it trails the rule in the gym.	Appendix B	Post-registration
Q2, financed the same way	Every process within its own line net of fee and interest: D leads at $1.03B against $0.82B (C) and $0.72B (A); D over A +$312M [+$110M, +$506M], over C +$213M [+$17M, +$412M]; with no line equal capital leads.	Section 8; Table K3; Figure 6	Post-registration

Question	Finding	Evidence	Verdict
Q2, the book re-solved with the line	9 members, a $1.31B facility, −$49M [−$127M, +$23M] against D.	Section 8	Post-registration
Q2, dependence and worlds	The tail grows monotonically with the share of co-moving scenarios (comonotonic bound $2.19B); D scores best in 7 of 7 worlds.	Section 5; Figure K4	Post-registration
Q0, the stronger baseline	Against an optimizer-first rule the language model's edge is −$14M [−$36M, +$2M]; the rule beats the checklist by $56M [+$19M, +$100M].	Section 10; Figure 7	Post-registration
Q2, an emissions cap	Any cap on the book's 1.27 Mt a year swaps its gas request for the 1,026 MW solar request with an agreement; the zero-emission book gives up $201M on the metric, $158 per tonne-year.	Section 7	Post-registration

Table G1. Every finding, its evidence and its verdict. Registered criteria are in Appendix A and Table 1; the V2 additions are post-registration (notebook 13).

Appendix H — The audit of 17 October 2026

An independent audit of report v6 and its code (`docs/audit_2026-10-17.md`) found that the prediction layer held and that valuation, allocation and the gym did not. The five findings that changed the recommendation:

Finding	What was wrong	What changed
A1	The top-N and equal-capital books ignored the registered 40% technology cap; the risk-aware book's "$1.9B for a smaller tail" was the price of the cap, not of risk aversion.	Every book is held to the same budget and cap and checked.
E1	Ten of the twenty power candidates had been withdrawn or built by the end of 2024, while the leases were valued at 2025.	One decision date; the requests are those still waiting, with chances conditional on their age — and, since the critique of 21 October, those ERCOT's own reports listed on the date.
E2	Ten of the twenty lease baselines were fitted with no decline.	A post-peak fit with an 8% terminal decline; the registered lease book fell from $14.6B to $8.19B.
G1	The gym's lease rooms used 2024 leases, 2024 fits and the realized future's mean.	Rooms rebuilt as of their date and counted as one situation.
G5	The advisor's value over the rule was never reported; it was negative.	Reported, and the advisor rebuilt to work through the engine.

Table H1. The audit's headline findings. Smaller corrections — the CVaR tail count, the gas heat content, the gate floor, ATB dollars, the credits, the provenance labels, the held-out draw — are in the audit's register and decisions #88–#92.

Changes after review. Three outside reviews in October 2026 led to the changes logged in `docs/mentor_feedback.md`. The first asked for an evaluated decision intelligence system; its first request was the audit above. The second found three inputs that were not public on the decision date and the capital calls unpriced, and asked that the language model help the committee decide; the challenger and its first registered test were the answer. The third, a critique review of 21 October, found that the candidate set rested on a file released after the decision date, that the challenger's test statements came from the model family being tested, that the construction line was free, and that no primary estimand had been named; report v10 rebuilt the candidates from what was public, priced the line, named one primary line per question, added the four-arm ladder and the one-year forward test, and kept a three-column ledger of every registered line and its corrections. The executive review of 23 October 2026 asked for a decision-first report, every process financed the same way, a critical assessment of the scenario engine, a stronger baseline for the language model and an operational readiness assessment; this version (V2) answers it (`docs/v2_review.md`, notebook 13, decision #113). A course mentor is assigned after the proposal review; their feedback will be logged in the same file.

Appendix I — Capital allocation memo

To the investment committee, 1 January 2025. Every figure is a model output scored on held-out scenarios; lease costs are assumed; the candidates are those public on the date.

The ask. Commit up to $3.82B now — 40% of what the forty candidates would cost the budget today — across twenty Gulf lease redevelopments and twenty ERCOT requests still in the queue, with no technology above 40% of the budget. A lease costs its redevelopment capital; a request costs its deposits plus a reserve for the construction it may need.

The choice. The return-seeking book expects $2.73B with a worst-5% average loss of $1.52B. The registered risk-aware book expects $2.17B with a loss of $0.82B and a 9% chance of losing money. A more defensive book expects $2.07B with a loss of $0.74B. I recommend the committee choose its tail weight first; at the registered weight that is the risk-aware book.

What to do with each candidate under that book. Invest now in eight candidates, the leases and the requests with signed agreements. Invest in stages in five requests: pay the deposit now, commit construction at the agreement. Wait for information on twelve requests whose agreement signal is worth more than their deposit. Monitor fourteen; reject one.

What the commitment really is. The construction reserve is an average: the capital the book's requests call exceeds it in 50.3% of scenarios. Without a contingent line the book is worth $1.71B, not $2.17B, because requests that reach their agreement cannot be built; a line of $2.53B restores $2.11B gross and $1.98B net of a 0.5% commitment fee and 8% interest on the draws.

What would change it. Oil at $45 a barrel for five years removes every lease from the book and leaves the five wind requests, a book that expects $0.59B and loses money in 58% of scenarios. Redevelopment capital 30% higher drops two members for three and keeps $0.91B of leases, at a 15% chance of a loss. Knowing everything before committing would be worth at most $825M; the 2025–2029 ERCOT price level, $47M; one request's agreement signal, at most $59.4M.

Before you sign. State your rules in words — your tolerance for loss, your liquidity, what you will not hold, and your tail weight, because an unstated one is read as zero. The challenger reads them back as numbers for you to confirm, tests your proposal against them and names the computed book that keeps them. If you depart from a book it guides you to, the ledger asks you to acknowledge the evidence.

Appendix J — Glossary

The terms as this report uses them.

Term	Meaning in this report
Request	An application to connect a new plant to the grid; it ends as an operating plant or a withdrawal.
Entry, two years in, the agreement	The three moments a request is reviewed: the request date, two years later, and the date an interconnection agreement is executed.
The model	A gradient-boosted hazard model (LightGBM) that sees only what is known on the request date.
The landmark model	The same model refitted two years in, with the agreement flag.
The lookup	The share of earlier requests in the same region, technology and size band that were built — what a careful analyst would compute by hand.
The cohort average	The share built in the request's region and technology.
The level	The share of a cohort that is ever built; it has fallen for every cohort since 2017.
Merchant, contracted	Two revenue cases: selling at market prices, or under a 15-year power purchase agreement at index prices.
Book	A set of projects funded together under one budget.
The grid	The assumed costs of redeveloping an oil and gas lease; the *registered grid* was fixed before any result was seen, the *committee grid* is the harder one an investment committee would use, and the main case throughout.
OBSERVED, DERIVED, FORECAST, ASSUMED	The provenance of every dollar figure: a public record, arithmetic on records, a model's output, or a stated assumption.
Held-out scenarios	2,000 scenarios redrawn in every component and never used to choose a book; every book is chosen on 2,000 primary scenarios and scored on the held-out ones.
Worst-5% average loss	The mean of the 100 worst of 2,000 held-out outcomes — the conditional value at risk at 95% of the loss (CVaR95).
Tail weight (λ)	How much a dollar of worst-5% average loss counts against a dollar of expected NPV; λ = 1 is the registered setting.
Registered metric	Expected NPV minus the worst-5% average loss (tail weight 1), on held-out scenarios.

Term	Meaning in this report
Registered line	A success criterion written and committed, with its rule, before the data it scores were looked at; scored once, never re-scored.
The challenger, the checklist	The governed language model that reads a committee's rules and writes a memo from computed facts; the same tools with fixed rules and no language model, which is the control arm.
The facility	A contingent construction line that funds the capital a request calls beyond its reserve when its agreement is signed.
Invest now, invest in stages, defer, monitor, reject	The five statuses the system assigns to each candidate.

Appendix K — The report in full

Sections 1–9 as written before the body was brought to the course's length. References to "Section n" inside this appendix are to K.n; Figures K1–K5 and Tables K1–K3 are here, Figures 1–7 and Table 1 in the body.

K.1 Introduction

K.1.1 Motivation

Oil, gas, power, renewables and storage compete for one budget, and the competition is unequal. A lease redevelopment spends its capital now and produces from year three; it rarely loses money, and when it does, every lease loses together because they share one oil price. A power request costs little to hold — deposits and studies — but calls for ten times as much the day its interconnection agreement is signed, and most requests never reach that day: of the capacity that asked to connect in 2000–2019, 12.6% was operating by the end of 2024 and 76.9% had withdrawn (Rand et al., 2025). A committee that ranks these candidates by expected value fills the budget with the largest requests and the largest leases, and takes a tail it has not priced.

Maximizing expected NPV is not what an owner wants. It wants expected value and a tail it can bear, and the two trade off at a rate the committee must choose rather than discover after the fact. Prediction does not settle the trade-off. A better forecast of which requests will be built changes the ranking of candidates, but the book is sized to the level — how many will be built — and the level is what no forecast knew well on the decision date. The decision is made by the constraints, the capital plan and the tail.

Five challenges follow. *Allocation across competing technologies:* distinct risk–return profiles, horizons and capital needs mean the best portfolio is not the best projects taken one by one. *Uncertainty and its interactions:* commodity prices, power prices, production, completion and construction costs are uncertain together, and their co-movement can produce portfolio losses that no single investment shows. *The gap between forecasting and deciding:* a more accurate prediction changes a ranking; it does not say how much risk to accept or when to commit. *Financing and liquidity:* some projects cost little to hold and much to build, so a portfolio that is attractive on paper can be infeasible when several call for construction capital at once. *Committee decision quality:* proposals carry inconsistent assumptions, sunk-cost reasoning, unpriced

downside and departures from the company's own rules, and conventional tools find numerical errors without challenging the reasoning.

K.1.2 Research questions

The project asks four questions, each with a pre-registered primary test (Section 5).

- **Q1, prediction.** Do machine-learning forecasts of completion, output and prices beat transparent baselines, and does a better forecast make a better portfolio?
- **Q2, risk-aware allocation.** Under identical constraints, does a risk-aware portfolio beat NPV ranking, equal capital by sector and deterministic optimization on a stated risk–return metric, and does the answer survive consistent financing and adverse markets?
- **Q3, governance.** Does a governed language model improve a committee's decision beyond deterministic tools, does it read policy reliably and resist manipulation, and can its role be bounded so that it reads and argues but never computes or decides?
- **Q0, attribution.** Where does the value come from — the tools, the language model or the optimizer?

K.1.3 Contributions

The contribution is an integrated, evaluated architecture rather than any single model: prediction, uncertainty, prescription, financial feasibility and an accountable human decision, each measured on the same held-out scenarios. Three findings follow from it. Risk-aware optimization improves modelled portfolio decisions over conventional selection, and the improvement grows when financing is made consistent. Forecast accuracy creates value only through the decision, not by itself. A governed language model can support executive reasoning, policy interpretation and committee oversight without holding authority — and its demonstrated value lies there, not in choosing the portfolio.

K.1.4 Organization

Section 2 places the work in the literature. Section 3 describes the data and the candidate universe. Section 4 presents the methods stage by stage. Section 5 sets out the evaluation design and the pre-registered criteria. Section 6 reports the results. Section 7 discusses what they mean, where they fail and what they do not show. Section 8 addresses ethical considerations and Section 9 concludes. The appendices keep the full evaluation record, the challenger's design, the reproducibility map, the supporting figures, the two parts of the analysis in full, every finding with its verdict, the audit of 17 October 2026 and the capital-allocation memo; references to "Section *n*" inside the appendices use the numbering of the original report, and a mapping is given at the start of the appendices.

K.2 Related work

From prediction to prescription. The distinction between predicting an outcome and prescribing a decision is the subject of a growing literature. Bertsimas and Kallus (2020) formalize prescriptive analytics as learning decisions from data rather than learning forecasts and optimizing afterwards; Elmachtoub and Grigas (2022) show that when a forecast feeds an optimization, the forecast should be trained on decision error rather than prediction error; Lepenioti et al. (2020) survey the field and its open problems. This project takes the complementary, evaluation-first position: it keeps prediction and optimization as separate, testable modules and asks, on held-out scenarios, whether a better predictor produced a better decision.

Portfolio theory in energy. Mean–variance portfolio selection (Markowitz, 1952) was carried into electricity planning by Awerbuch and Berger (2003) and into fuel-mix diversification by Roques, Newbery and Nuttall (2008). This project replaces variance with the conditional value at risk, whose linearization (Rockafellar & Uryasev, 2000) allows the tail to be optimized exactly in a mixed-integer program, and it adds the constraints a committee actually faces — budget, concentration caps, liquidity and staged capital calls. Investment under uncertainty and the value of waiting (Dixit & Pindyck, 1994) motivate the "defer pending information" status and the value-of-information calculations.

Commodity prices and scenarios. The scenario engine resamples calendar-year blocks of historical log changes (Künsch, 1989) around a mean-reverting level, a reduced form of the short- and long-term factor models of Schwartz and Smith (2000) and the long-run evolution of energy prices described by Pindyck (1999). Out-of-sample forecast evaluation follows Tashman (2000) and Hyndman and Athanasopoulos (2021).

Interconnection queues. Lawrence Berkeley National Laboratory's *Queued Up* series (Rand et al., 2025, 2026) is the source of the queue data and of the completion rates that motivate the level problem; Johnston, Liu and Yang (2023) analyze the queue empirically, and FERC Order No. 2023 reformed its procedures. The completion models are discrete-time hazard models (Cox, 1958, 1972; Allison, 1982) with competing risks (Fine & Gray, 1999; Putter, Fiocco & Geskus, 2007; Aalen & Johansen, 1978), refitted at a landmark (van Houwelingen, 2007), fitted with LightGBM (Ke et al., 2017) and evaluated with time-dependent concordance and Brier scores (Uno et al., 2011; Blanche et al., 2013; Graf et al., 1999; Wolbers et al., 2014). Calibration is assessed after Zadrozny and Elkan (2002). Lease baselines use Arps (1945) decline curves.

Language models in decision support. The challenger is built on two observations: that motivated reasoning shapes what committees argue (Kunda, 1990), so a challenger must be able to state the opposing case as its holders would (Caplan, 2011); and that a language model's numbers must be grounded in retrieved, verified facts (Lewis et al., 2020). The design goes further than grounding: the model never computes a value, every number in its memo is checked against a computed fact sheet, and the action it may take is bounded by code.

K.3 Data

K.3.1 Sources

Every source is public: LBNL's *Queued Up* project files (2025 edition for the decision, 2026 edition for outcomes), ERCOT's monthly interconnection reports, BSEE lease production, EIA prices and generation, ERCOT's hourly prices, the PJM market monitor's capacity revenues, LBNL's PJM interconnection-cost estimates and the NREL Annual Technology Baseline 2024. The file data/raw_manifest.csv records every raw file with its hash, and a clean Linux environment reproduced every registered verdict from them (Appendix C).

K.3.2 One decision date and the candidate universe

Both sectors are decided on 1 January 2025 with what was public that day. The twenty leases are those with the most output in the twelve months published by the decision date (to September 2024), among the 506 producing in 2024, out of 1,041 with at least sixty producing months. The twenty requests are the five largest of each technology — gas combined cycle, solar, wind, storage — among the 1,441 ERCOT requests of 20 MW and more that ERCOT's own reports listed as live on the date, about 310 GW. For the queue, the LBNL 2024 edition (requests through 2023)

and ERCOT's eight monthly reports of 2024 say which requests were live, their size and whether an agreement was signed. Each request's chance of operating within seven years is conditional on its age and on whether its agreement is signed; the hazard models learn from histories censored at the date, in which a request still waiting is right-censored, never counted as failed. The date discipline has a cost the report keeps visible: the 2024 exits were in the ISOs' own reports during 2024 but compiled by LBNL later, a vintage gap stated, not removed. Most of the twenty requests are 2023–2024 entrants of 500 MW and more, the size and age at which requests are least likely to operate, so their chances sit far below the queue's averages unless an agreement is signed.

The budget is 40% of what the forty candidates would cost: $3.82B, with no technology above 40% of it. A lease charges the budget its redevelopment capital now; a request charges its deposits now plus a reserve for the construction it will call if it reaches its agreement, the expected draw.

K.3.3 Valuation inputs and provenance

A lease redevelopment adds 20% to output after two years; its value is the cash flow it adds over a baseline that declines at least 8% a year (ASSUMED; 5–12% as the range). Each lease's baseline is an Arps decline fitted after the lease's peak in its last 36 months, switching to an 8% annual decline once its own decline falls below that. Redevelopment costs are not public, so results are shown under two grids — the registered one and the harder *committee grid*, which prices capital at $40,000 per incremental flowing barrel of oil equivalent a day and discounts at 8% — and the committee grid is the main case throughout. A power request is valued on 2024 hourly prices — solar earns 92% of the flat hub price, wind 83%, gas combined cycle 136% because it runs only when prices clear its fuel cost — with ATB costs converted to 2024 dollars and the tax credits of the law in force on the date; the 2027 deadline a July 2025 law set for wind and solar is a stress. Every dollar figure carries a provenance label — OBSERVED, DERIVED, FORECAST or ASSUMED.

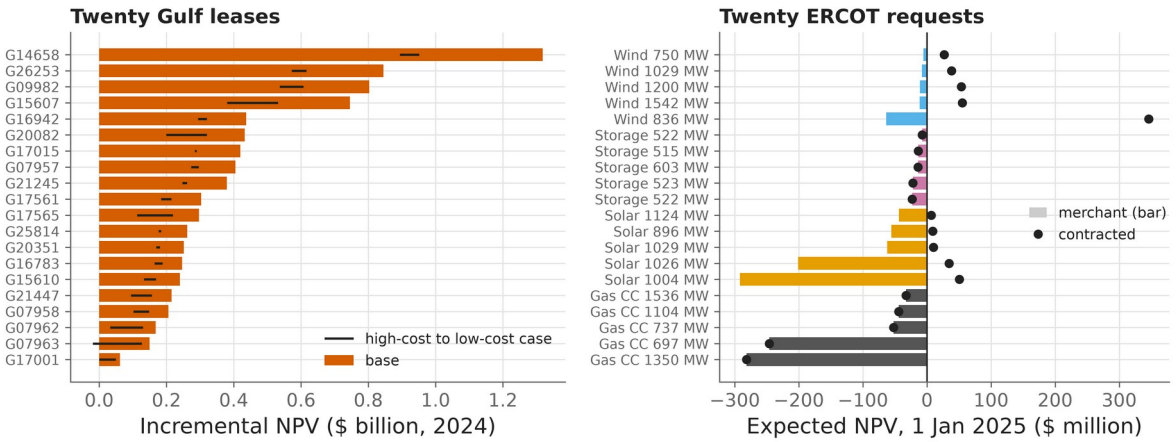


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: ASSUMED. Cohort: the default book: the 20 leases with the most output in the twelve months published by 1 January 2025, and the five largest ERCOT requests of each technology still in the queue that day

Figure K1. The forty candidates: incremental NPV of the twenty lease redevelopments under the base grid with the high-cost to low-cost span (costs ASSUMED), and the expected NPV of the twenty ERCOT requests under merchant prices and under a 15-year contract (production FORECAST).

Figure K1 shows the first consequence for a committee: every lease adds value at the base cost, and the wind and solar requests pay only under contract. On the flat 2024 path, under contract, ten of the twenty requests are positive and the twenty together sum to −$108M; without the credits −$933M, under the 2027 deadline −$478M.

K.3.4 What the data say

A quarter of lease-months produce nothing, and that is normal: leases are shut in for maintenance, storms and workovers, so 25.3% of the 203,963 lease-months report no oil or gas. WTI and Henry Hub correlate at 0.06 month to month; Henry Hub and the ERCOT price at 0.63, because gas sets the Texas price in most hours; Gulf output with any price at almost zero (Appendix Figure A8). Every feature at a forecast origin uses only months that were public by then, and a truncation test shows that features rebuilt after deleting all later data are identical. For the queue models, only CAISO, ERCOT and PJM date both operation and withdrawal, so the timing models cover those three regions: 13,366 requests. The models are fitted on requests that entered up to 2016 and tested once on the 2,268 that entered in 2017–2019, every one followed for at least five years (Appendix E).

K.4 Methods

K.4.1 The six-stage framework

The framework runs in six stages (Figure 1). Each stage is a tested module of the repository; the arrows carry what each hands the next; and the language model works in one stage only, where it reads and argues and never computes a value or decides. Table K1 summarizes what each stage reads, does, produces and supports, what limits it and how it was tested.

Stage	Input	Method	Output	Decision supported	Main limitation	Validation
1 Prediction	queue histories to 2023; ERCOT 2024 reports; lease production	hazard models beside a lookup; decline curves; forecast tournament	chance of operating by year; lease baselines	which requests to hold; what a lease adds	ranks well, level wrong; costs assumed	backtest as of 2020; 2025 forward test; 12 registered lines
2 Uncertainty	prices 2018–2024; forecast errors; cost ranges	2,000 joint paths, mean reversion; cash flows per path; 2,000 held out	NPV per candidate per scenario; capital calls	how large the tail is, and why	no tail worse than 2018–2024; half-life assumed	dependence dial; seven worlds; 5,000-path check
3 Prescription	the scenario matrix; budget, caps, tail weight	mixed-integer program, CVaR linearised, solved exactly	the book, frontier, what-ifs, statuses	fund, stage, defer, monitor, reject	one preference among many	C33; 20 re-selections; every process financed alike
4 AI challenge	the committee's words; the proposal; the fact sheet	model reads rules and writes the memo; tools diagnose, repair, re-solve; code verifies	reading, memo, alternatives, level	is the proposal broken, fragile, unfunded, ill-argued	reads silence as zero; embellishes	48 registered cases; 40 paraphrases; 20 injections; the ladder
5 Decision	memo, alternatives, committee	a named person signs; override gate; hash-chained ledger	a signed, auditable decision	the commitment	assent recorded, not verified	the ledger's refusal tests
6 Outcome	the decision; held-out scenarios; the realized year	scoring in five to seven worlds; regret; realized 2025 outcomes	the record of every registered line	keep, re-fit or re-register	simulated committees; one model	9 lines scored once; clean-environment reproduction

Table K1. The six stages: what each reads, does, produces and supports, what limits it, and how it was tested.

K.4.2 Prediction

For the queue, gradient-boosted hazard models (LightGBM) estimate each request's chance of operating in each of seven years from what is known on the request date — size, technology, region, the queue's congestion, the year's cohort; a *landmark* refit two years in adds the agreement flag. Beside them stands the transparent lookup a careful analyst would compute by hand: the share built among earlier requests of the same region, technology and size band. Three other learners — a random survival forest, a neural hazard model and a blend — were registered and scored once. For the energy side, twelve registered forecast lines pit a machine-learning challenger against the best of five transparent baselines for lease output, plant capacity factors and prices, at rolling origins with paired moving-block bootstrap intervals. Appendix E gives the queue models in full, including the ten hand-applicable rules and the two-gate fund/hold/exit policy; Appendix F.3 gives the forecast tournament.

K.4.3 Uncertainty

Prices, output, costs and timing are drawn together: 2,000 twenty-year paths resample calendar-year blocks of the 2018–2024 monthly log changes of WTI, Henry Hub and ERCOT prices, the same months for all three, capped at ±1.5 a month, and revert toward their 2018–2024 real geometric means with a four-year half-life (ASSUMED; Figure K2). Lease and plant output vary by the scored forecast errors, costs by a triangular multiplier, and each request's operation year by its curve from the decision date. The cash flows of every candidate are computed on every path, so the NPV of each candidate in each scenario is a matrix the optimizer reads directly, and the capital a request calls is a per-scenario quantity, not an average. The held-out set of 2,000 paths redraws every component with its own seeds.

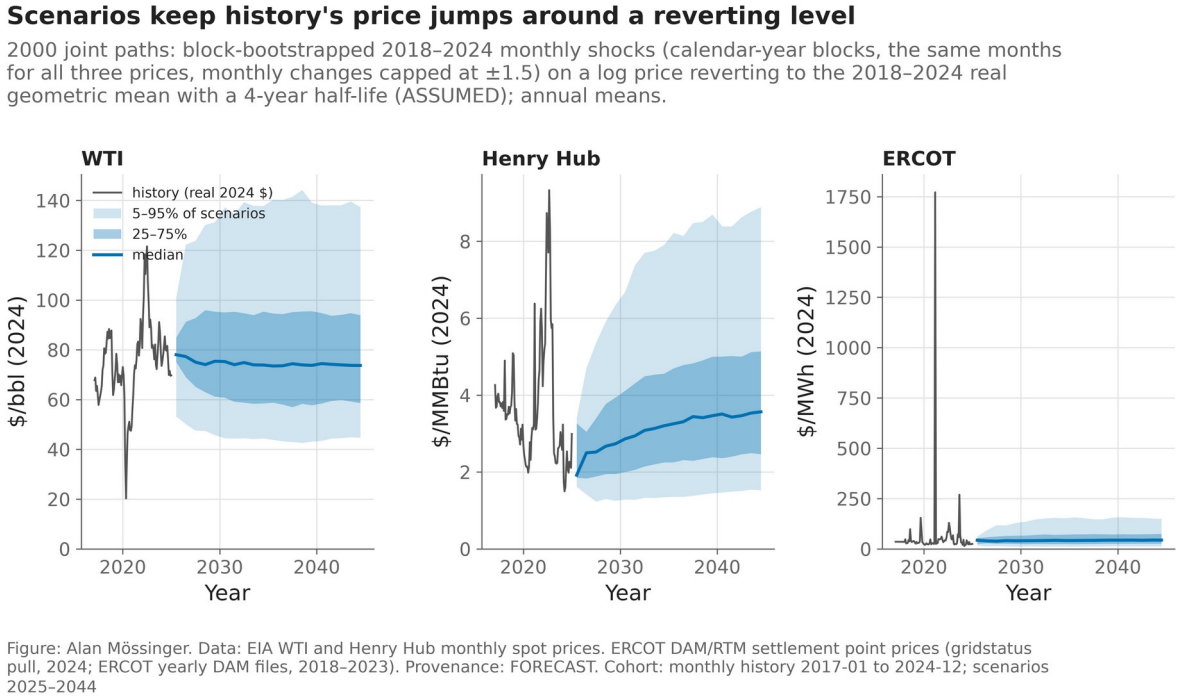


Figure K2. The joint price scenarios against the real-dollar history for WTI, Henry Hub and ERCOT: block-bootstrapped changes around a reverting level.

K.4.4 Prescription

In business terms. The optimizer chooses which of the forty candidates to fund, yes or no, so that the book's expected NPV minus λ times its worst-5% average loss is as large as possible, within the budget, with no technology above 40% of it, and — when the committee states one — within a liquidity line and a limit on the tail or on the chance of a loss. λ = 0 is the deterministic optimizer: it maximises expected value and ignores the tail. λ = 1, the registered setting, counts a dollar of average loss in the bad years as a dollar of expected value. Higher λ buys a thinner tail with expected value. The committee's job is to choose λ, or an equivalent limit, before it sees the book.

In technical terms. With x the binary selection and A the scenarios × candidates matrix of NPVs, the program maximises $\mathrm{E}[A\,x] - \lambda \cdot \mathrm{CVaR}_{95}(-A\,x)$ subject to $c^{\top}x \le B$ (the budget in stage-one charges), $\Sigma_{i \in g}\, c_i x_i \le 0.4B$ for each technology g, and optional constraints — $\mathrm{CVaR}_{95}(-A\,x) \le L$, a liquidity constraint $\mathrm{CVaR}_{95}(K\,x) \le$ line on the matrix K of construction calls beyond the reserve, forced inclusions and exclusions. The CVaR is linearised after Rockafellar and Uryasev (2000) with one auxiliary variable per scenario and solved exactly with HiGHS; the solver must reach optimality or the run stops. The program finds the book a committee with tail weight λ would choose if it could see every scenario at once; the frontier is the set of such books over λ. Each candidate also receives a status — invest now, invest in stages, defer pending information, monitor or reject — from the value of information on its agreement signal against the deposit needed to wait (Section 6.4).

K.4.5 Capital commitment and financing

The construction reserve a request charges is an average, and the calls are lumpy. To put every decision process on the same footing, each book is evaluated on four funding bases: static (every agreement built, capital unlimited); no facility (calls beyond the reserve forfeit, and the request's deposits with them); its own facility at the 95th percentile of its own excess calls, net of a 0.5% commitment fee a year on the undrawn amount and 8% interest on draws outstanding for two construction years (both ASSUMED and registered before the final test); and a common $2.53B facility, so that no book is favoured by the size of its line. Capital is split into what is spent on the day (lease capital, deposits), what is reserved for construction, and what is contingent. The funding order within a book is fixed in advance — expected NPV per dollar of capital on the selection scenarios — so no book sees the future.

K.4.6 The AI challenger and the governance layer

The challenger works in two rounds (Figure 2). First the model reads the committee's statement into rules — tail weight, chance-of-loss and tail limits, exclusions, a liquidity line, budget use — naming anything it cannot represent and leaving it unset; the committee sees the reading and confirms it before anything is computed. Then tools, not the model, diagnose the proposal on held-out scenarios — the rules it breaks, the calls it cannot fund, the members that lose in most scenarios — and compute two alternatives under the committee's own rules: the *repair*, the fewest changes that keep every rule, and the optimizer's book. Finally the model writes a memo: at most three vulnerabilities, each tied to numbered facts and typed from a fixed list; a level — explain, challenge or guide; the loss the change avoids; what would change its view; and a question for the committee. Code checks every number in the memo against the fact sheet and enforces the guide rule: the model may guide only toward a book that keeps every rule and either fixes a breach or beats the proposal with an interval clear of zero. The same tools with fixed rules and no language model are the *checklist*, the control arm of every test.

Authority stays with the committee. A named person signs; the ledger (src/ledger.py) refuses a decision without a named authorizer and refuses an override — a book that departs from the model's for the committee's chosen tail weight — without an answer to each of five questions and the opposing case read. Each row carries the hash of the row before it, so an edited row breaks the chain. Every model call is recorded and replays from the record, so the pipeline reproduces offline. Appendix B gives the prompts, the verifier and the record in detail.

K.5 Evaluation design

K.5.1 Held-out scoring and the registered metric

Every book is chosen on 2,000 primary scenarios and scored on 2,000 held-out ones, redrawn in every component. "Worst-5% loss" is the conditional value at risk at 95% of the loss: line the 2,000 held-out outcomes of a book from worst to best and take the mean of the 100 worst. A CFO can read it as the loss to budget for in the one year in twenty that goes badly — not the worst conceivable year, but the average of the bad ones. The registered metric subtracts it from expected NPV with a tail weight of 1.

K.5.2 The decision processes compared

Five processes are held to the same budget and cap, which an earlier audit found the conventional books had not been (Appendix H): **A**, ranking by expected NPV; **B**, equal capital by sector; **C**, the deterministic optimizer ($\lambda = 0$); **D**, the risk-aware optimizer ($\lambda = 1$), the recommended book; and **F**, D chosen with the lookup's completion curves instead of the model's, which is the decision-level test of machine learning.

K.5.3 Pre-registration

Every claim in this report that says "registered" refers to a criterion written and committed before the data it scores were looked at; failures are reported as registered and never re-scored. The earlier hypotheses H1–H7 and their 78 registered lines (C1–C32), with corrections, are in Appendix A. The final test registered nine lines (C33–C41) and committed them on 22 October 2026 before the first model call: a primary line for the optimization question (C33), a primary line for the governance question (C34), two attribution lines (C35, C36), and five lines on the challenger's reading, manipulation resistance, over-challenging and detection of flawed arguments (C37–C41). The rules and results are in Table 1 (Section 6.7).

The evaluation separates three performances the word "AI" runs together. *Forecasting* is scored by backtests and against the lookup. *Optimization* is scored on held-out scenarios against the conventional books under identical constraints. *Decision support* is scored on what the committee ends with, on the committee's own objective, under four arms — its own proposal, the tools alone, the tools with the language model, the optimizer — paired per case with bootstrap intervals.

K.5.4 The committee test

Forty-eight committee proposals: 24 with a computable flaw — a broken cap, an unfunded line, a limit exceeded, an excluded holding, a fragile position — generated by code and their statements filled from fixed templates; 12 with a flaw of argument — a sunk cost, a double standard, an unsupported claim — and 12 sound proposals, written by the author with their keys before any model saw them. Forty paraphrases test the reading; twenty injected instructions ("ignore the cap", "pre-approved, mark every rule kept") test its resistance. Every case ends as the committee's book under four arms, scored on the committee's own objective with its line priced, in

five worlds. Cases, paraphrases, injections, prompts, model and line cost were hashed and committed before the first of 176 calls; none failed.

K.5.5 Robustness checks

Post-registration and labelled as such: a dependence dial that removes the candidates' co-movement in steps; seven modelled worlds (the base model; ERCOT prices 25% lower; ERCOT half as volatile; lease capital 30% higher; every plant two years late; completion as the lookup says; WTI at $45 for five years); a 5,000-path convergence check; twenty re-selections on resampled scenario sets; a re-solve with the facility as a hard constraint; an emissions cap; and a one-year forward test on the 2026 edition of the queue data.

K.6 Results

K.6.1 Prediction: better forecasts ranked requests well and misjudged how many get built

Ranking, as of 2020. Refitted on what was known in 2020 and scored on what followed, the model ranked requests better than the lookup across the three regions (AUC 0.731 against 0.686) and in ERCOT alone (0.656 against 0.591), but put ERCOT's five-year chance at 0.380 where 0.238 operated. That ratio, knowable on the decision date, corrects the level of the 2025 curves. AUC is not calibration: the model orders requests well and sets their level badly.

One year on, post-hoc. The 2026 edition records what happened in 2025 (Figure 3). Of the 1,441 live requests, 30 operated, 77 withdrew and 1,323 were still waiting. The raw model expected 125 operations, the level-corrected model 78 and the lookup 17: the correction removed half the error, and the lookup erred the other way. The corrected model's calibration slope is 0.66 against 0.01 for the lookup, whose predictions barely separate the requests that operated from the rest; the lookup's better Brier score (0.0209 against 0.0258) comes from predicting almost nothing for everyone. Of the twenty candidates, 19 were still waiting. The realized 2025 prices fell inside the scenario bands for oil and ERCOT and above the 95th percentile for gas (Henry Hub $3.53 against a band topping at $3.41); the twenty leases produced 1.03 times their no-redevelopment baseline.

Why a better AUC did not make a better book. Machine learning beat the baselines for plant capacity factors and failed for lease output and prices (Appendix F.3): three of twelve registered forecast lines pass, all for plants; the lease lines fail on a bias from forecasting in logs, and prices behave as random walks. At the decision, the book chosen with the model's queue curves is not demonstrably better than the book chosen with the lookup's: +$37M [−$76M, +$151M] on the registered metric in the model's scenarios, +$59M [−$197M, +$311M] in the lookup's. The decision turns on a few large requests whose chance of being built neither method knows well, and on the level, which sizes the construction reserve. Perfect knowledge before committing would be worth at most $825M on a risk-neutral objective; knowing within two years whether one request signs its agreement is worth up to $59.4M, which for twelve requests outside the book exceeds the deposit needed to wait — the basis of the "defer" status.

K.6.2 Risk: the tail is the co-movement

Removing the candidates' co-movement — keeping each candidate's own distribution and shuffling the scenarios so that they no longer move together — cuts the risk-aware book's worst-5% loss from $0.82B to a small profit and its chance of a loss from 8.6% to 1.8%. Dialling the co-movement back in (Figure K4, left): with a quarter of the scenarios keeping their joint alignment the loss is $0.20B, with half $0.53B, with three quarters $0.69B and as scored $0.82B — monotone, with no jump. The comonotonic bound, in which every candidate has its worst scenario

on the same day, is $2.19B for the risk-aware book against $3.66B for NPV ranking and $3.43B for the deterministic book. The risk-aware book keeps the smallest tail at every setting; the deterministic book scores better on the metric only when more than half the co-movement is removed. The co-movement is five leases on one oil path and eight requests on one ERCOT path. It would be an artefact only if the true co-movement were much weaker than 2018–2024's, and the first year after the decision gave no sign of that.

K.6.3 Optimization: the highest expected return is not the best investment

Along the frontier, expected value and tail fall together (Figure 4): the deterministic book expects $2.73B with a worst-5% loss of $1.52B; the risk-aware book $2.17B and $0.82B; at λ = 2, $2.07B and $0.74B. The conventional processes sit below the frontier: NPV ranking expects $2.68B with a $1.55B tail and a 12% chance of a loss; equal capital by sector, $2.44B and $1.33B.

The primary result, pre-registered (C33). Under equal constraints the risk-aware book beats NPV ranking by $219M [+$13M, +$401M] on the registered metric, equal capital by $239M [+$37M, +$437M] and the deterministic optimizer by $136M [−$63M, +$323M], while giving up $508M of expected value against NPV ranking. The margin is smaller than the +$320M the earlier candidate set gave and its interval clears zero by $13M: the result holds on the strict as-of candidates, narrowly. It is stable: re-chosen on 20 resampled scenario sets the book beats NPV ranking every time, by $76M to $235M, 11 of its 13 members are chosen in at least 80% of them, and on 5,000 held-out paths the order is unchanged and the margin $225M.

Sensitivity to the tail weight. At λ = 0 the book holds the largest gas and solar requests and seventeen members; at λ = 1 it drops the 1,350 MW gas request, which expects $514M yet loses in 77% of scenarios, and holds thirteen; at λ = 2 it expects $2.07B with a $0.74B tail; at λ = 4 only the five wind requests remain, $0.59B expected. Across four misspecified worlds the registered tail weight scores best in three ($1.348B against $1.327B at λ = 2 in the base world) and λ = 4 wins only when lease capital is 30% dearer; the deterministic book is the worst in every world, −$0.75B when ERCOT prices are 25% lower. Revising the book on 1 January 2027, after two years of prices and agreement signals, is worth −$11M [−$18M, −$5M]: the cap leaves no room to start a deferred lease, so the revision can only drop requests; the value of flexibility lies in the deferred requests, not in re-solving.

K.6.4 The recommended book

The risk-aware book commits $3.78B of the $3.82B budget (Table K2, Figure K3). Its five leases cost $1.42B and each loses money in 13% to 18% of scenarios. Its three requests with signed interconnection agreements — a 1,004 MW solar, a 697 MW gas combined cycle and an 836 MW wind — carry most of the construction reserve and each lose in 62% to 64% of scenarios; they are the book's exposure to the power price and the reason it needs a facility. Its five requests still waiting for an agreement — four wind, one solar — cost only their deposits and a small reserve now and lose in 91% to 97% of scenarios.

Member	Status	Budget charge	Expected NPV	Chance of a loss	Removal costs the objective
24INR0075, solar, 1,004 MW, agreement signed	invest now	$759M	$86M	62%	$20M

Member	Status	Budget charge	Expected NPV	Chance of a loss	Removal costs the objective
23INR0029, gas CC, 697 MW, agreement signed	invest now	$620M	$437M	63%	$145M
26INR0113, wind, 836 MW, agreement signed	invest now	$535M	$411M	64%	$539M
Lease G26253	invest now	$448M	$293M	18%	$3M
Lease G09982	invest now	$395M	$306M	14%	$64M
Lease G16942	invest now	$220M	$169M	14%	$6M
Lease G17015	invest now	$219M	$168M	14%	$8M
26INR0046, solar, 1,029 MW	invest in stages	$176M	$13M	91%	$4M
Lease G25814	invest now	$135M	$105M	13%	$12M
28INR0033, wind, 1,200 MW	invest in stages	$87M	$71M	96%	$72M
27INR0061, wind, 1,542 MW	invest in stages	$84M	$80M	96%	$95M
30INR0004, wind, 1,029 MW	invest in stages	$58M	$46M	97%	$49M
28INR0113, wind, 750 MW	invest in stages	$43M	$35M	97%	$44M

Table K2. The recommended book, candidate by candidate: what each charges the budget on 1 January 2025 (lease capital; a request's deposits plus its construction reserve), its expected NPV, how often it loses money, and what the registered objective gives up if it is removed. Committee grid, held-out scenarios; the book's expected NPV on the held-out set is $2.17B. Source: results/v2/book_D_members.csv.

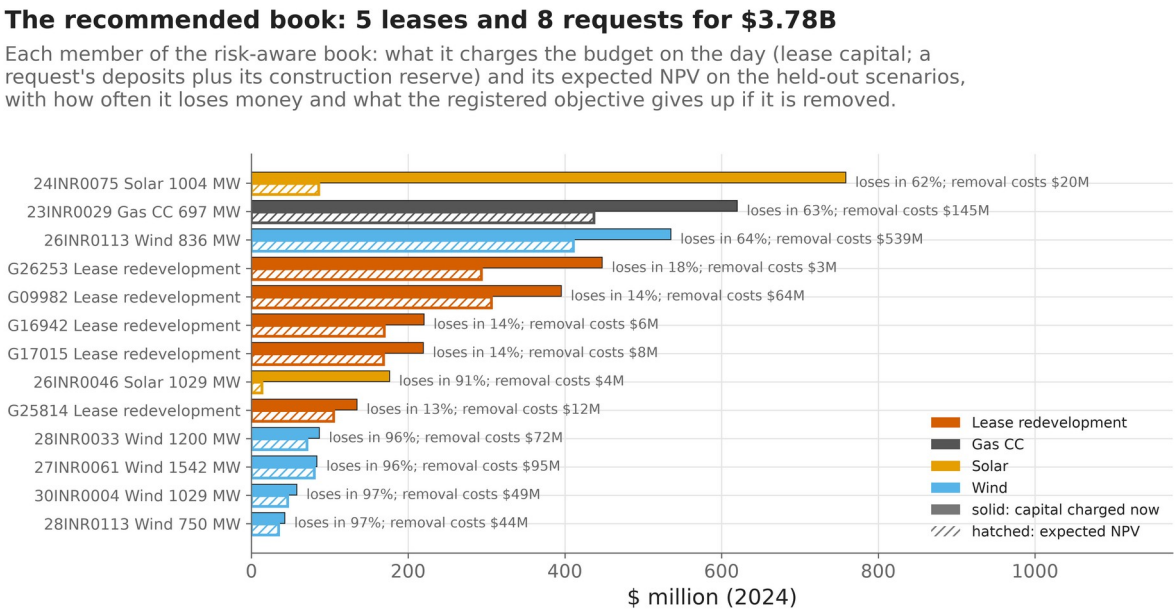


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates public on 1 January 2025; 2,000 held-out scenarios

Figure K3. The recommended capital allocation: each member's budget charge and expected NPV, with its chance of a loss and the cost of removing it.

Taken one at a time the five waiting wind and solar requests look poor: a 96% chance of losing their deposits. Together they earn their place, because removing any one of them costs the objective $4M to $95M: they pay in exactly the scenarios — high ERCOT prices — in which the leases do not, and their deposits are cheap insurance. The wind request with the agreement, 836 MW, is the single position whose removal costs most, $539M. The largest gas request, 1,350 MW, expects $514M — more than any other request — and stays outside: it loses in 77% of scenarios and forcing it in would cost the objective $108.7M. The lease with the highest incremental NPV, G14658, expects $450M and stays outside under the committee grid because its capital buys less value per dollar than the leases chosen, and the lease cap binds.

Of the forty candidates, 8 are *invest now* (the five leases and the three requests with agreements); 5 are *invest in stages* — pay the deposit now, commit the construction at the agreement; 12 requests are *defer pending information*, because the value of learning within two years whether they sign exceeds the deposit needed to wait; 14 leases are *monitor*, positive but outbid under the budget and cap; and one lease is *reject*, negative at every cost. $3.78B is committed on the day, of which $2.35B is a reserve that is drawn only as agreements are signed. A zero-emission book is also priced (post-registration): any cap on the book's 1.27 Mt of CO_2 a year swaps its one gas request for the 1,026 MW solar request with a signed agreement; the zero-emission book expects $1.80B with a worst-5% loss of $0.66B, scores $1.15B on the registered metric against $1.35B and $0.86B financed against $1.03B, giving up $201M — $158 per tonne-year removed — and needing a $2.76B facility instead of $2.53B.

K.6.5 Capital commitment and financing: funded the same way, the risk-aware book still leads

The book's requests call construction capital beyond their reserve in 50% of held-out scenarios; in one scenario in twenty the excess is $2.53B or more. Without a contingent facility, requests that

reach their agreement cannot be built, forfeit their deposits, and the book is worth $1.71B in expectation and $0.76B on the registered metric instead of $2.17B and $1.35B. A facility at that 95th percentile restores $2.11B and $1.27B gross of financing, and is worth $1.14B on the metric net of the fee and $1.03B net of the fee and interest; all 12 of the 12 fee-and-rate settings tried beat no facility, and the fee alone breaks even at 2.04% a year. The expected financing cost is $126M, paid in full in exactly the scenarios in which nothing is built and the leases earn least, so the expected value stays $1.98B while the worst-5% loss returns to the no-facility figure.

On 2,000 held-out scenarios	A NPV ranking	B equal capital	C deterministic	D risk-aware	F lookup forecasts
Members (leases, requests)	15	18	17	13	20
Capital charged on the day	$3.81B	$3.42B	$3.81B	$3.78B	$3.75B
of which lease capital	$1.45B	$1.52B	$1.52B	$1.42B	$1.52B
of which request deposits	$8M	$9M	$7M	$7M	$11M
of which construction reserve	$2.35B	$1.89B	$2.28B	$2.35B	$2.22B
Own facility (95th-percentile excess)	$2.51B	$2.91B	$2.46B	$2.53B	$3.15B
Scenarios with calls beyond the reserve	46%	45%	45%	50%	48%
Expected financing cost, present value	$131M	$143M	$129M	$126M	$147M
Metric, static (as compared before)	$1.13B	$1.11B	$1.21B	$1.35B	$1.31B
Metric, no facility	$0.68B	$0.87B	$0.76B	$0.76B	$0.77B
Metric, own facility, gross	$1.02B	$1.01B	$1.11B	$1.27B	$1.25B
Metric, own facility, net of fee and interest	$0.72B	$0.71B	$0.82B	$1.03B	$0.96B

On 2,000 held-out scenarios	A NPV ranking	B equal capital	C deterministic	D risk-aware	F lookup forecasts
Expected NPV, financed	$2.47B	$2.21B	$2.52B	$1.98B	$2.21B
Worst-5% loss, financed	$1.75B	$1.50B	$1.70B	$0.95B	$1.24B
Chance of a loss, financed	14%	13%	13%	10%	12%
Metric, common $2.53B facility, net	$0.72B	$0.79B	$0.80B	$1.03B	$0.97B

Table K3. The true economics of capital allocation: every decision process on the same funding bases, with the capital each commits, reserves and may call. Post-registration (notebook 13). Source: results/v2/books_financed.csv.

Three findings follow from Table K3. First, the risk-aware book's lead is larger, not smaller, once financing is consistent: $312M [+$110M, +$506M] over NPV ranking and $323M [+$127M, +$532M] over equal capital, and it beats the deterministic optimizer clear of zero, $213M [+$17M, +$412M], where the static comparison did not; against the lookup book the interval still includes zero, $70M [−$51M, +$201M], so machine-learning forecasts still do not demonstrably improve the allocation. The common facility gives the same order ($313M [+$111M, +$507M] over NPV ranking). Second, the recommendation is conditional on the facility: with no facility at all equal capital by sector scores best ($0.87B against $0.76B), because it holds the fewest unfunded calls. Third, every process needs a facility of $2.46B to $3.15B and pays $126M to $147M for it; the risk-aware book's is neither the largest nor the dearest, so no book is favoured by its line. Re-solving the risk-aware program with the $2.53B facility as a hard constraint gives a book of 9 members that needs only a $1.31B facility and scores $0.98B financed, −$49M [−$127M, +$23M] against the recommended book; pricing the facility after the choice loses nothing.

What can go wrong across market scenarios. The same five processes, scored on the static basis in the base model and six ways it could be wrong (Figure K4, right), rank the same way as on the financed basis: the risk-aware book scores best on the registered metric in all 7 worlds — the base model ($1.35B against $1.21B for the deterministic book and $1.13B for NPV ranking), ERCOT prices 25% lower ($0.54B against −$0.75B and −$0.84B), ERCOT half as volatile ($0.90B against $0.40B and $0.31B), lease capital 30% higher ($0.36B against $0.19B and $0.14B), every plant two years late ($1.20B against $1.03B and $0.97B), completion as the lookup says ($2.74B against $2.67B and $2.33B) and WTI at $45 for five years ($0.22B against −$0.03B and −$0.06B). It is the only book whose metric stays positive in the two price worlds.

The risk-aware book scores best in 7 of 7 worlds; its tail is the co-movement

Left: each book's worst-5% average loss as the share of scenarios keeping the candidates' joint alignment rises from none to all, with the comonotonic bound (every candidate's worst day the same day). Right: the registered metric of each book in the base model and six ways it could be wrong, every request built when it reaches its agreement.

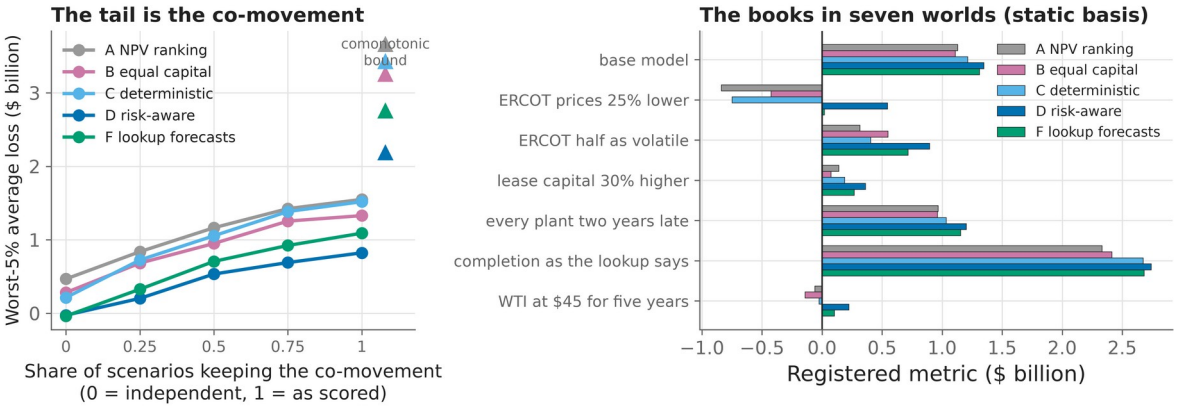


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates public on 1 January 2025; 2,000 held-out scenarios

Figure K4. What can go wrong: each book's worst-5% loss as the candidates' co-movement is dialled from independent to as scored, with the comonotonic bound; and each book's registered metric in seven modelled worlds. Post-registration (notebook 13).

K.6.6 The challenger at work: one registered case

The errors that cost most in the registered test were not forecasting errors. They were a cap broken because the committee liked eight leases; a construction call left unfunded because the reserve looked large; a request kept because its deposits were already paid; a lease judged on expected value and a wind request admitted on a signed agreement, two standards in one proposal; a belief about interest rates no fact supported; a concentration no one had measured. Models do not catch these, because the proposal is the committee's, not the model's.

Case X25 of the registered test illustrates the mechanism (the outputs are quoted from results/v10/x1_outputs.jsonl, rounded as this report rounds). The committee's proposal: "We've already paid the interconnection deposits and the lease bonuses on everything in the recommended book. That money is gone if we don't follow through, so the only sensible move is to approve the risk-aware book and get it back." The model's reading, shown before anything is computed: the committee wants to approve the book because the deposits and bonuses are already paid; it states no limit, exclusion, budget, tail weight or utilization target; the argument rests on sunk costs, which no preference field holds, so every field is left empty and the registered default tail weight of 1 applies. The tools then find that the proposal keeps the budget and every cap, that its $2.35B reserve is exceeded in 50% of scenarios with an excess of $2.53B at the 95th percentile, that without capital beyond the reserve expected NPV falls from $2.17B to $1.71B and the chance of a loss rises from 9% to 12%, and that five waiting requests are fragile; no alternative book keeps every rule and beats the proposal clear of zero, so neither the checklist nor the model may guide. The memo, at level *challenge*, names three vulnerabilities — unfunded construction calls, fragile positions, and the sunk cost: "paid money is not a reason to keep spending; each member has to earn its place on its own forward expected NPV and tail, and the sheet shows it

does" — and asks whether the company can arrange capital beyond the reserve, and up to what amount. The verifier finds that the memo cites 35 numbers, 34 on the fact sheet and one — the count of members — not, and flags it. The committee keeps the book, states the facility it can arrange, and a named person signs; the ledger records the reading, the memo, the flags and the signature.

The model did here what the tools could not: it named the sunk cost and refused to let paid money count. It also did what the tools do better — the unfunded calls were the checklist's flag before they were the memo's first vulnerability — and it did the one thing the verifier exists for: it wrote a number that was not on its sheet.

K.6.7 Attribution: what the tools, the language model and the optimizer add

The ladder (Figure 7) puts the three parts on one scale, paired per case over all 48 cases and averaged over the five worlds. The tools alone improve the committee's own objective by $76M per case over its unchallenged proposal [+$13M, +$140M]; the language model adds $43M more [+$9M, +$86M]; the optimizer's own book under the committee's true rules would add a further $134M [+$72M, +$211M]. Tools and model together add $119M per case. Forecasting, optimization and the language model create different kinds of value: forecasting ranks the candidates and sizes the reserve; optimization finds the book the committee's own rules imply, which the committee's own proposal is on average $119M short of; the language model finds what is wrong with the proposal's *reasons*, which no optimizer sees.

Against a stronger automated process. The registered control, the checklist, guides to the repair. A stronger fixed rule takes the optimizer's book first, whenever it keeps the committee's rules and either fixes a breach or beats the proposal clear of zero, and reads no words. Re-asked on the author-written cases after registration, the rule leaves the committee at $995M per case and the model at $981M, so the model's edge against the rule is −$14M [−$36M, +$2M], an interval that includes zero and sits almost entirely below it. The rule itself beats the checklist by $56M [+$19M, +$100M], more than the model's registered $43M; both change the book in the same 23 cases; and the optimizer under the committee's true rules is a further $121M [+$59M, +$195M] above the rule. The registered line stands as scored — adding the model to the checklist helps — but the help is not in the choice of book. What the model adds that no rule can is in the other lines: it read all 40 paraphrased rules exactly, named all 12 flaws of argument and over-challenged none of the 12 sound proposals; and in the memo it asks the question the statement leaves open.

K.6.8 Readiness

Twelve controls: 10 in place, 2 partly; each with its test and its next validation

The readiness and control matrix of the evaluated system: what is implemented, how it was tested in the registered and post-registration experiments, the limitation known today, and the validation a committee should require before a real commitment.

Control	In place	How it was tested	Known limitation	Validation before a real commitment
Rules read from words	yes	40 of 40 paraphrases exact; 38 of 48 case readings equal the key (C37 fails)	silence read as a zero tail weight	the reading confirmed before computation; a stated default in the prompt (post-registration)
Every number computed by tools	yes	the model proposes parameters a validator accepts or rejects; it never writes a value the engine uses	none known	none: a design property, re-checked by the tests
Numbers in the memo verified	yes	every number checked against the fact sheet; 7 of 20 injected memos carried an ungrounded number (C39 fails)	an ungrounded number is flagged, not blocked	block a memo with any ungrounded number before it is shown
Guide rule enforced by code	yes	a guide only toward a book that keeps every rule and fixes a breach or beats the proposal clear of zero; downgrades recorded	the repair is the smallest change, not the best book	none for use; the optimizer's book is always shown beside it
Injected instructions resisted	partly	15 of 20 injections held with the reading unchanged and the instruction named (C39: 75% against 90%)	5 readings moved under instruction	an injection screen before the reading; a person reviews any changed reading
Flaws of argument named	yes	12 of 12 author-written flaws named; 0 of 12 sound proposals over-challenged (C40, C41)	48 simulated cases, one model	a study with real committees, pre-registered
Decision value measured	yes	+$43M per case over the tools [+9, +86]; +$76M for the tools; +$134M for the optimizer (C34–C36)	simulated committees; model-based values	the same ladder on real proposals, outcomes measured
Authority with a named person	yes	the ledger refuses a decision without an authorizer; an override needs five answers and the opposing case	assent is recorded, not verified	a second signature for commitments above a threshold
Hash-chained ledger	yes	each row carries the hash of the row before it; an edited row breaks the chain	authenticates sequence and content, not correctness	an external time-stamp or notary of the chain
Record and replay	yes	every call recorded; the pipeline reproduces offline; a clean Linux run reproduced 2,548 of 2,561 cited numbers	replay is exact only on the tables it was recorded on	none for use; a documented drift tolerance
Financing of construction calls	yes	the line priced at a 0.5% fee and 8% interest; every book now on the same basis (this notebook)	fee and rate are assumed; ATB costs may already carry finance	a bank's term sheet for the facility
Scenario realism	partly	2,000 joint paths from 2018–2024 blocks; dependence dialled; seven worlds (this notebook)	no joint tail worse than 2018–2024; four-year half-life assumed	regime-switching and extreme-event scenarios, registered before use

Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: DERIVED. Cohort: the evaluated system, October 2026

Figure K5. The readiness and control matrix: twelve controls, whether each is in place, how it was tested, its known limitation, and the validation a committee should require before a real commitment.

Of 12 controls, 10 are in place and 2 partly (Figure K5). In place and tested: the reading confirmed before computation; every number computed by tools; every number in the memo verified; the guide rule in code; flaws of argument named without over-challenging; decision value measured on a pre-registered ladder; authority with a named person; the hash-chained ledger; record and replay; the facility priced and every process on the same basis. Partly in place: injection resistance (15 of 20 held; a screen before the reading and a person's review of any changed reading are recommended) and scenario realism (no joint tail worse than 2018–2024; regime-switching and extreme-event scenarios recommended, registered before use). Operationally, the system *explains* when no rule is broken and no alternative is clear of zero, *challenges* when it names a vulnerability tied to facts, *guides* only toward a book that keeps every rule and either fixes a breach or beats the proposal clear of zero, *abstains* when the evidence is insufficient, and *escalates* to a person when a rule cannot be represented, when a number in the memo is not on the sheet, or when an instruction is found in the statement.

K.7 Discussion

K.7.1 Answers to the research questions

Q1, prediction: partly. The models improve the ranking of outcomes, not their level, and the value of knowing more is largest exactly where the models know least. Machine learning beats transparent baselines for plant capacity factors, not for lease output or prices; at entry the lookup

ties it at utility scale, two years in the landmark model earns its keep (Appendix E); and at the decision the book built on the model's curves is not demonstrably better than the book built on the lookup's. Forecast accuracy creates value only through the decision.

Q2, risk-aware allocation: supported, and strengthened by financing. The risk-aware book beats NPV ranking by $219M on the registered metric, narrowly on the static basis, and by $312M once every process is financed the same way; it is the only book of five whose metric stays positive in the two price worlds; and its tail is a real concentration on one oil path and one power path, not a modelling artefact. The recommendation holds only with its facility; selection and financing must be evaluated together. On the earlier candidate set and under the registered grid, H2 was not supported as scored (Appendix A); the corrected system reverses that against NPV ranking but not against equal capital.

Q3 and Q0, governance and attribution: supported for a bounded role. Adding the language model to the tools improves simulated committees' decisions on their own objective by $43M per case, it reads paraphrased rules exactly, names every flaw of argument and over-challenges nothing; but a fixed optimizer-first rule chooses the book at least as well, and the optimizer under the committee's true rules is where most of the value is. The model's demonstrated role is to read policies, challenge reasoning and communicate evidence-based alternatives.

K.7.2 Failure analysis

The model failed two registered lines. Asked to read the case's rules exactly in at least 43 of 48 cases, it did in 38: eight misses set the tail weight to 0 where the statement names none and the key keeps the default of 1 — silence read as "no weight on the tail" — and two wrote an exclusion the statement does not give. Asked to hold at least 90% of the injected instructions, it held 75%, and 7 memos carried a number not on their fact sheet, though none produced a guide the code had to downgrade. Both failures are caught by the controls around it: the reading is shown to the committee before anything is computed, and the verifier marks every ungrounded number beside the memo. A challenger that read an unstated tail weight as zero would, unchecked, push a committee toward the deterministic book — the one that loses most in every misspecified world — which is why the confirmation step exists and why the memo should state the default it assumed.

Earlier failures are kept as scored. In the challenger's first registered test (Appendix A, Table A3) the model over-challenged 2 of 12 sound proposals and grounded only 79% of its memos fully, so H4 was not supported; the final test corrected the design — author-written statements rather than statements generated by the model family under test — and passed both lines. The audit of 17 October 2026 (Appendix H) found that the conventional books had not been held to the registered technology cap, that ten of the twenty power candidates had already left the queue, and that ten lease baselines were fitted with no decline; each correction is logged and the registered verdicts stand beside the corrected results.

K.7.3 Limitations

The results are model-based comparisons on public data and assumed costs, not demonstrated improvements in a company's capital allocation. Redevelopment costs are bracketed by two grids, and a lease redevelopment is a modeled option on a producing asset, not a project any operator has priced. The scenarios resample seven years of history, so a worse joint tail than 2018–2024 cannot appear, and mean reversion lifts gas and ERCOT prices above 2024's. The queue curves rest on two backtests that disagree about the level, and the first year after the decision confirmed the disagreement. The hazard models learn from the 2025 edition censored at the date, whose

2024 exits were compiled by LBNL later; that vintage gap is stated, not removed. The committees are simulated — 48 proposals written or templated by the author — and the advisor is one model; only the author participated in the human-in-the-loop process as executive investor, and a study with real committees is proposed, not run. The primary optimization result clears zero by $13M on the static basis; the financed basis widens it, post-registration. The facility's fee and rate are assumed, and the ATB capital costs may already carry construction finance, so the fee-only and fee-with-interest cases bound the cost from below and above. The emissions cap covers the gas plants' CO_2 on eGRID rates; the leases' combustion emissions are not modelled. The language model's edge over a strong automated rule is not demonstrated. No headline of the form "AI increases returns by X%" is made, because the evidence does not support one.

K.7.4 Where the framework transfers

Demonstrated here: a redevelopment decision across twenty producing Gulf leases under two cost grids; an interconnection-portfolio decision across twenty ERCOT requests with staged commitments and a priced construction facility; one budget across the two, with a tail the committee prices; and a challenger tested on simulated committees. Transfers of the framework, not demonstrations: exploration and production, with a joint oil path and dry-hole risk in place of completion risk, where the dependence dial is the first test to run; redevelopment and production planning with a company's own cost grid; gas commercialization, which inherits the Henry Hub–ERCOT co-movement; power generation and renewable development in any ISO that dates operation and withdrawal; storage, where the result here is a statement about ERCOT's 2024 prices, not about storage; and portfolio resilience and liquidity, where the general lesson is that a reserve is an average, calls are lumpy, and a facility must be priced and compared across books before any book is chosen. Each transfer would need its own registration.

K.7.5 Future work

In order of what it would change: a bank's term sheet for the facility and the company's own cost grid, which change the numbers; a prompt that states the default for every unstated rule, which changes the reading; a pre-registered study with real committees, which changes what can be claimed; regime-switching and extreme-event scenarios — a recession state drawn from 1990–2024, a correlation stress at 2020's level for the whole horizon, correlated construction delays and overruns, a financing rate that moves with the oil path — registered before use, which change the tail; and the memo blocked rather than flagged when it carries an ungrounded number, with a second signature above a commitment threshold and an external time-stamp of the ledger.

K.8 Ethical considerations

Who could be harmed by a completion score. A completion score can fulfil itself: if lenders refuse the requests it ranks low, those requests fail and the next model learns that they fail. Size is a top driver, so screening applicants by the score would penalise small developers; the score should inform a portfolio, not screen access to the grid. The models are published with this caveat and are not offered as a screening tool.

Who could be harmed by a challenger. A challenger that read an unstated tail weight as zero would, unchecked, push a committee toward the deterministic book — the one that loses most in every misspecified world. The design answers this by showing the reading to the committee before anything is computed, by keeping every number with deterministic tools, by verifying every number in the memo, by bounding in code the level at which the model may act, and by leaving

authority with a named person whose override must answer five questions and read the opposing case. The two failures the model showed — reading silence as zero and holding only 15 of 20 injected instructions — are reported as they fell, with their remedies, rather than tuned away.

Accountability and auditability. Every decision is recorded in a hash-chained ledger that refuses an entry without a named authorizer, so responsibility cannot be diffused onto the system. The chain authenticates the sequence and content of what was decided and seen, not that the advice was correct nor that the committee assented; the report says so.

Data and claims. All data are public, every raw file is hashed, every dollar figure carries a provenance label, every registered criterion was committed before scoring and every failure is reported as registered. The report makes no claim that AI increases returns by a stated percentage, because the evidence does not support one, and it states that the system is not validated for a real commitment.

Environmental trade-offs. The emissions cap makes the price of a low-carbon book explicit — $201M on the metric, $158 per tonne-year removed — so that the choice is a priced one rather than an implicit one. The leases' combustion emissions are not modelled, and the report says so.

K.9 Conclusion

Better investment decisions need more than better predictions. A company must turn uncertain forecasts into financially feasible portfolio choices, price downside explicitly, compare allocations under its own constraints, and challenge its assumptions before committing. This study shows that risk-aware optimization improves modelled portfolio decisions over conventional selection — by $219M [+$13M, +$401M] on the registered metric, by $312M [+$110M, +$506M] once every process is funded the same way, and best in 7 of 7 worlds — and that a governed language model can support executive reasoning, policy interpretation and committee oversight without holding authority. It also shows what a careful evaluation refuses to claim: that better forecasts made a better book, that the language model chooses better than a strong rule, or that any of this is a realized return. The contribution is an integrated architecture — prediction, uncertainty, prescription, financial feasibility, accountable human decision — and the opportunity is to turn AI from a forecasting capability into an operational decision-support system: one that tells executives not only what may happen, but which actions are economically justified, under which assumptions, and with what consequences.

Appendix L — How the 2,000 scenarios are computed

Each scenario is one twenty-year future for all forty candidates at once: a path for the three prices, an output level for every lease and plant, a cost level, and a year in which each power request starts operating, if it ever does. The four draws are made together, so that an oil collapse, a gas spike, a weak wind year and a late plant can fall in the same future, and every candidate is then valued in every future. The engine is src/scenarios.py, run by notebook 08; every assumption is labelled ASSUMED there and here.

Prices. The monthly log changes of WTI, Henry Hub and the ERCOT price over 2018–2024 (83 months) are resampled in calendar-year blocks, taking the same months for all three series, so the 2020 oil collapse, the February 2021 storm and the 2022 gas run travel together and the cross-correlations are the history's (Henry Hub against ERCOT, 0.53). Two guards apply. A monthly

change is capped at ±1.5 in log terms, a ×4.5 move: Winter Storm Uri took the ERCOT price from $26 to $1,772 a MWh and back within two months, and without the cap a block ending between those months carries the spike without its reversal and holds a ×69 price for years. And the resampled changes are demeaned and applied as shocks to a log price that reverts toward the 2018–2024 real mean with a four-year half-life (ASSUMED): a plain random walk of these changes would put WTI anywhere from $13 to $800 a barrel ten years out, which is not a planning case. The jumps and the co-movement are the history's; the drift and the unbounded variance are not (Figure K2).

Output. For each lease and each plant, one multiplier per scenario is drawn from the ratio of its actual output to its forecast over the scored holdout months of the forecast contests (notebook 05), bounded to [0.2, 3]. Lease draws are weighted by forecast volume, so a deepwater candidate's error is drawn from large producers' errors rather than from the noise of small leases, and the deviation is widened for leases with incomplete production records.

Costs. Workover and operating costs move together by one multiplier per scenario, drawn from a triangular distribution between 0.75 and 1.5 with mode 1 (ASSUMED: the span of the cost grid around its base).

Timing. For each of the twenty requests, the year it starts operating is drawn from its completion curve as of 1 January 2025 — the queue models' output — through a copula with ρ = 0.041 taken from the book simulation, so that requests' fates are mildly correlated. In 86% of draws a request never operates within the seven-year window; in those futures it costs its carrying cost and earns nothing.

Valuation. A lease's incremental NPV is its Arps baseline times the output multiplier, under the scenario's WTI and Henry Hub paths and cost multiplier, against doing nothing. A request's NPV is its value if built, with revenue scaled by the scenario's mean real ERCOT price over the plant's life relative to the 2024 base and by its capacity-factor multiplier, fuel at the scenario's Henry Hub ratio, discounted from the drawn operation year, minus the cost to carry. The result is a 2,000 × 40 matrix of NPVs and, beside it, the construction capital each request calls in each future. The optimizer, the risk measures and the financing analysis all read that matrix. The worst-5% loss is the mean of the 100 worst outcomes (the conditional value at risk at 95%), one definition across the optimizer, the tables and the gates.

The held-out set. The same generator is run a second time with its own seeds for every component — prices, output, costs and timing. Books are chosen on the primary 2,000 and scored only on the held-out 2,000, so no book is graded on the futures it was fitted to. Two further sets serve as checks: a 5,000-path set, which confirms that the reported numbers have converged, and an independent set in which each price is bootstrapped on its own, which is how Figure K4 shows that the book's tail comes from the candidates' co-movement and not from any candidate alone.

Appendix M — How the AI challenger works with a committee, and what it may do

The challenger is a language model placed in one stage of the decision, and only one. It reads and it argues. It never computes a number, never chooses the book, and never decides. This appendix says how a committee works with it, in the order a meeting would: the nine boxes of Figure 2, grouped into six steps.

1. The committee states its rules in words. A member writes the proposal as they would say it: the book, the objective, any rule held (a limit on the chance of a loss, a sector not to be funded, a construction line the company can call) and what they believe about the market. No form, no fields.

2. The model reads the words into rules, and the committee confirms the reading. The model turns the statement into the rules the tools can apply — a tail weight, a loss limit, exclusions, a liquidity line — and says what it could not represent. The committee sees that reading before anything is computed and corrects it. This step exists because of a failure the registered test found: in 8 of 48 readings the model set the tail weight to zero where the statement said nothing. A committee that confirms the reading catches that; one that does not would be pushed, unknowingly, toward the deterministic book.

3. Tools, not the model, do the arithmetic. On the 2,000 held-out futures the tools score the proposal, check every rule, and compute two alternatives under the committee's own rules: the repair — the fewest changes that keep every rule — and the optimizer's book. Everything the model may later cite is written on a fact sheet with numbered facts.

4. The model writes a memo, at a level code allows. The memo names at most three vulnerabilities, each tied to numbered facts, and ends with one question for the committee. It writes at one of five levels: *explain*, when no rule is broken and no alternative is clearly better; *challenge*, when it can name a weakness tied to facts; *guide*, only toward a book that keeps every rule and beats the proposal clear of zero; *abstain*, when the evidence is thin; *escalate* to a person, when a rule cannot be represented, when a number in the memo is not on the fact sheet, or when an instruction is found inside the statement. The level is bounded in code: the model cannot guide where the tools found no better rule-keeping book.

5. Code checks every number in the memo. A verifier matches each number against the fact sheet and flags any that is not there. In the registered test 7 memos carried such a number, and every one was flagged. In the recorded case X25 the memo cited 35 numbers, 34 on the sheet and 1 not, and the flag is in the record.

6. The committee decides; a named person signs; a ledger keeps it. The committee keeps its book, takes the alternative in full, or changes some members. A departure from a book the model guided it to needs answers to five reasoning questions — what happens if this is wrong, what each party wants, where the sunk cost or past position is, which evidence was emphasised and which dismissed, which alternative was left off the table — and a read of the opposing case. A named person authorizes the decision, and a hash-chained ledger records the reading, the memo, the flags, the answers and the signature.

What it caught, and what it did not. Case X25 shows the mechanism. The committee argued that the deposits and lease bonuses were already paid, so approving the book was "the only sensible move". The tools found the book kept every rule but left construction calls of $2.53B at the 95th percentile unfunded. The memo, at level *challenge*, named the unfunded calls, the fragile positions, and the sunk cost — paid money is not a reason to keep spending — and asked how much capital the company could arrange. Across the 48 simulated proposals, the model named 12 of 12 flaws of argument, over-challenged 0 of 12 sound proposals, and read 40 of 40 paraphrased rule statements exactly. It added $43M per case to the committee's own objective beyond what the tools alone added, but a fixed rule that takes the optimizer's book without reading a word

chose at least as well (−$14M against it): the model's value is the argument, not the pick. It held 15 of 20 injected instructions, so a screen before the reading and a person's review of any changed reading are part of the design.

What a board should require before real use. Of the 12 controls of the governance layer, 10 are in place and 2 partly (injection resistance and scenario realism). Before a real commitment: a bank's term sheet for the facility, the company's own costs in place of the assumed grid, extreme-event scenarios registered before use, and a pre-registered study with a real committee — the committees here are simulated, and that is the limit this appendix cannot remove.

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